Liu, Chapter 2: General properties of schemes

Solutions to the exercises

Conventions. Rings are commutative with 1. Results of the text are cited by their numbers in Liu ("Prop. 3.14", "Lemma 2.7", ...). Exercise labels are those of the book, numbered within each section of the chapter: for instance Exercise 3.12 is the twelfth exercise at the end of Section 2.3. Each exercise is restated before its solution; sub-items that the book displays as nested lists are written as running prose.

Honest notes. Several printed statements need a correction or an interpretation, and each solution says what is proved: Ex. 1.4 (a nonnoetherian counterexample to nilpotency of $\mathfrak{p}A_{\mathfrak{p}}$), Ex. 2.3(c) (false for a non-separated presheaf, proved for separated ones, with a counterexample), Ex. 3.3(b) and Ex. 3.10 (a closure bar and the case $A = 0$), Section 2.4 (the empty open set in a statement with $U = \emptyset$, and the meaning of "indecomposable"), Ex. 5.5 (false if the ideal contains nonzero elements of degree 0). Some solutions rely on standard facts named where used.

2.1 Spectrum of a ring

Throughout, rings are commutative with 1. Numbers such as "Lemma 1.6", "Cor. 1.12" refer to the results of Section 2.1 itself (Chapter 2). We use freely that a finitely generated algebra over a field is noetherian (Hilbert basis theorem), and that the integral elements of a ring extension form a subring.

Exercise 1.1

> **Problem.** Let $A = k[[T]]$ be the ring of formal power series with coefficients in a field k. Determine $\operatorname{Spec} A$.

Solution. *Answer.* $\operatorname{Spec} A = \{0,\ TA\}$ has exactly two points: the generic point 0 and the closed point TA; the closed subsets are $\emptyset$, $\{TA\}$ and $\operatorname{Spec} A$ (so the nonempty open sets are $\{0\}$ and $\operatorname{Spec} A$).

Units. Let $a = \sum a_i T^i \in A$ with $a_0 \neq 0$. We solve $ab = 1$ for $b = \sum b_i T^i$: put $b_0 = a_0^{-1}$ and, recursively, $b_n = -a_0^{-1} \sum_{i=1}^{n} a_i b_{n-i}$. Thus a is a unit. Conversely a unit has nonzero constant term. Hence the non-units are exactly the series with $a_0 = 0$, i.e. the elements of TA.

Nonzero elements. Every $a \neq 0$ can be written $a = T^n u$ with $n \geq 0$ and u a unit (take n the smallest index with $a_n \neq 0$ and divide by T^n). So every ideal $I \neq 0$ equals $T^n A$ with n the

minimal order of its elements: indeed $T^n u \in I$ gives $T^n \in I$, and every element of I is a multiple of T^n.

Primes. 0 is prime since A is a domain. TA is prime (maximal), since $A/TA \simeq k$. If $n \geq 2$, then $T^n A$ is not prime ($T \cdot T^{n-1} \in T^n A$ but $T, T^{n-1} \notin T^n A$), and $A = T^0 A$ is not prime by convention. So $\operatorname{Spec} A = \{0, TA\}$.

Topology. $V(0) = \operatorname{Spec} A$, $V(A) = \emptyset$, and $V(T^n A) = \{TA\}$ for $n \geq 1$ because every prime containing T^n contains T. Hence the closed sets are $\emptyset, \{TA\}, \operatorname{Spec} A$, and $\overline{\{0\}} = \operatorname{Spec} A$ while $\{TA\}$ is closed. □

Exercise 1.2

> **Problem.** Let $\varphi : A \to B$ be a homomorphism of finitely generated algebras over a field. Show that the image of a closed point under $\operatorname{Spec} \varphi$ is a closed point.

Solution. *Remark on the hypothesis.* "Homomorphism of algebras over a field" must mean a homomorphism of algebras over the same field k, i.e. a k-linear ring homomorphism. For a mere ring homomorphism the statement is false: $\mathbf{Q}[x]$ and $\mathbf{Q}(t)$ are finitely generated algebras over the fields $\mathbf{Q}$ and $\mathbf{Q}(t)$ respectively, and the ring homomorphism $\mathbf{Q}[x] \to \mathbf{Q}(t)$, $x \mapsto t$, sends the closed point 0 of $\operatorname{Spec} \mathbf{Q}(t)$ to the non-closed point 0 of $\operatorname{Spec} \mathbf{Q}[x]$.

So let φ be a k-algebra homomorphism, let $\mathfrak{m} \subset B$ be a maximal ideal (a closed point, Remark 1.3) and $\mathfrak{p} = \varphi^{-1}(\mathfrak{m})$, which is a prime ideal. By Cor. 1.12, $B/\mathfrak{m}$ is a finite extension of k, so $\dim_k B/\mathfrak{m} < \infty$. The map φ induces an injective k-algebra homomorphism $A/\mathfrak{p} \hookrightarrow B/\mathfrak{m}$. Hence $D := A/\mathfrak{p}$ is a domain which is a finite dimensional k-vector space. For $0 \neq d \in D$, multiplication by d is an injective k-linear endomorphism of D, hence surjective, so $de = 1$ for some e. Thus D is a field and $\mathfrak{p}$ is maximal, i.e. $\mathfrak{p}$ is a closed point of $\operatorname{Spec} A$. □

Exercise 1.3

> **Problem.** Let $k = \mathbf{R}$ and $A = k[X, Y]/(X^2 + Y^2 + 1)$. Let x, y be the images of X, Y in A. (a) Let $\mathfrak{m}$ be a maximal ideal of A. Show that there exist $a, b, c, d \in k$ such that $x^2 + ax + b,\ y^2 + cy + d \in \mathfrak{m}$. Using $x^2 + y^2 + 1 = 0$, show that $\mathfrak{m}$ contains an element $f = \alpha x + \beta y + \gamma$ with $(\alpha, \beta) \neq (0, 0)$. Deduce $\mathfrak{m} = fA$. (b) Show that $(\alpha, \beta, \gamma) \mapsto (\alpha x + \beta y + \gamma)A$ is a bijection from $\mathbf{P}(k^3) \setminus \{(0, 0, 1)\}$ onto the set of maximal ideals of A. (c) Let $\mathfrak{p}$ be a non-maximal prime ideal of A. Show that $k[X] \to A$ is finite and injective, and deduce $\mathfrak{p} \cap k[X] = 0$. For $g \in \mathfrak{p}$ with integral equation $g^n + a_{n-1} g^{n-1} + \cdots + a_0 = 0$, $a_i \in k[X]$, show $a_0 = 0$. Conclude $\mathfrak{p} = 0$.

Solution. Write $K = A/\mathfrak{m}$ for a maximal ideal $\mathfrak{m}$; by Cor. 1.12, $\dim_{\mathbf{R}} K \leq 2$ (so $K \simeq \mathbf{R}$ or $\mathbf{C}$). Since $X^2 + Y^2 + 1$ is irreducible in $\mathbf{R}[X, Y]$ (see (c)), A is a domain, and $A \neq 0$.

(a) Since $\dim_{\mathbf{R}} K \leq 2$, the elements $1, \bar{x}, \bar{x}^2$ of K are linearly dependent over $\mathbf{R}$: there are real numbers $(\lambda_0, \lambda_1, \lambda_2) \neq 0$ with $\lambda_0 + \lambda_1 \bar{x} + \lambda_2 \bar{x}^2 = 0$. If $\lambda_2 \neq 0$ we divide by it and obtain a, b with $x^2 + ax + b \in \mathfrak{m}$. If $\lambda_2 = 0$ then $\bar{x} \in \mathbf{R}$ (when $\lambda_1 \neq 0$; and $\lambda_1 = \lambda_2 = 0$ forces $\lambda_0 = 0$), say $\bar{x} = r$, and $x^2 - rx = x(x - r) \in \mathfrak{m}$, which has the required form with $a = -r$, $b = 0$. The same argument for y gives c, d.

Now $1, \bar{x}, \bar{y} \in K$ are linearly dependent over $\mathbf{R}$ because $\dim_{\mathbf{R}} K \leq 2$: there is $(\alpha, \beta, \gamma) \neq (0,0,0)$ with $\alpha\bar{x} + \beta\bar{y} + \gamma = 0$. If $\alpha = \beta = 0$ then $\gamma = 0$, a contradiction. So $f = \alpha x + \beta y + \gamma \in \mathfrak{m}$ with $(\alpha, \beta) \neq (0,0)$. (This is the stated claim; the relation $x^2 + y^2 + 1 = 0$ would give $ax + cy + (b + d - 1) \in \mathfrak{m}$, but this element could be a constant, so we use the linear dependence directly.)

fA is maximal. Let $f = \alpha x + \beta y + \gamma$ be any element with $(\alpha, \beta) \neq (0,0)$. By symmetry between x and y assume $\alpha \neq 0$. In $A/fA = \mathbf{R}[X,Y]/(X^2 + Y^2 + 1, \alpha X + \beta Y + \gamma)$ we can eliminate $X = -(\beta Y + \gamma)/\alpha$, so $A/fA \simeq \mathbf{R}[Y]/(g)$ with

$$g = \left(\tfrac{\beta^2}{\alpha^2} + 1\right) Y^2 + \tfrac{2\beta\gamma}{\alpha^2} Y + \tfrac{\gamma^2}{\alpha^2} + 1.$$

This is a polynomial of degree 2 with discriminant

$$\tfrac{4\beta^2\gamma^2}{\alpha^4} - 4\,\tfrac{(\alpha^2+\beta^2)(\alpha^2+\gamma^2)}{\alpha^4} = -\tfrac{4(\alpha^2+\beta^2+\gamma^2)}{\alpha^2} < 0,$$

so g is irreducible over $\mathbf{R}$ and $A/fA \simeq \mathbf{R}[Y]/(g) \simeq \mathbf{C}$ is a field. Thus fA is maximal. Also $1, \bar{y}$ form an $\mathbf{R}$-basis of A/fA.

For our $\mathfrak{m}$ we now have $fA \subseteq \mathfrak{m}$ with fA maximal and $\mathfrak{m} \neq A$, so $\mathfrak{m} = fA$.

(b) The point $(\alpha : \beta : \gamma)$ with $(\alpha, \beta) = (0,0)$ is $(0:0:1)$, and $f = 1$ then generates A. For $(\alpha : \beta : \gamma) \in \mathbf{P}(k^3) \setminus \{(0:0:1)\}$ we have $(\alpha, \beta) \neq (0,0)$, and the ideal fA does not change if (α, β, γ) is multiplied by a nonzero scalar; by (a), fA is a maximal ideal. So the map is well defined, and surjective by (a).

Injectivity. Let $f = \alpha x + \beta y + \gamma$ and $f' = \alpha' x + \beta' y + \gamma'$ with $f'A = fA$ (both (α, β), (α', β') nonzero). Consider the $\mathbf{R}$-linear map $\theta : \mathbf{R}^3 \to A/fA$, $(u, v, w) \mapsto u\bar{x} + v\bar{y} + w$. Assume $\alpha \neq 0$ (the case $\beta \neq 0$ is symmetric). Then $\bar{x} \in \mathbf{R} + \mathbf{R}\bar{y}$ in A/fA and $\{1, \bar{y}\}$ is a basis of A/fA, so θ is surjective onto a 2-dimensional space and $\dim \operatorname{Ker} \theta = 1$. Since $(\alpha, \beta, \gamma) \in \operatorname{Ker} \theta$ is nonzero, $\operatorname{Ker} \theta = \mathbf{R}(\alpha, \beta, \gamma)$. As $f' \in fA$, $(\alpha', \beta', \gamma') \in \operatorname{Ker} \theta$, so it equals $\lambda(\alpha, \beta, \gamma)$ for some $\lambda \neq 0$. The two points of $\mathbf{P}(k^3)$ coincide. So the map is a bijection.

(c) Finite and injective. In A we have $y^2 + (x^2 + 1) = 0$, so y is integral over $\mathbf{R}[x]$ (the image of $\mathbf{R}[X]$) and $A = \mathbf{R}[x] + \mathbf{R}[x]y$ is generated by $1, y$ as an $\mathbf{R}[X]$-module: finite. Injective: if $P(X) \in \mathbf{R}[X]$ lies in $(X^2 + Y^2 + 1)\mathbf{R}[X,Y]$, say $P = (X^2 + Y^2 + 1)Q$ with $Q \neq 0$, then the Y-degree of the right side is ≥ 2 while that of P is 0; so $Q = 0$ and $P = 0$.

A is a domain. The polynomial $Y^2 + (X^2 + 1) \in \mathbf{R}[X][Y]$ is monic of degree 2 in Y; it is reducible iff $-(X^2 + 1)$ is a square in $\mathbf{R}[X]$, which is impossible (the leading coefficient of a square is positive). Hence it is irreducible, so it is a prime element of the UFD $\mathbf{R}[X,Y]$, and A is a domain.

$\mathfrak{p} \cap \mathbf{R}[X] = 0$. Let $\mathfrak{q} = \mathfrak{p} \cap \mathbf{R}[X]$, a prime ideal of the PID $\mathbf{R}[X]$. Suppose $\mathfrak{q} \neq 0$. Then $\mathfrak{q}$ is maximal and $\mathbf{R}[X]/\mathfrak{q}$ is a finite extension of $\mathbf{R}$. The ring $A/\mathfrak{p}$ is generated as an $\mathbf{R}[X]/\mathfrak{q}$-module by $1, \bar{y}$, so $\dim_{\mathbf{R}} A/\mathfrak{p} < \infty$; it is a domain, hence a field (as in Ex. 1.2), and $\mathfrak{p}$ is maximal, contrary to assumption. So $\mathfrak{q} = 0$.

Conclusion. Let $g \in \mathfrak{p}$. Since A is finite over $\mathbf{R}[X]$, g satisfies an integral equation $g^n + a_{n-1}g^{n-1} + \cdots + a_0 = 0$ with $a_i \in \mathbf{R}[X]$ (images in A); choose one with $n \geq 1$ minimal. Then $a_0 = -g(g^{n-1} + a_{n-1}g^{n-2} + \cdots + a_1) \in \mathfrak{p} \cap \mathbf{R}[X] = 0$. If $n = 1$ then $g = -a_0 = 0$. If $n \geq 2$, then $g\,(g^{n-1} + \cdots + a_1) = 0$ in the domain A; if $g \neq 0$ this gives $g^{n-1} + a_{n-1}g^{n-2} + \cdots + a_1 = 0$, an integral equation of smaller degree, contradicting minimality; hence $g = 0$. Therefore $\mathfrak{p} = 0$.

Summary. $\operatorname{Spec} A = \{0\} \cup \{(\alpha x + \beta y + \gamma)A\}$, the nonzero points being the maximal ideals in bijection with $\mathbf{P}(\mathbf{R}^3) \setminus \{(0:0:1)\}$ as in (b); 0 is prime since A is a domain. □

Exercise 1.4

> **Problem.** Let A be a ring. (a) Let $\mathfrak{p}$ be a minimal prime ideal of A. Show that $\mathfrak{p}A_\mathfrak{p}$ is nilpotent. Deduce that every element of $\mathfrak{p}$ is a zero divisor in A. (b) Show that if A is reduced, then any zero divisor in A lies in a minimal prime ideal. Show by an example (use Lemma 1.6) that this is false if A is not reduced.

Solution. *(a)* The primes of $A_\mathfrak{p}$ correspond to the primes of A contained in $\mathfrak{p}$ (Lemma 1.7(c)); by minimality of $\mathfrak{p}$ the only one is $\mathfrak{p}$. So $A_\mathfrak{p}$ has the unique prime ideal $\mathfrak{p}A_\mathfrak{p}$, and by Lemma 1.6(a) (applied to $A_\mathfrak{p}$ and $I = 0$) $\mathfrak{p}A_\mathfrak{p}$ is the nilradical of $A_\mathfrak{p}$. Thus *every element* of $\mathfrak{p}A_\mathfrak{p}$ is nilpotent.

Correction. The ideal $\mathfrak{p}A_\mathfrak{p}$ itself need not be nilpotent when A is not noetherian: take $A = k[x_1, x_2, \dots]/(x_1, x_2^2, x_3^3, \dots)$. Every x_i is nilpotent, so the only prime is $\mathfrak{m} = (x_i)_i$ (with $A/\mathfrak{m} = k$), $A = A_\mathfrak{m}$, and $\mathfrak{m}^N \ni x_{N+1}^N \neq 0$ for all N. What is true is: $\mathfrak{p}A_\mathfrak{p}$ is a nil ideal, and it is nilpotent if it is finitely generated (e.g. if A is noetherian): if $\mathfrak{p}A_\mathfrak{p} = (g_1, \dots, g_r)$ with $g_i^{n_i} = 0$, then $(\mathfrak{p}A_\mathfrak{p})^N = 0$ for $N = \sum(n_i - 1) + 1$, since each monomial of degree N in the g_i contains some $g_i^{n_i}$.

Zero divisors. Let $a \in \mathfrak{p}$. Then $a/1 \in \mathfrak{p}A_\mathfrak{p}$ is nilpotent, so $a^n/1 = 0$ for some $n \geq 1$, i.e. there is $s \in A \setminus \mathfrak{p}$ with $sa^n = 0$. Choose $n \geq 1$ minimal with this property (for some s fixed). Then $b := sa^{n-1} \neq 0$: for $n = 1$, $b = s \neq 0$ because $0 \in \mathfrak{p}$ and $s \notin \mathfrak{p}$; for $n \geq 2$ by minimality of n. And $ab = sa^n = 0$. So a is a zero divisor.

(b) Let A be reduced and $ab = 0$ with $b \neq 0$ (the zero ring has no zero divisors). First, every prime $\mathfrak{q}$ contains a minimal prime: the set of primes contained in $\mathfrak{q}$, ordered by reverse inclusion, is inductive, since the intersection of a chain $(\mathfrak{p}_i)$ of primes is prime (if $uv \in \bigcap \mathfrak{p}_i$ and $u \notin \mathfrak{p}_i$, then $u \notin \mathfrak{p}_j$ for all $\mathfrak{p}_j \subseteq \mathfrak{p}_i$, so $v \in \mathfrak{p}_j$ for those, and $v \in \mathfrak{p}_i \subseteq \mathfrak{p}_j$ for the others); Zorn's lemma gives a minimal element. Hence the intersection of all primes equals the intersection of the minimal primes; by Lemma 1.6(a) it is $\sqrt{0} = 0$. Since $b \neq 0$, some minimal prime $\mathfrak{p}$ does not contain b. From $ab = 0 \in \mathfrak{p}$ we get $a \in \mathfrak{p}$.

Counterexample in the non-reduced case. Let $A = k[X, Y]/(X^2, XY)$ and x, y the images. Then $x \neq 0$ (the ideal (X^2, XY) consists of polynomials whose terms have degree ≥ 2), $x^2 = 0$, and $A/xA \simeq k[Y]$ is a domain, so xA is prime. Every prime contains the nilpotent x, hence contains xA; so xA is the unique minimal prime (equivalently $\sqrt{0} = xA$, Lemma 1.6(a)). The element y is a zero divisor ($yx = 0$, $x \neq 0$), but $y \notin xA$ (its image in $A/xA = k[Y]$ is $Y \neq 0$). So y lies in no minimal prime. □

Exercise 1.5

> **Problem.** Let k be a field and $\mathfrak{m}$ a maximal ideal of $k[T_1, \dots, T_n]$. (a) Let $P_1(T_1)$ be a generator of $\mathfrak{m} \cap k[T_1]$ and $k_1 = k[T_1]/(P_1)$. Show that there is an exact sequence $0 \to P_1(T_1)k[T_1, \dots, T_n] \to k[T_1, \dots, T_n] \to k_1[T_2, \dots, T_n] \to 0$. (b) Show that there exist n polynomials $P_1(T_1), P_2(T_1, T_2), \dots, P_n(T_1, \dots, T_n)$ such that $k[T_1, \dots, T_i] \cap \mathfrak{m} = (P_1, \dots, P_i)$ for all $i \leq n$ (the ideal being taken in $k[T_1, \dots, T_i]$). In particular $\mathfrak{m}$ is generated by n elements.

Solution. *(a)* Let $R = k[T_1, \ldots, T_n]$. The ideal $\mathfrak{m} \cap k[T_1]$ is prime and proper ($1 \notin \mathfrak{m}$). It is nonzero: by Cor. 1.12, $R/\mathfrak{m}$ is a finite, hence algebraic, extension of k, so the image of T_1 satisfies a nonzero polynomial over k, which is then in $\mathfrak{m} \cap k[T_1]$. So $P_1 \neq 0$ is irreducible (a nonzero prime of the PID $k[T_1]$), and k_1 is a field. Write $R = k[T_1][T_2, \ldots, T_n]$ and let $\pi : R \to k_1[T_2, \ldots, T_n]$ reduce coefficients modulo P_1. It is a surjective ring homomorphism, and $\operatorname{Ker} \pi$ is the set of polynomials all of whose coefficients lie in $P_1 k[T_1]$, i.e. $\operatorname{Ker} \pi = P_1 R$. Multiplication by P_1 is injective since R is a domain. This is the exact sequence.

(b) We prove by induction on n, for all fields k simultaneously, that such P_i exist. For $n = 1$ take P_1 as in (a): $\mathfrak{m} = \mathfrak{m} \cap k[T_1] = (P_1)$. Let $n \geq 2$ and take P_1, k_1, π as in (a). Since $P_1 \in \mathfrak{m} = \pi^{-1}(\pi(\mathfrak{m}))$ (because $\operatorname{Ker} \pi = P_1 R \subseteq \mathfrak{m}$), the ideal $\mathfrak{m}' = \pi(\mathfrak{m})$ is a maximal ideal of $k_1[T_2, \ldots, T_n]$ (it corresponds to $\mathfrak{m}/P_1 R$ in $R/P_1 R$), and $\mathfrak{m} = \pi^{-1}(\mathfrak{m}')$. By the induction hypothesis (field k_1, $n-1$ variables $T_2, \ldots, T_n$) there are $Q_2(T_2), \ldots, Q_n(T_2, \ldots, T_n)$ with $k_1[T_2, \ldots, T_j] \cap \mathfrak{m}' = (Q_2, \ldots, Q_j)$ for $2 \leq j \leq n$. Lift each Q_j to $P_j \in k[T_1, \ldots, T_j]$ with $\pi(P_j) = Q_j$ (lift each coefficient in k_1 to $k[T_1]$). Then $P_j \in \pi^{-1}(\mathfrak{m}') = \mathfrak{m}$.

Fix $i \leq n$ and put $R_i = k[T_1, \ldots, T_i]$. The inclusion $(P_1, \ldots, P_i)R_i \subseteq \mathfrak{m} \cap R_i$ is clear. Conversely let $F \in \mathfrak{m} \cap R_i$. Then $\pi(F) \in \mathfrak{m}' \cap k_1[T_2, \ldots, T_i]$, hence $\pi(F) = \sum_{j=2}^{i} C_j Q_j$ with $C_j \in k_1[T_2, \ldots, T_i]$ (for $i = 1$ the sum is empty). Lift C_j to $D_j \in R_i$. Then $F - \sum_j D_j P_j \in R_i$ is killed by π, so it lies in $P_1 R \cap R_i = P_1 R_i$ (a polynomial in $T_1, \ldots, T_i$ divisible by P_1 in R is divisible in R_i, by comparing coefficients of the monomials in $T_2, \ldots, T_n$). Hence $F \in (P_1, \ldots, P_i)R_i$.

For $i = n$ this gives $\mathfrak{m} = (P_1, \ldots, P_n)$. □

Exercise 1.6

> **Problem.** A topological space X is quasi-compact if every open covering has a finite subcovering. Show that any closed subspace of a quasi-compact space is quasi-compact. Show that $\operatorname{Spec} A$ is quasi-compact.

Solution. *Closed subspaces.* Let X be quasi-compact, $Z \subseteq X$ closed, and $Z \subseteq \bigcup_i U_i$ be a covering of Z by sets $U_i \cap Z$ with U_i open in X (every open of Z has this form). Then $\{U_i\}_i \cup \{X \setminus Z\}$ is an open covering of X; take a finite subcovering. Discarding $X \setminus Z$ (which does not meet Z) we get finitely many U_i covering Z.

$\operatorname{Spec} A$. Let $\operatorname{Spec} A = \bigcup_i U_i$ with U_i open. By Prop. 1.1 each U_i is a union of principal open sets $D(f)$, so the family $\mathcal{F}$ of all $D(f)$ contained in some U_i covers $\operatorname{Spec} A$. It suffices to find finitely many members of $\mathcal{F}$ covering $\operatorname{Spec} A$: each lies in some U_i, and those finitely many U_i then cover.

Write $\mathcal{F} = \{D(f_j)\}_{j \in J}$ and let I be the ideal generated by the f_j. Then $V(I) = \bigcap_j V(f_j A) = \operatorname{Spec} A \setminus \bigcup_j D(f_j) = \emptyset$ (Prop. 1.1(b)). So I is contained in no prime ideal, hence in no maximal ideal, so $I = A$ (otherwise I lies in a maximal ideal by Zorn's lemma). Thus $1 = \sum_{l=1}^{m} a_l f_{j_l}$ for finitely many indices. The ideal generated by $f_{j_1}, \ldots, f_{j_m}$ is A, so no prime contains all of them, i.e. $\operatorname{Spec} A = \bigcup_l D(f_{j_l})$. □

Exercise 1.7

> **Problem.** Describe the closed subsets of $\operatorname{Spec} \mathbf{Z}[T]$. Show that they are not all principal.

Solution. *Step 1: the points of* $\operatorname{Spec}\mathbf{Z}[T]$. Let P be a prime of $\mathbf{Z}[T]$ and $P\cap\mathbf{Z}=p\mathbf{Z}$ with $p=0$ or a prime number.

If $p\neq 0$: $P/p\mathbf{Z}[T]$ is a prime of $\mathbf{Z}[T]/(p)=\mathbf{F}_p[T]$, so it is 0 or $(\bar{g})$ with $\bar{g}$ irreducible. Thus $P=(p)$ or $P=(p,g)$ with $g\in\mathbf{Z}[T]$ a lift of an irreducible $\bar{g}\in\mathbf{F}_p[T]$; the latter are maximal because $\mathbf{F}_p[T]/(\bar{g})$ is a field.

If $p=0$: $P\cap(\mathbf{Z}\setminus 0)=\emptyset$, so by Lemma 1.7(c) P corresponds to the prime $P\mathbf{Q}[T]$ of $\mathbf{Q}[T]$, and $P=P\mathbf{Q}[T]\cap\mathbf{Z}[T]$. Now $P\mathbf{Q}[T]=0$ or $=h\mathbf{Q}[T]$ with h irreducible; scaling, take $h\in\mathbf{Z}[T]$ primitive; by Gauss's lemma h is irreducible in $\mathbf{Z}[T]$ and $h\mathbf{Q}[T]\cap\mathbf{Z}[T]=h\mathbf{Z}[T]$. So $P=0$ or $P=(h)$ with h a primitive irreducible polynomial of degree ≥ 1.

So the points are: 0; the primes (h), h primitive irreducible of positive degree; the primes (q), q a prime number (these two kinds are the "height one" points); and the maximal ideals (q,g).

Step 2: irreducible closed sets. For a point P the closure of $\{P\}$ is $V(P)$. We have $V(0)=\operatorname{Spec}\mathbf{Z}[T]$. For $P=(h)$ or $P=(q)$ we claim that $V(P)=\{P\}\cup\{\text{maximal ideals containing } P\}$. Indeed let $P'\supsetneq P$ be prime. If $P=(h)$: $P'\cap\mathbf{Z}=0$ would force $P'=(h')$ (it is nonzero since it contains h), and $(h)\subseteq(h')$ gives $h'\mid h$, so h' is an associate of h (as h is irreducible), contradicting $P'\neq P$; so $P'\cap\mathbf{Z}=q\mathbf{Z}$ with q prime, and $P'=(q)$ or (q,g); but $(q)\supseteq(h)$ would give $h=qu$ with h primitive, so q a unit, impossible; so $P'=(q,g)$ is maximal. If $P=(q)$: $P'\ni q$ so $P'\cap\mathbf{Z}=q\mathbf{Z}$ and $P'=(q)$ or (q,g), i.e. P' is maximal. For P maximal, $V(P)=\{P\}$.

Step 3: all closed sets. $\mathbf{Z}[T]$ is noetherian (Hilbert basis), so every radical ideal is a finite intersection of primes $P_1\cap\cdots\cap P_r$; by Lemma 1.6(b) and Prop. 1.1(a), $V(I)=V(\sqrt{I})=V(P_1\cap\cdots\cap P_r)=V(P_1)\cup\cdots\cup V(P_r)$ (if $\sqrt{I}=A$ then $V(I)=\emptyset$, the empty union). Conversely a finite union of closed sets is closed. Hence:

The closed subsets of $\operatorname{Spec}\mathbf{Z}[T]$ *are exactly the finite unions of sets* $V(P)$ *with* P *prime.* Concretely: $\operatorname{Spec}\mathbf{Z}[T]$ itself, and the proper closed subsets, which are the sets

$$V(h_1)\cup\cdots\cup V(h_r)\cup V(q_1)\cup\cdots\cup V(q_s)\cup\{\mathfrak{m}_1,\ldots,\mathfrak{m}_t\}$$

with $r,s,t\geq 0$ finite, h_i primitive irreducible nonconstant polynomials, q_j prime numbers and $\mathfrak{m}_l$ maximal ideals. Here $V(h)$ consists of the point (h) and the maximal ideals (q,g) with $h\in(q,g)$, and $V(q)$ of (q) and all (q,g). In words: a proper closed subset is a finite set of closed points together with finitely many "curves" (horizontal $V(q)$, vertical $V(h)$, each with its generic point).

Not all closed sets are principal. Consider $C=\{\mathfrak{m}\}$ with $\mathfrak{m}=(2,T)$; it is closed, $C=V(2,T)$ (indeed $\mathfrak{m}$ is maximal, Remark 1.3). Suppose $C=V(f)$. Then $f\neq 0$ (else $V(f)=\operatorname{Spec}\mathbf{Z}[T]$), and f is not a unit (else $V(f)=\emptyset$). Take a prime element π dividing f in the UFD $\mathbf{Z}[T]$. Then (π) is a prime containing f, so $(\pi)\in V(f)$, but (π) is not maximal: if π is a constant prime number q, then $(q)\subsetneq(q,T)$; if $\deg\pi\geq 1$, choose a prime number q not dividing the leading coefficient of π; then the image of π in $\mathbf{F}_q[T]$ has the same positive degree, so $(\pi,q)\neq\mathbf{Z}[T]$ (as $\mathbf{Z}[T]/(\pi,q)\simeq\mathbf{F}_q[T]/(\bar{\pi})\neq 0$), and $q\notin(\pi)$ (a nonzero multiple of π has positive degree); hence $(\pi)\subsetneq(\pi,q)$. Thus $V(f)$ contains a non-closed point, so $V(f)\neq C$. Hence C is a closed subset that is not principal. □

Exercise 1.8

Problem. Let $\varphi : A \to B$ be an integral homomorphism. (a) Show that $\operatorname{Spec}\varphi : \operatorname{Spec} B \to \operatorname{Spec} A$ maps a closed point to a closed point, and that any preimage of a closed point is a closed point. (b) Let $\mathfrak{p} \in \operatorname{Spec} A$. Show that the canonical homomorphism $A_{\mathfrak{p}} \to B \otimes_A A_{\mathfrak{p}}$ is integral. (c) Let $T = \varphi(A \setminus \mathfrak{p})$. Suppose φ is injective. Show that T is a multiplicative subset of B, and $B \otimes_A A_{\mathfrak{p}} = T^{-1}B \neq 0$. Deduce that $\operatorname{Spec}\varphi$ is surjective if φ is integral and injective.

Solution. *(a)* Let $\mathfrak{q} \in \operatorname{Spec} B$ and $\mathfrak{p} = \varphi^{-1}(\mathfrak{q}) = (\operatorname{Spec}\varphi)(\mathfrak{q})$. The map φ induces an injective homomorphism of domains $\bar{\varphi} : D := A/\mathfrak{p} \hookrightarrow E := B/\mathfrak{q}$, and E is integral over D (reduce an integral equation modulo $\mathfrak{q}$). We show that D is a field iff E is a field; by Remark 1.3 this is exactly the claim (the image of a closed point is closed, and a point whose image is closed is closed).

(E field $\Rightarrow$ D field.) Let $0 \neq a \in D$. Then $a^{-1} \in E$ is integral over D: $a^{-n} + c_{n-1}a^{-(n-1)} + \cdots + c_0 = 0$ with $c_i \in D$. Multiplying by a^{n-1}: $a^{-1} = -(c_{n-1} + c_{n-2}a + \cdots + c_0 a^{n-1}) \in D$.

(D field $\Rightarrow$ E field.) Let $0 \neq b \in E$ and take an integral equation $b^n + c_{n-1}b^{n-1} + \cdots + c_0 = 0$ over D of minimal degree $n \geq 1$. Then $c_0 \neq 0$: otherwise $b(b^{n-1} + \cdots + c_1) = 0$ gives (as E is a domain) either $n = 1$ and $b = 0$, or an equation of degree $n - 1$. So $b \cdot (b^{n-1} + \cdots + c_1) = -c_0$ with c_0 invertible in D, and b is invertible in E.

(b) Put $S = A \setminus \mathfrak{p}$ and $C = B \otimes_A A_{\mathfrak{p}}$, with the ring homomorphisms $\beta : B \to C$, $b \mapsto b \otimes 1$, and $\iota : A_{\mathfrak{p}} \to C$, $a/s \mapsto 1 \otimes a/s$. Every element of C is a finite sum of elements $b \otimes \frac{a}{s} = \beta(b)\,\iota(a/s)$. For $b \in B$ choose $b^n + \sum_{i<n} \varphi(a_i)b^i = 0$; applying β gives $\beta(b)^n + \sum_{i<n} \iota(a_i/1)\beta(b)^i = 0$ (since $\beta\varphi(a) = a \cdot (1 \otimes 1) = \iota(a/1)$). So $\beta(b)$ is integral over $\iota(A_{\mathfrak{p}})$, and $\iota(a/s)$ trivially is. The integral elements form a subring, so every element of C is integral over $A_{\mathfrak{p}}$.

(c) T is multiplicative: $1 = \varphi(1) \in T$, and $\varphi(s)\varphi(t) = \varphi(st)$ with $st \in A \setminus \mathfrak{p}$ since $\mathfrak{p}$ is prime. Moreover $0 \notin T$: $\varphi(s) = 0$ with φ injective gives $s = 0 \in \mathfrak{p}$. The canonical map $B \otimes_A S^{-1}A \to T^{-1}B$, $b \otimes a/s \mapsto \varphi(a)b/\varphi(s)$, is the standard isomorphism $B \otimes_A S^{-1}A \simeq T^{-1}B$ (localization of the A-module B at S, which is the localization of the ring B at T). Since $0 \notin T$, $1/1 \neq 0$ in $T^{-1}B$ (a relation $t \cdot 1 = 0$ with $t \in T$ would give $0 \in T$). So $C = T^{-1}B \neq 0$.

Surjectivity. Let $\mathfrak{p} \in \operatorname{Spec} A$. As $C \neq 0$, C has a maximal ideal $\mathfrak{n}$. By (b), $\iota : A_{\mathfrak{p}} \to C$ is integral, so by (a) the prime $\iota^{-1}(\mathfrak{n})$ is a maximal ideal of the local ring $A_{\mathfrak{p}}$, i.e. $\iota^{-1}(\mathfrak{n}) = \mathfrak{p}A_{\mathfrak{p}}$. Let $\mathfrak{q} = \beta^{-1}(\mathfrak{n}) \in \operatorname{Spec} B$. Since $\beta \circ \varphi = \iota \circ (A \to A_{\mathfrak{p}})$, we get

$$\varphi^{-1}(\mathfrak{q}) = \varphi^{-1}\beta^{-1}(\mathfrak{n}) = (A \to A_{\mathfrak{p}})^{-1}\iota^{-1}(\mathfrak{n}) = (A \to A_{\mathfrak{p}})^{-1}(\mathfrak{p}A_{\mathfrak{p}}) = \mathfrak{p}.$$

So $\mathfrak{p}$ is in the image of $\operatorname{Spec}\varphi$. □

Exercise 1.9

Problem. Let A be a finitely generated algebra over a field k. (a) Suppose A is finite over k. Show that $\operatorname{Spec} A$ is a finite set of cardinality at most $\dim_k A$ (construct a strictly descending chain of ideals by taking intersections of maximal ideals). Show that every prime ideal of A is maximal. (b) Show that $\operatorname{Spec} k[T_1, \ldots, T_d]$ is infinite if $d \geq 1$. (c) Show that $\operatorname{Spec} A$ is finite iff A is finite over k.

Solution. *(a)* Let $n = \dim_k A < \infty$. For a prime $\mathfrak{p}$, $A/\mathfrak{p}$ is a domain of finite dimension over k, hence a field (as in Ex. 1.2), so $\mathfrak{p}$ is maximal: $\operatorname{Spec} A = \operatorname{Max} A$.

Let $\mathfrak{m}_1, \dots, \mathfrak{m}_r$ be distinct maximal ideals and $I_0 = A$, $I_i = \mathfrak{m}_1 \cap \cdots \cap \mathfrak{m}_i$. We claim $I_i \supsetneq I_{i+1}$ for $0 \le i < r$. Otherwise $I_i \subseteq \mathfrak{m}_{i+1}$; for $i = 0$ this contradicts $\mathfrak{m}_1 \neq A$. For $i \ge 1$, $\mathfrak{m}_1 \cdots \mathfrak{m}_i \subseteq I_i \subseteq \mathfrak{m}_{i+1}$, and as $\mathfrak{m}_{i+1}$ is prime, some $\mathfrak{m}_j \subseteq \mathfrak{m}_{i+1}$ with $j \le i$, so $\mathfrak{m}_j = \mathfrak{m}_{i+1}$ by maximality, contradicting distinctness. The I_i are k-subspaces, so $n = \dim I_0 > \dim I_1 > \cdots > \dim I_r \ge 0$ forces $r \le n$. Hence $\#\operatorname{Spec} A \le n$.

(b) Let $d \ge 1$ and let $P \in k[T_1]$ be monic irreducible. The ideal $\mathfrak{m}_P = (P(T_1), T_2, \dots, T_d)$ has quotient $k[T_1]/(P)$, a field, so $\mathfrak{m}_P$ is maximal. Moreover $\mathfrak{m}_P \cap k[T_1] = (P)$: the composite $k[T_1] \to k[T_1, \dots, T_d] \to k[T_1, \dots, T_d]/\mathfrak{m}_P \simeq k[T_1]/(P)$ is the canonical surjection, with kernel (P). Hence distinct monic irreducible P give distinct $\mathfrak{m}_P$. There are infinitely many monic irreducible polynomials in $k[T_1]$ for any field k (Euclid): given a finite list $P_1, \dots, P_r$, the polynomial $P_1 \cdots P_r + 1$ has degree ≥ 1, hence a monic irreducible factor, which is none of the P_i. So $\operatorname{Spec} k[T_1, \dots, T_d]$ is infinite.

(c) ($\Leftarrow$) is (a). ($\Rightarrow$) Suppose $\operatorname{Spec} A$ finite. If $A = 0$ there is nothing to show, so let $A \neq 0$. Let $\mathfrak{m}_1, \dots, \mathfrak{m}_r$ be the maximal ideals (finitely many). By Lemma 1.18, $\mathfrak{N} := \sqrt{0} = \bigcap_i \mathfrak{m}_i$. Each $A/\mathfrak{m}_i$ is finite over k by Cor. 1.12, and the natural map $A/\mathfrak{N} \to \prod_i A/\mathfrak{m}_i$ is injective, so $\dim_k A/\mathfrak{N} < \infty$.

Now A is noetherian (Hilbert basis), so $\mathfrak{N}$ is finitely generated, say by $g_1, \dots, g_s$ with $g_j^{n_j} = 0$; as in Ex. 1.4(a), $\mathfrak{N}^N = 0$ for $N = \sum(n_j - 1) + 1$. Consider the filtration $A \supseteq \mathfrak{N} \supseteq \mathfrak{N}^2 \supseteq \cdots \supseteq \mathfrak{N}^N = 0$. Each $\mathfrak{N}^i$ is a finitely generated A-module (noetherian), hence $\mathfrak{N}^i/\mathfrak{N}^{i+1}$ is a finitely generated module over $A/\mathfrak{N}$, which is finite dimensional over k; therefore $\dim_k \mathfrak{N}^i/\mathfrak{N}^{i+1} < \infty$. Summing, $\dim_k A = \dim_k A/\mathfrak{N} + \sum_{i=1}^{N-1} \dim_k \mathfrak{N}^i/\mathfrak{N}^{i+1} < \infty$. So A is finite over k. □

2.2 Ringed topological spaces

Throughout, sheaves and presheaves are of abelian groups on a topological space X, and results of the text are cited by their numbers in Liu ("Def. 2.2", "Prop. 2.15", ...). A morphism of presheaves is injective if all $\alpha(U)$ are, and surjective if all stalk maps α_x are (Def. 2.10).

Exercise 2.1

> **Problem.** Determine the sheaf associated to the presheaf A_X of Example 2.4 (a non-trivial abelian group A; $A_X(U) = A$ for $U \neq \emptyset$, $A_X(\emptyset) = 0$, restrictions Id_A between non-empty open sets). Such a sheaf is called a constant sheaf. Show that A_X is a sheaf if and only if every non-empty open subset of X is connected.

Solution. *The stalks.* For $x \in X$ the direct limit of $A_X(U) = A$ over the neighbourhoods $U \ni x$ with identity transition maps is A (all $U \ni x$ are non-empty); we write $(A_X)_x = A$, and the germ at x of $a \in A_X(U)$ is a.

The associated sheaf. By the construction in Prop. 2.15, $A_X^\dagger(U)$ is the group of functions $f : U \to \bigsqcup_{x \in U} (A_X)_x = U \times A$, i.e. functions $f : U \to A$, such that every $x \in U$ has an open neighbourhood $V \subseteq U$ and a section $a \in A_X(V) = A$ with $f(y) = a$ for all $y \in V$. These are exactly the locally constant functions $U \to A$ (equivalently the continuous maps $U \to A$ when A has the discrete topology). Restrictions are restrictions of functions, and

$\theta(U) : A_X(U) \to A_X^\dagger(U)$ sends a to the constant function with value a (for $U \neq \emptyset$). So *the sheaf associated to A_X is the sheaf $U \mapsto \{$locally constant functions $U \to A\}$*. (It is a sheaf, as Prop. 2.15 asserts: being locally constant and equality of functions are local conditions, so uniqueness and glueing of Def. 2.2 hold for it as a subpresheaf of the sheaf of all functions into A.) Its stalk at every x is A and $\theta_x = \mathrm{Id}_A$.

Criterion. Let $U \neq \emptyset$ be open. The map $\theta(U)$ is injective, because a constant function with value a on $U \neq \emptyset$ is zero only if $a = 0$. We claim $\theta(U)$ is surjective if and only if U is connected.

If U is connected and $f : U \to A$ is locally constant, then for each $a \in A$ the set $f^{-1}(a)$ is open (local constancy), and U is the disjoint union of these sets. Since $U \neq \emptyset$ is connected, only one of them is non-empty, so f is constant, $f = \theta(U)(a)$. If U is not connected, write $U = U_1 \sqcup U_2$ with U_1, U_2 non-empty, open and disjoint. Choose $a \neq 0$ in A and let $f = a$ on U_1, $f = 0$ on U_2. Then f is locally constant but not constant, so it is not in the image of $\theta(U)$.

For $U = \emptyset$ both groups are 0. Hence $\theta : A_X \to A_X^\dagger$ is an isomorphism of presheaves iff every non-empty open set is connected.

Conclusion. If every non-empty open set is connected, $A_X \simeq A_X^\dagger$ is a sheaf. Conversely, if A_X is a sheaf, then $\theta : A_X \to A_X^\dagger$ is a morphism of sheaves which is an isomorphism on all stalks (Prop. 2.15), hence an isomorphism by Prop. 2.12; so every non-empty open set is connected by the criterion. □

Exercise 2.2

> **Problem.** Let $\mathcal{F}$ be a sheaf on X and $s, t \in \mathcal{F}(X)$. Show that the set of $x \in X$ such that $s_x = t_x$ is open in X.

Solution. Put $u = s - t \in \mathcal{F}(X)$. Since taking germs is a group homomorphism, $s_x = t_x$ if and only if $u_x = 0$. Let x be such that $u_x = 0$. By the definition of the direct limit $\mathcal{F}_x = \varinjlim_{U \ni x} \mathcal{F}(U)$, the germ of u at x is zero iff there is an open neighbourhood V of x with $u|_V = 0$. Then for every $y \in V$ we have $u_y = (u|_V)_y = 0$. Thus V is contained in the set $\{x : s_x = t_x\}$, which is therefore a neighbourhood of each of its points, i.e. open. (Only the presheaf property was used.) □

Exercise 2.3

> **Problem.** (Sheaf associated to a presheaf) Fix a topological space X and a presheaf $\mathcal{F}$ on X; keep the notation before Lemma 2.7. (a) Let $\mathcal{U} = \{U_i\}_i$ be an open covering of X and $\mathcal{F}_{\mathcal{U}}(X) = \operatorname{Ker} d^1$. For any open $W \subseteq X$ define in the same way a group $\mathcal{F}_{\mathcal{U}}(W)$ relative to the covering $\{W \cap U_i\}_i$ of W. Show that $\mathcal{F}_{\mathcal{U}}$ is a presheaf and that there is a morphism of presheaves $\mathcal{F} \to \mathcal{F}_{\mathcal{U}}$. (b) Let $\mathcal{V} = \{V_k\}_k$ be another open covering of X; write $\mathcal{V} \preceq \mathcal{U}$ if every V_k is contained in some U_i. Show that there is a canonical homomorphism $\mathcal{F}_{\mathcal{U}}(W) \to \mathcal{F}_{\mathcal{V}}(W)$ for every open W if $\mathcal{V} \preceq \mathcal{U}$. (c) Show that $\preceq$ makes the set of open coverings of X into a direct system. Let $\mathcal{F}'(W) = \varinjlim_{\mathcal{V}} \mathcal{F}_{\mathcal{V}}(W)$, the limit over the open coverings of W. Show that $\mathcal{F}'$ is the sheaf associated to $\mathcal{F}$.

Solution. Honest note. Parts (a), (b) are proved as stated. Part (c) is *false as stated for an arbitrary presheaf*: $\mathcal{F}'$ (the "plus construction") is always a separated presheaf, and it is a

sheaf, hence the sheaf associated to $\mathcal{F}$, *if $\mathcal{F}$ satisfies the uniqueness axiom (4)*; for general $\mathcal{F}$ one must apply the construction twice. A counterexample is given at the end. I prove: (c1) the direct system; (c2) $\theta_x : \mathcal{F}_x \to \mathcal{F}'_x$ is an isomorphism for all x; (c3) $\mathcal{F}'$ is separated for every $\mathcal{F}$, and is a sheaf if $\mathcal{F}$ is separated; (c4) the corrected statement that $\mathcal{F}'$ is the associated sheaf iff it is a sheaf (in particular if $\mathcal{F}$ is separated), and $(\mathcal{F}')'$ is always the associated sheaf.

Conventions. An open covering of an open set W is a *set* $\mathcal{U}$ of open subsets of W with union W (so the coverings of W form a set, and we index a covering by its members). For such $\mathcal{U}$ put

$$C_W(\mathcal{U}) = \Big\{(s_A)_{A\in\mathcal{U}} \in \prod_{A\in\mathcal{U}} \mathcal{F}(A) \ : \ s_A|_{A\cap B} = s_B|_{A\cap B} \text{ for all } A, B \in \mathcal{U}\Big\} = \operatorname{Ker} d^1.$$

For a covering $\mathcal{U}$ of X and open $W \subseteq X$, let $\mathcal{U}|_W = \{W \cap A : A \in \mathcal{U}\}$, a covering of W; then $\mathcal{F}_{\mathcal{U}}(W)$ of the exercise is $C_W(\mathcal{U}|_W)$. (Using indexed families $\{U_i\}_i$ as in the book changes nothing below, since only the sets U_i and the compatibility of the s_i matter.)

(a) For $W' \subseteq W$ and $s = (s_B)_{B\in\mathcal{U}|_W} \in \mathcal{F}_{\mathcal{U}}(W)$ define $\rho(s) \in \prod_{B'\in\mathcal{U}|_{W'}} \mathcal{F}(B')$ as follows: a member of $\mathcal{U}|_{W'}$ has the form $B' = W' \cap A$ with $A \in \mathcal{U}$, and we put $\rho(s)_{B'} = s_{W\cap A}|_{B'}$. This is well defined: if also $B' = W' \cap A_1$, then $B' \subseteq W \cap A \cap A_1$, and $s_{W\cap A}$, $s_{W\cap A_1}$ agree on $W \cap A \cap A_1$ by compatibility. The family $\rho(s)$ is compatible because restriction is transitive. Functoriality $\rho_{W''W'}\rho_{W'W} = \rho_{W''W}$ and $\rho_{WW} = \mathrm{Id}$ is clear from transitivity of restriction in $\mathcal{F}$. Moreover $\mathcal{F}_{\mathcal{U}}(\emptyset) = 0$ because $\mathcal{F}(\emptyset) = 0$. So $\mathcal{F}_{\mathcal{U}}$ is a presheaf. The map $\theta_{\mathcal{U}}(W) : \mathcal{F}(W) \to \mathcal{F}_{\mathcal{U}}(W)$, $s \mapsto (s|_{W\cap A})_A$, lands in $\operatorname{Ker} d^1$ since $(s|_{W\cap A})|_{W\cap A\cap B} = s|_{W\cap A\cap B}$ is symmetric in A, B, i.e. $d^1 d^0 = 0$; it is a group homomorphism and commutes with restrictions by transitivity. This is the morphism of presheaves $\mathcal{F} \to \mathcal{F}_{\mathcal{U}}$.

(b) Work with coverings $\mathcal{U}, \mathcal{V}$ of an open set W (apply to $\mathcal{U}|_W$, $\mathcal{V}|_W$ for the statement as given; $\mathcal{V} \preceq \mathcal{U}$ implies $\mathcal{V}|_W \preceq \mathcal{U}|_W$). Let $\mathcal{V} \preceq \mathcal{U}$ and choose a map $\tau : \mathcal{V} \to \mathcal{U}$ with $V \subseteq \tau(V)$ for all $V \in \mathcal{V}$. Define

$$c_\tau : C_W(\mathcal{U}) \to C_W(\mathcal{V}), \qquad (s_A)_A \mapsto (s_{\tau(V)}|_V)_{V\in\mathcal{V}}.$$

It is a homomorphism, and its image is compatible: for $V, V' \in \mathcal{V}$ both $s_{\tau(V)}|_{V\cap V'}$ and $s_{\tau(V')}|_{V\cap V'}$ are restrictions to $V \cap V' \subseteq \tau(V) \cap \tau(V')$ of two elements which agree on $\tau(V) \cap \tau(V')$. *Independence of τ:* if τ' is another choice, then $V \subseteq \tau(V) \cap \tau'(V)$ and $s_{\tau(V)}|_V = s_{\tau'(V)}|_V$ by the compatibility of (s_A). So $c_{\mathcal{V}\mathcal{U}} := c_\tau$ is canonical. It is compatible with restriction to a smaller open set W' (both ways around the square give $(s_{\tau(V)}|_{W'\cap V})_V$, by the independence just proved), and with the maps θ of (a): $c(\theta(s)) = (s|_V)_V = \theta_{\mathcal{V}}(s)$. Also $c_{\mathcal{U}\mathcal{U}} = \mathrm{Id}$ (take $\tau = \mathrm{Id}$) and, for $\mathcal{V}' \preceq \mathcal{V} \preceq \mathcal{U}$, $c_{\mathcal{V}'\mathcal{V}}c_{\mathcal{V}\mathcal{U}} = c_{\mathcal{V}'\mathcal{U}}$ (compose the choice functions and use independence).

(c1) The direct system. The relation $\preceq$ is reflexive and transitive (a preorder), and any two coverings $\mathcal{U}, \mathcal{V}$ of W have the common refinement $\{A \cap B : A \in \mathcal{U}, B \in \mathcal{V}\} \preceq \mathcal{U}, \mathcal{V}$. So the coverings of W form a directed (pre)ordered set and, by (b), $\mathcal{U} \mapsto C_W(\mathcal{U})$ with the maps $c_{\mathcal{V}\mathcal{U}}$ is a direct system. Put $\mathcal{F}'(W) = \varinjlim_{\mathcal{U}} C_W(\mathcal{U})$ (this is the limit meant in (c)). Write $[\mathcal{U}; (s_A)]$ for the class of $(s_A) \in C_W(\mathcal{U})$. Two elements $(s) \in C_W(\mathcal{U})$, $(s') \in C_W(\mathcal{U}')$ define the same class iff they have the same image in $C_W(\mathcal{V})$ for some common refinement $\mathcal{V}$; in particular $[\mathcal{U}; (s)] = 0$ iff $c_{\mathcal{V}\mathcal{U}}(s) = 0$ for some $\mathcal{V} \preceq \mathcal{U}$. For $W' \subseteq W$ the maps $\rho : C_W(\mathcal{U}) \to C_{W'}(\mathcal{U}|_{W'})$, $(s_A) \mapsto (s_A|_{W'\cap A})$ commute with the c's, hence induce $\mathcal{F}'(W) \to \mathcal{F}'(W')$; these are functorial, and $\mathcal{F}'(\emptyset) = 0$. So $\mathcal{F}'$ is a presheaf, and $\theta(W) : \mathcal{F}(W) \to \mathcal{F}'(W)$, $s \mapsto [\{W\}; (s)]$, is a morphism of presheaves. Note: for $t = [\mathcal{U}; (s_A)] \in \mathcal{F}'(W)$ and $A_0 \in \mathcal{U}$ we have

$$t|_{A_0} = \theta(s_{A_0}) \ \text{ in } \mathcal{F}'(A_0) \qquad (*)$$

Indeed $t|_{A_0} = [\mathcal{U}|_{A_0}; (s_A|_{A_0\cap A})_A]$, while $\{A_0 \cap A\}_A \preceq \{A_0\}$ and the image of s_{A_0} in $C_{A_0}(\mathcal{U}|_{A_0})$ is $(s_{A_0}|_{A_0\cap A})_A$, which equals $(s_A|_{A_0\cap A})_A$ by compatibility.

(c2) $\theta_x : \mathcal{F}_x \to \mathcal{F}'_x$ is an isomorphism for every x. *Surjective:* a germ in $\mathcal{F}'_x$ is t_x with $t = [\mathcal{U}; (s_A)] \in \mathcal{F}'(W)$, $x \in W$. Pick $A_0 \in \mathcal{U}$ containing x. By $(*)$, $t_x = (t|_{A_0})_x = \theta(s_{A_0})_x = \theta_x((s_{A_0})_x)$. *Injective:* let $\sigma = s_x$ with $s \in \mathcal{F}(V)$ and suppose $\theta_x(\sigma) = \theta(s)_x = 0$. Then there is an open $V' \ni x$, $V' \subseteq V$, with $\theta(s|_{V'}) = \theta(s)|_{V'} = 0$ in $\mathcal{F}'(V')$, i.e. some covering $\mathcal{V} \preceq \{V'\}$ of V' satisfies $s|_B = 0$ for all $B \in \mathcal{V}$. Choose $B \in \mathcal{V}$ containing x: then $\sigma = (s|_B)_x = 0$.

(c3) $\mathcal{F}'$ is separated; it is a sheaf if $\mathcal{F}$ is separated.

Separated. Let $W = \bigcup_j W_j$ be an open covering and $t = [\mathcal{U}; (s_A)] \in \mathcal{F}'(W)$ with $t|_{W_j} = 0$ for all j. Then for each j there is a covering $\mathcal{V}_j \preceq \mathcal{U}|_{W_j}$ of W_j whose image of $(s_A|_{W_j\cap A})$ is 0: for every $B \in \mathcal{V}_j$ and any $A \in \mathcal{U}$ with $B \subseteq A$ we get $s_A|_B = 0$. Let $\mathcal{V} = \bigcup_j \mathcal{V}_j$, a covering of W refining $\mathcal{U}$. For $B \in \mathcal{V}$ and $A \in \mathcal{U}$ with $B \subseteq A$ we have $s_A|_B = 0$ (choose j with $B \in \mathcal{V}_j$; any choice of A gives the same restriction to B by compatibility). Hence $c_{\mathcal{V}\mathcal{U}}(s) = 0$ and $t = 0$. This is exactly axiom (4) for $\mathcal{F}'$.

Glueing when $\mathcal{F}$ is separated. Let $W = \bigcup_j W_j$ and $t_j = [\mathcal{U}_j; (s_{j,A})_{A\in\mathcal{U}_j}] \in \mathcal{F}'(W_j)$ with $t_j|_{W_{jl}} = t_l|_{W_{jl}}$, $W_{jl} = W_j \cap W_l$.

Claim. For j, l, $A \in \mathcal{U}_j$, $B \in \mathcal{U}_l$: $s_{j,A}|_{A\cap B} = s_{l,B}|_{A\cap B}$. Indeed, $t_j|_{W_{jl}} = t_l|_{W_{jl}}$ gives a covering $\mathcal{Z}$ of W_{jl}, refining both $\mathcal{U}_j|_{W_{jl}}$ and $\mathcal{U}_l|_{W_{jl}}$, such that for every $Z \in \mathcal{Z}$ and all $A' \in \mathcal{U}_j$, $B' \in \mathcal{U}_l$ with $Z \subseteq A'$, $Z \subseteq B'$ we have $s_{j,A'}|_Z = s_{l,B'}|_Z$. Now the sets $Z \cap A \cap B$ ($Z \in \mathcal{Z}$) cover $A \cap B \subseteq W_{jl}$; for each of them, by compatibility of $(s_{j,\cdot})$ and $(s_{l,\cdot})$, and using A', B' with $Z \subseteq A', B'$,

$$s_{j,A}|_{Z\cap A\cap B} = s_{j,A'}|_{Z\cap A\cap B} = s_{l,B'}|_{Z\cap A\cap B} = s_{l,B}|_{Z\cap A\cap B}.$$

Since $\mathcal{F}$ is separated, $s_{j,A}|_{A\cap B} = s_{l,B}|_{A\cap B}$. This proves the claim.

Now let $\mathcal{U} = \bigcup_j \mathcal{U}_j$, a covering of W; for each $A \in \mathcal{U}$ choose $j(A)$ with $A \in \mathcal{U}_{j(A)}$ and put $\sigma_A = s_{j(A),A}$. By the claim $(\sigma_A) \in C_W(\mathcal{U})$; let $t = [\mathcal{U}; (\sigma_A)]$. To see $t|_{W_j} = t_j$: let $\mathcal{Y} = \{A \cap A' : A \in \mathcal{U}, A' \in \mathcal{U}_j\}$, a covering of W_j refining $\mathcal{U}|_{W_j}$ and $\mathcal{U}_j$. The image of $t|_{W_j}$ in $C_{W_j}(\mathcal{Y})$ is $(\sigma_A|_Y)$ and that of t_j is $(s_{j,A'}|_Y)$ (for $Y \subseteq A \cap A'$); they agree by the claim ($\sigma_A = s_{j(A),A}$, applied to $j(A)$ and j). So $t|_{W_j} = t_j$. Hence $\mathcal{F}'$ satisfies (5), and is a sheaf.

(c4) Conclusion. Suppose $\mathcal{F}'$ is a sheaf. Let $(\mathcal{F}^\dagger, \theta^\dagger)$ be the associated sheaf (Prop. 2.15). By the universal property there is a morphism $\psi : \mathcal{F}^\dagger \to \mathcal{F}'$ with $\psi\theta^\dagger = \theta$. On stalks $\theta_x = \psi_x\theta^\dagger_x$ and θ_x (by (c2)) and $\theta^\dagger_x$ (Prop. 2.15) are isomorphisms, so each ψ_x is an isomorphism, and ψ is an isomorphism by Prop. 2.12. Thus $(\mathcal{F}', \theta)$ is the sheaf associated to $\mathcal{F}$. This applies when $\mathcal{F}$ is separated, by (c3). Conversely, if $\mathcal{F}'$ is the associated sheaf, it is a sheaf. For arbitrary $\mathcal{F}$, $\mathcal{F}'$ is separated by (c3), so $(\mathcal{F}')'$ is a sheaf by (c3) again, and $\mathcal{F} \to \mathcal{F}' \to (\mathcal{F}')'$ induces isomorphisms on all stalks by (c2) applied twice; the argument above, applied to the morphism $\mathcal{F} \to (\mathcal{F}')'$ (a composite with stalk isomorphisms into a sheaf), shows that $(\mathcal{F}')'$ is the sheaf associated to $\mathcal{F}$.

Counterexample to (c) for non-separated $\mathcal{F}$. Let $X = \{a, b, p, q\}$ with the open sets $\emptyset, \{a\}, \{b\}, W = \{a, b\}, U_p = \{a, b, p\}, U_q = \{a, b, q\}, X$ (closed under unions and intersections). Define $\mathcal{F}(W) = \mathcal{F}(U_p) = \mathcal{F}(U_q) = \mathbf{Z}$, $\mathcal{F}(\emptyset) = \mathcal{F}(\{a\}) = \mathcal{F}(\{b\}) = \mathcal{F}(X) = 0$, restrictions among the three copies of $\mathbf{Z}$ the identity, all others zero (a presheaf, not separated, since $\mathcal{F}(W) \to \mathcal{F}(\{a\}) \times \mathcal{F}(\{b\})$ is not injective). Every covering of U_p contains U_p, so $\{U_p\}$ is equivalent to every covering of U_p and $\mathcal{F}'(U_p) = \mathbf{Z}$; likewise $\mathcal{F}'(U_q) = \mathbf{Z}$. The covering $\{\{a\}, \{b\}\}$ of W refines every covering of W, so $\mathcal{F}'(W) = \mathcal{F}(\{a\}) \times \mathcal{F}(\{b\}) = 0$. Every covering of X contains X or

both U_p, U_q, and $\{U_p, U_q\}$ refines every covering of X; so $\mathcal{F}'(X) = \{(m,n) \in \mathbf{Z}^2 : m = n\} \simeq \mathbf{Z}$, mapping onto $\mathcal{F}'(U_p)$ and $\mathcal{F}'(U_q)$ by the diagonal. The sections $1 \in \mathcal{F}'(U_p)$ and $2 \in \mathcal{F}'(U_q)$ agree on W (both restrict to $0 \in \mathcal{F}'(W)$) but do not glue, so $\mathcal{F}'$ is not a sheaf. (The associated sheaf has $\mathbf{Z} \times \mathbf{Z}$ as sections over X.) □

Exercise 2.4

> **Problem.** Let $0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}''$ be an exact sequence of sheaves on X. Show that $0 \to \mathcal{F}'(X) \to \mathcal{F}(X) \to \mathcal{F}''(X)$ is an exact sequence of abelian groups.

Solution. Call the maps $\alpha : \mathcal{F}' \to \mathcal{F}$ and $\beta : \mathcal{F} \to \mathcal{F}''$. By Prop. 2.18 the sequence of stalks $0 \to \mathcal{F}'_x \to \mathcal{F}_x \to \mathcal{F}''_x$ is exact for every $x \in X$, i.e. α_x is injective and $\operatorname{Ker} \beta_x = \operatorname{Im} \alpha_x$.

Exact at $\mathcal{F}'(X)$. If $s \in \mathcal{F}'(X)$ and $\alpha(X)(s) = 0$, then $\alpha_x(s_x) = (\alpha(X)(s))_x = 0$, so $s_x = 0$ for all x, and $s = 0$ by Lemma 2.9.

$\beta\alpha = 0$ on X. For $s \in \mathcal{F}'(X)$ the germs of $\beta(X)\alpha(X)(s)$ are $\beta_x\alpha_x(s_x) = 0$ (since $\operatorname{Im} \alpha_x \subseteq \operatorname{Ker} \beta_x$), so it vanishes by Lemma 2.9.

$\operatorname{Ker} \beta(X) \subseteq \operatorname{Im} \alpha(X)$. Let $u \in \mathcal{F}(X)$ with $\beta(X)(u) = 0$. For each $x \in X$, $\beta_x(u_x) = 0$, so $u_x = \alpha_x(\sigma)$ for some $\sigma \in \mathcal{F}'_x$. Represent $\sigma = (s_x)_x$ by a section $s_x \in \mathcal{F}'(U_x)$ on an open neighbourhood U_x of x. The germs of $\alpha(U_x)(s_x)$ and $u|_{U_x}$ at x coincide, hence after shrinking U_x (and replacing s_x by its restriction) we may assume $\alpha(U_x)(s_x) = u|_{U_x}$. On $U_x \cap U_y$,

$$\alpha(U_x \cap U_y)\big(s_x|_{U_x \cap U_y} - s_y|_{U_x \cap U_y}\big) = u|_{U_x \cap U_y} - u|_{U_x \cap U_y} = 0.$$

By injectivity of α on stalks, hence on every open set (the same argument as in the first step with Lemma 2.9 on $U_x \cap U_y$), s_x and s_y agree on $U_x \cap U_y$. As $\mathcal{F}'$ is a sheaf they glue to $s \in \mathcal{F}'(X)$ with $s|_{U_x} = s_x$. Then $\alpha(X)(s)$ and u agree on every U_x, hence are equal by (4). So $u \in \operatorname{Im} \alpha(X)$. □

Exercise 2.5

> **Problem.** (Supports of sheaves) Let $\mathcal{F}$ be a sheaf on X and $\operatorname{Supp} \mathcal{F} = \{x \in X \mid \mathcal{F}_x \neq 0\}$. We want to show that in general $\operatorname{Supp} \mathcal{F}$ is not a closed subset. Fix a sheaf $\mathcal{G}$ on X and a closed point $x_0 \in X$. Define a presheaf $\mathcal{F}$ by $\mathcal{F}(U) = \mathcal{G}(U)$ if $x_0 \notin U$ and $\mathcal{F}(U) = \{s \in \mathcal{G}(U) \mid s_{x_0} = 0\}$ otherwise. Show that $\mathcal{F}$ is a sheaf and that $\operatorname{Supp} \mathcal{F} = \operatorname{Supp} \mathcal{G} \setminus \{x_0\}$.

Solution. *Presheaf.* $\mathcal{F}(U)$ is a subgroup of $\mathcal{G}(U)$, and $\mathcal{F}(\emptyset) = \mathcal{G}(\emptyset) = 0$. Let $V \subseteq U$ and $s \in \mathcal{F}(U)$. If $x_0 \notin V$, then $s|_V \in \mathcal{G}(V) = \mathcal{F}(V)$. If $x_0 \in V$, then $x_0 \in U$, so $s_{x_0} = 0$ and $(s|_V)_{x_0} = s_{x_0} = 0$, hence $s|_V \in \mathcal{F}(V)$. So the restrictions of $\mathcal{G}$ induce restrictions on $\mathcal{F}$, and $\mathcal{F}$ is a subpresheaf of $\mathcal{G}$.

Sheaf. Axiom (4) holds for $\mathcal{F}$ because it holds for $\mathcal{G}$ and $\mathcal{F}(U) \subseteq \mathcal{G}(U)$. For (5), let $U = \bigcup U_i$ and $s_i \in \mathcal{F}(U_i)$ with $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$. Since $\mathcal{G}$ is a sheaf, there is $s \in \mathcal{G}(U)$ with $s|_{U_i} = s_i$. If $x_0 \notin U$, then $s \in \mathcal{G}(U) = \mathcal{F}(U)$. If $x_0 \in U$, pick i_0 with $x_0 \in U_{i_0}$; then $s_{x_0} = (s|_{U_{i_0}})_{x_0} = (s_{i_0})_{x_0} = 0$ because $s_{i_0} \in \mathcal{F}(U_{i_0})$. So $s \in \mathcal{F}(U)$, and $s|_{U_i} = s_i$ in $\mathcal{F}$.

Stalks. Let $x \neq x_0$. Since x_0 is closed, $X \setminus \{x_0\}$ is an open neighbourhood of x, and the open neighbourhoods $U \subseteq X \setminus \{x_0\}$ of x are cofinal among all neighbourhoods. For such U, $\mathcal{F}(U) = \mathcal{G}(U)$ with the same restrictions. Hence $\mathcal{F}_x = \mathcal{G}_x$. At x_0: a germ in $\mathcal{F}_{x_0}$ is s_{x_0} with $s \in \mathcal{F}(U)$, $x_0 \in U$, hence $s_{x_0} = 0$ in $\mathcal{G}_{x_0}$, so there is an open $V \ni x_0$, $V \subseteq U$, with $s|_V = 0$ in $\mathcal{G}(V)$, hence in $\mathcal{F}(V)$. Thus the germ of s at x_0 in $\mathcal{F}_{x_0}$ is 0, and $\mathcal{F}_{x_0} = 0$.

Therefore $\operatorname{Supp}\mathcal{F} = \{x \neq x_0 : \mathcal{G}_x \neq 0\} = \operatorname{Supp}\mathcal{G} \setminus \{x_0\}$. For example, if $\mathcal{G}$ is the sheaf of continuous real functions on $X = \mathbf{R}$ and $x_0 = 0$, then $\operatorname{Supp}\mathcal{G} = \mathbf{R}$ and $\operatorname{Supp}\mathcal{F} = \mathbf{R} \setminus \{0\}$ is not closed. □

Exercise 2.6

> **Problem.** Let $f : X \to Y$, $g : Y \to Z$ be continuous maps, $\mathcal{F}$ a sheaf on X and $\mathcal{H}$ a sheaf on Z. Show that $f^{-1}(g^{-1}\mathcal{H}) = (g \circ f)^{-1}\mathcal{H}$ and $g_*(f_*\mathcal{F}) = (g \circ f)_*\mathcal{F}$.

Solution. Direct images. For $W \subseteq Z$ open, $(g_*(f_*\mathcal{F}))(W) = (f_*\mathcal{F})(g^{-1}W) = \mathcal{F}(f^{-1}(g^{-1}W)) = \mathcal{F}((g \circ f)^{-1}W) = ((g \circ f)_*\mathcal{F})(W)$, and the restriction maps of both sheaves are those of $\mathcal{F}$ (for $W' \subseteq W$ they are the restrictions $\mathcal{F}(f^{-1}g^{-1}W) \to \mathcal{F}(f^{-1}g^{-1}W')$). So the two sheaves are equal.

Inverse images. Since f^{-1} is defined by a sheafification, the equality means a canonical isomorphism. For a presheaf $\mathcal{P}$ on Y let $P_f\mathcal{P}$ be the presheaf $U \mapsto \varinjlim_{V \supseteq f(U)} \mathcal{P}(V)$ on X (V open in Y); so $f^{-1}\mathcal{G} = (P_f\mathcal{G})^\dagger$. It is functorial in $\mathcal{P}$.

Stalks. For every presheaf $\mathcal{P}$ on Y and $x \in X$ there is an isomorphism $\varepsilon : (P_f\mathcal{P})_x \to \mathcal{P}_{f(x)}$ sending the germ of the class $[V, s]$ ($s \in \mathcal{P}(V)$, $V \supseteq f(U)$, $x \in U$) to $s_{f(x)}$ (note $f(x) \in V$). It is well defined by the compatibility of the direct limits. It is surjective: given $s \in \mathcal{P}(V)$ with $f(x) \in V$, take $U = f^{-1}(V) \ni x$. It is injective: if $s_{f(x)} = 0$, there is an open $V' \ni f(x)$, $V' \subseteq V$, with $s|_{V'} = 0$; on $U' = U \cap f^{-1}(V') \ni x$ the class $[V, s]|_{U'} = [V', s|_{V'}]$ is 0. These isomorphisms are natural in $\mathcal{P}$.

The comparison map. Let $\theta : P_g\mathcal{H} \to g^{-1}\mathcal{H}$ be the canonical morphism. Define $\mu : P_{g\circ f}\mathcal{H} \to P_f(P_g\mathcal{H})$ by sending the class $[W, s]$ ($s \in \mathcal{H}(W)$, $W \supseteq gf(U)$) to $[g^{-1}(W), [W, s]]$; here $V = g^{-1}(W) \supseteq f(U)$ and $[W, s] \in (P_g\mathcal{H})(V)$ because $W \supseteq g(V)$. This is compatible with the identifications in the direct limits and with restrictions, so it is a morphism of presheaves on X. Let

$$\varphi = P_f(\theta) \circ \mu : \ P_{g\circ f}\mathcal{H} \longrightarrow P_f(g^{-1}\mathcal{H}).$$

On stalks at x we have, with ε as above, $\varepsilon_x^{g\circ f} = \varepsilon_{f(x)}^g \circ \varepsilon_x^f \circ \mu_x$ (both send the germ of $[W, s]$ to $s_{gf(x)}$), hence μ_x is an isomorphism; and $\varepsilon_x^f \circ P_f(\theta)_x = \theta_{f(x)} \circ \varepsilon_x^f$, where $\theta_{f(x)}$ is an isomorphism by Prop. 2.15. So φ_x is an isomorphism for all x.

Sheafifying. Let $\vartheta_1 : P_{g\circ f}\mathcal{H} \to (g \circ f)^{-1}\mathcal{H}$ and $\vartheta_2 : P_f(g^{-1}\mathcal{H}) \to f^{-1}g^{-1}\mathcal{H}$ be the canonical maps. By the universal property (Def. 2.14) the morphism $\vartheta_2\varphi$ factors uniquely as $\Phi\vartheta_1$ with $\Phi : (g \circ f)^{-1}\mathcal{H} \to f^{-1}(g^{-1}\mathcal{H})$ a morphism of sheaves. On stalks $\vartheta_{2,x}\varphi_x = \Phi_x\vartheta_{1,x}$, where $\vartheta_{1,x}, \vartheta_{2,x}, \varphi_x$ are isomorphisms (Prop. 2.15), so Φ_x is an isomorphism for every x, and Φ is an isomorphism by Prop. 2.12. This is the canonical identification $f^{-1}(g^{-1}\mathcal{H}) \simeq (g \circ f)^{-1}\mathcal{H}$ (compatible with the stalk identifications $\mathcal{H}_{gf(x)}$). □

Exercise 2.7

Problem. Let $\mathcal{B}$ be a base of open subsets of a topological space X (stable under finite intersection, cf. Rem. 2.6). Let $\mathcal{F}, \mathcal{G}$ be two sheaves on X. Suppose that for every $U \in \mathcal{B}$ there is a homomorphism $\alpha(U) : \mathcal{F}(U) \to \mathcal{G}(U)$ compatible with restrictions (between members of $\mathcal{B}$). Show that this extends in a unique way to a homomorphism of sheaves $\alpha : \mathcal{F} \to \mathcal{G}$. Show that if $\alpha(U)$ is surjective (resp. injective) for every $U \in \mathcal{B}$, then α is surjective (resp. injective).

Solution. Characterisation. For an open U and $s \in \mathcal{F}(U)$ we look for $t \in \mathcal{G}(U)$ with

$$t|_V = \alpha(V)(s|_V) \qquad \text{for every } V \in \mathcal{B},\ V \subseteq U. \qquad (\dagger)$$

Uniqueness. The members of $\mathcal{B}$ contained in U cover U ($\mathcal{B}$ is a base), so by (4) at most one t satisfies (†).

Existence. Write $U = \bigcup_i U_i$ with $U_i \in \mathcal{B}$ and set $t_i = \alpha(U_i)(s|_{U_i}) \in \mathcal{G}(U_i)$. Since $U_i \cap U_j \in \mathcal{B}$ and α commutes with restrictions between members of $\mathcal{B}$,

$$t_i|_{U_i \cap U_j} = \alpha(U_i \cap U_j)(s|_{U_i \cap U_j}) = t_j|_{U_i \cap U_j}.$$

By (5) there is $t \in \mathcal{G}(U)$ with $t|_{U_i} = t_i$. Let $V \in \mathcal{B}$, $V \subseteq U$. For every i, $V \cap U_i \in \mathcal{B}$ and

$$t|_{V \cap U_i} = t_i|_{V \cap U_i} = \alpha(V \cap U_i)(s|_{V \cap U_i}) = \big(\alpha(V)(s|_V)\big)|_{V \cap U_i}.$$

The sets $V \cap U_i$ cover V, so $t|_V = \alpha(V)(s|_V)$ by (4). Thus t satisfies (†).

Definition of α. Let $\alpha(U)(s)$ be this unique t. Since (†) is linear in (s,t), $\alpha(U)$ is a group homomorphism. For $U' \subseteq U$ open, $\alpha(U)(s)|_{U'}$ satisfies (†) for $s|_{U'}$ (for $V \in \mathcal{B}$, $V \subseteq U'$: $(\alpha(U)(s)|_{U'})|_V = \alpha(U)(s)|_V = \alpha(V)(s|_V)$), so it equals $\alpha(U')(s|_{U'})$; hence α commutes with all restrictions and is a morphism of sheaves. If $U \in \mathcal{B}$, the given $\alpha(U)(s)$ satisfies (†) by the hypothesis that the given maps commute with restrictions between members of $\mathcal{B}$, so by uniqueness it equals the t defined above. Thus α extends the given family.

Uniqueness of the extension. If α' is a morphism of sheaves with $\alpha'(V) = \alpha(V)$ for $V \in \mathcal{B}$, then for every U and $s \in \mathcal{F}(U)$ the section $t = \alpha'(U)(s)$ satisfies $t|_V = \alpha'(V)(s|_V) = \alpha(V)(s|_V)$ for $V \in \mathcal{B}$, $V \subseteq U$, i.e. (†). So $t = \alpha(U)(s)$ by uniqueness.

Injectivity. Suppose all $\alpha(V)$, $V \in \mathcal{B}$, are injective. Let $x \in X$ and $\sigma \in \mathcal{F}_x$ with $\alpha_x(\sigma) = 0$. Write $\sigma = s_x$ with $s \in \mathcal{F}(U)$; since $\mathcal{B}$ is a base we may shrink U and assume $U \in \mathcal{B}$. As $(\alpha(U)(s))_x = 0$, there is an open $U' \ni x$, $U' \subseteq U$, with $\alpha(U)(s)|_{U'} = 0$; choose $V \in \mathcal{B}$ with $x \in V \subseteq U'$. Then $\alpha(V)(s|_V) = \alpha(U)(s)|_V = 0$ so $s|_V = 0$ and $\sigma = 0$. Hence every α_x is injective. Now if U is open and $\alpha(U)(s) = 0$, then $\alpha_x(s_x) = 0$, so $s_x = 0$ for all $x \in U$, and $s = 0$ by Lemma 2.9. So $\alpha(U)$ is injective for every open U.

Surjectivity. Suppose all $\alpha(V)$, $V \in \mathcal{B}$, are surjective. Let $x \in X$ and $\tau \in \mathcal{G}_x$; write $\tau = t_x$ with $t \in \mathcal{G}(U)$ and shrink to $U \in \mathcal{B}$ containing x. Choose $s \in \mathcal{F}(U)$ with $\alpha(U)(s) = t$. Then $\tau = (\alpha(U)(s))_x = \alpha_x(s_x)$. So every α_x is surjective, i.e. α is surjective in the sense of Def. 2.10. (Surjectivity of $\alpha(U)$ for *all* open U cannot be expected, cf. Example 2.11.) □

Conventions for this part. Sheaves are sheaves of abelian groups unless said otherwise; all arguments work verbatim for sheaves of rings. For a morphism $f : X \to Y$ of ringed spaces and $V \subseteq Y$ open, $f^\#(V) : \mathcal{O}_Y(V) \to \mathcal{O}_X(f^{-1}V)$ is the ring homomorphism given by $f^\#$. For morphisms $f : X \to Y$, $h : Y \to Z$ the composition is $(h \circ f)^\# = h_*(f^\#) \circ h^\#$, that is,

$(h\circ f)^{\#}(W) = f^{\#}(h^{-1}W)\circ h^{\#}(W)$. For a morphism $g: U\to Y$ and an open set $U'\subseteq U$, the restriction $g|_{U'}$ is the composite of g with the open immersion $U'\to U$, so $(g|_{U'})^{\#}(V)$ is $g^{\#}(V)$ followed by restriction $\mathcal{O}_X(g^{-1}V)\to\mathcal{O}_X(g^{-1}V\cap U')$.

Exercise 2.8

> **Problem.** (Glueing sheaves) Let $\{U_i\}_i$ be an open covering of X. Suppose given sheaves $\mathcal{F}_i$ on U_i and isomorphisms $\varphi_{ij}:\mathcal{F}_i|_{U_i\cap U_j}\simeq\mathcal{F}_j|_{U_i\cap U_j}$ such that $\varphi_{ii}=\mathrm{Id}$ and $\varphi_{ik}=\varphi_{jk}\circ\varphi_{ij}$ on $U_i\cap U_j\cap U_k$. Show that there exists a unique sheaf $\mathcal{F}$ on X with isomorphisms $\psi_i:\mathcal{F}|_{U_i}\simeq\mathcal{F}_i$ such that $\psi_j=\varphi_{ij}\circ\psi_i$ on $U_i\cap U_j$.

Solution. "Unique" means unique up to a unique isomorphism compatible with the ψ_i; this is what we prove. Taking $k=i$ in the cocycle condition gives $\varphi_{ji}\circ\varphi_{ij}=\varphi_{ii}=\mathrm{Id}$, so $\varphi_{ji}=\varphi_{ij}^{-1}$.

Construction. For $U\subseteq X$ open, let $\mathcal{F}(U)$ be the set of families $s=(s_i)_i$ with $s_i\in\mathcal{F}_i(U\cap U_i)$ such that

$$\varphi_{ij}\big(s_i|_{U\cap U_i\cap U_j}\big)=s_j|_{U\cap U_i\cap U_j}\qquad\text{for all } i,j.$$

This is a subgroup of $\prod_i\mathcal{F}_i(U\cap U_i)$ because the φ_{ij} and the restrictions are homomorphisms. For $U'\subseteq U$ define the restriction componentwise, $s|_{U'}=(s_i|_{U'\cap U_i})_i$. It preserves the displayed condition because φ_{ij} commutes with restrictions, and it is clearly functorial. Also $\mathcal{F}(\emptyset)=0$. So $\mathcal{F}$ is a presheaf.

$\mathcal{F}$ is a sheaf. Let $U=\bigcup_a W_a$ be an open covering. (Uniqueness) If $s\in\mathcal{F}(U)$ and $s|_{W_a}=0$ for all a, then for every i the sets $W_a\cap U_i$ cover $U\cap U_i$ and $s_i|_{W_a\cap U_i}=0$, so $s_i=0$ because $\mathcal{F}_i$ is a sheaf. (Glueing) Let $s^a\in\mathcal{F}(W_a)$ with $s^a|_{W_a\cap W_b}=s^b|_{W_a\cap W_b}$. Fix i. The sections $s_i^a\in\mathcal{F}_i(W_a\cap U_i)$ agree on the overlaps $W_a\cap W_b\cap U_i$, hence glue to $s_i\in\mathcal{F}_i(U\cap U_i)$ with $s_i|_{W_a\cap U_i}=s_i^a$. We check that $s=(s_i)$ lies in $\mathcal{F}(U)$. The two sections $\varphi_{ij}(s_i|_{U\cap U_i\cap U_j})$ and $s_j|_{U\cap U_i\cap U_j}$ of $\mathcal{F}_j$ over $U\cap U_i\cap U_j$ have the same restriction to each $W_a\cap U_i\cap U_j$, namely $\varphi_{ij}(s_i^a|)=s_j^a|$. By the uniqueness property of $\mathcal{F}_j$ they are equal. Clearly $s|_{W_a}=s^a$.

The isomorphisms ψ_i. For $U\subseteq U_i$ open, put $\psi_i(U):\mathcal{F}(U)\to\mathcal{F}_i(U)$, $s\mapsto s_i$ (note $U\cap U_i=U$). It is a homomorphism compatible with restrictions. It is bijective: define $\chi:\mathcal{F}_i(U)\to\mathcal{F}(U)$ by $\chi(t)=(\varphi_{ij}(t|_{U\cap U_j}))_j$. This lies in $\mathcal{F}(U)$: for j,k, on $U\cap U_j\cap U_k$ we have $\varphi_{jk}(\varphi_{ij}(t|))=\varphi_{ik}(t|)$ by the cocycle condition, which is the required relation between the j-th and k-th components. Then $\psi_i\chi(t)=\varphi_{ii}(t)=t$. Conversely if $s\in\mathcal{F}(U)$ then the j-th component of s is $s_j=\varphi_{ij}(s_i|_{U\cap U_j})$ by definition of $\mathcal{F}(U)$, so $\chi\psi_i(s)=s$. Hence $\psi_i:\mathcal{F}|_{U_i}\simeq\mathcal{F}_i$. Finally, for $U\subseteq U_i\cap U_j$ and $s\in\mathcal{F}(U)$ we have $\psi_j(s)=s_j=\varphi_{ij}(s_i)=\varphi_{ij}(\psi_i(s))$, i.e. $\psi_j=\varphi_{ij}\circ\psi_i$ on $U_i\cap U_j$.

Uniqueness. Let $\mathcal{F}'$ be a sheaf with isomorphisms $\psi_i':\mathcal{F}'|_{U_i}\simeq\mathcal{F}_i$ such that $\psi_j'=\varphi_{ij}\psi_i'$ on $U_i\cap U_j$. Define $\Theta(U):\mathcal{F}'(U)\to\mathcal{F}(U)$ by $\Theta(U)(s)=(\psi_i'(s|_{U\cap U_i}))_i$. It lies in $\mathcal{F}(U)$ since $\varphi_{ij}\psi_i'(s|)=\psi_j'(s|)$ on $U\cap U_i\cap U_j$. It is a morphism of sheaves. It is injective: if all $\psi_i'(s|_{U\cap U_i})=0$ then $s|_{U\cap U_i}=0$ for all i, so $s=0$. It is surjective: given $(s_i)\in\mathcal{F}(U)$, put $t_i=\psi_i'^{-1}(s_i)\in\mathcal{F}'(U\cap U_i)$. On $U\cap U_i\cap U_j$ we have $\psi_j'(t_i|)=\varphi_{ij}\psi_i'(t_i|)=\varphi_{ij}(s_i|)=s_j|=\psi_j'(t_j|)$, so $t_i|=t_j|$ by injectivity of ψ_j'. The t_i glue to $t\in\mathcal{F}'(U)$ and $\Theta(U)(t)=(s_i)$. So $\Theta:\mathcal{F}'\simeq\mathcal{F}$, and $\psi_i\circ\Theta|_{U_i}=\psi_i'$ by construction. Any isomorphism Θ' with $\psi_i\Theta'=\psi_i'$ on U_i has $\Theta'(s)_i=\psi_i'(s|_{U\cap U_i})$, so $\Theta'=\Theta$. □

Exercise 2.9

Problem. Let $\mathcal{F}$ be a sheaf on X whose support $Y = \operatorname{Supp}\mathcal{F} = \{x : \mathcal{F}_x \neq 0\}$ is a finite set of closed points (a skyscraper sheaf). Show that for every open subset U of X we have $\mathcal{F}(U) = \bigoplus_{x\in U} \mathcal{F}_x$. In particular, $\mathcal{F} = 0$ if and only if $\mathcal{F}(X) = 0$.

Solution. Since $\mathcal{F}_x = 0$ for $x \notin Y$, the sum $\bigoplus_{x\in U} \mathcal{F}_x$ equals $\bigoplus_{y\in U\cap Y} \mathcal{F}_y$, a finite direct sum (equal to the product). The precise statement is that the canonical map

$$\Phi_U : \mathcal{F}(U) \to \bigoplus_{x\in U} \mathcal{F}_x, \qquad s \mapsto (s_x)_{x\in U},$$

is an isomorphism. It is a homomorphism, and $s_x = 0$ for $x \notin Y$ so the image lies in the direct sum.

Injective. If $s_y = 0$ for all $y \in U \cap Y$, then $s_x = 0$ for all $x \in U$, hence $s = 0$ by Lemma 2.9 (applied to the sheaf $\mathcal{F}|_U$).

Surjective. Let $(t_y)_{y\in U\cap Y}$ be given, $t_y \in \mathcal{F}_y$. For each y choose an open neighbourhood $V'_y \subseteq U$ of y and $s^y \in \mathcal{F}(V'_y)$ with germ t_y at y. The set $Y \setminus \{y\}$ is finite, consisting of closed points, hence closed, so $V_y = V'_y \setminus (Y \setminus \{y\})$ is open, contains y, and $V_y \cap Y = \{y\}$. Replace s^y by $s^y|_{V_y}$. Put $V_0 = U \setminus Y$, open because Y is closed (finite union of closed points), and $s^0 = 0 \in \mathcal{F}(V_0)$. The sets V_0 and V_y ($y \in U \cap Y$) cover U. For two distinct members of this family, the intersection contains no point of Y (each V_y meets Y only in y, V_0 misses Y, and $V_y \cap V_z$ misses y and z for $y \neq z$). Since $\mathcal{F}_w = 0$ for $w \notin Y$, the two sections have all germs zero on the intersection, and so coincide there by Lemma 2.9. Hence they glue to $s \in \mathcal{F}(U)$ with $s_y = s^y_y = t_y$ for $y \in U \cap Y$. So $\Phi_U(s) = (t_y)$.

The last statement. If $\mathcal{F} = 0$ then $\mathcal{F}(X) = 0$. Conversely, if $\mathcal{F}(X) = 0$, then $\bigoplus_{y\in Y} \mathcal{F}_y = 0$ by the first part, so $\mathcal{F}_y = 0$ for each $y \in Y$. But $\mathcal{F}_y \neq 0$ for $y \in Y$ by definition of the support, so $Y = \emptyset$, all stalks vanish, and $\mathcal{F} = 0$ by Prop. 2.12 (the morphism $\mathcal{F} \to 0$ is an isomorphism on stalks). □

Exercise 2.10

Problem. Let $(X, \mathcal{O}_X)$ be a ringed topological space. Let Y be a subset of X endowed with the induced topology, and let $i : Y \to X$ be the canonical inclusion. Show that $(Y, i^{-1}\mathcal{O}_X)$ is a ringed topological space.

Solution. Recall the construction: for $U \subseteq Y$ open, $P(U) = \varinjlim_{V\supseteq i(U)} \mathcal{O}_X(V)$ (limit over open $V \subseteq X$ containing U) is a presheaf on Y. Because $\mathcal{O}_X(V)$ are rings and restrictions are ring homomorphisms, $P(U)$ is a ring (a filtered direct limit of rings) and the restriction maps of P are ring homomorphisms. So P is a presheaf of rings. The sheaf $P^\dagger = i^{-1}\mathcal{O}_X$ of Prop. 2.15 consists of functions $f : U \to \coprod_{y\in U} P_y$ locally given by sections of P; the pointwise operations make $P^\dagger(U)$ a ring, since the stalks P_y are rings (Def. 2.8) and the local description is stable under sums and products (take the intersection of two neighbourhoods). The restrictions are ring homomorphisms, and $\theta : P \to P^\dagger$ is a ring homomorphism. So $i^{-1}\mathcal{O}_X$ is a sheaf of rings.

It remains to show that the stalks are local rings. We have $(i^{-1}\mathcal{O}_X)_y \simeq P_y$ as rings (Prop. 2.15), and $P_y = \varinjlim_{U\ni y} \varinjlim_{V\supseteq U} \mathcal{O}_X(V)$. The pairs (U, V) with U open in Y, V open in X, $y \in U \subseteq V$

form a directed system (order by reverse inclusion in both) and the pairs $(V \cap Y, V)$ are cofinal in it (given (U, V), write $U = V' \cap Y$ with V' open in X; then $(V \cap V' \cap Y, V \cap V')$ lies below (U, V)). Therefore $P_y = \varinjlim_{V \ni y} \mathcal{O}_X(V) = \mathcal{O}_{X,y}$, an isomorphism of rings. This is the formula $(i^{-1}\mathcal{O}_X)_y = \mathcal{O}_{X,i(y)}$ recalled in the text. Since $\mathcal{O}_{X,y}$ is local, so is $(i^{-1}\mathcal{O}_X)_y$. Hence $(Y, i^{-1}\mathcal{O}_X)$ is a ringed topological space. □

Exercise 2.11

> **Problem.** Let X, Y be two ringed topological spaces. Suppose given an open covering $\{U_i\}_i$ of X and morphisms $f_i : U_i \to Y$ such that $f_i|_{U_i \cap U_j} = f_j|_{U_i \cap U_j}$ (as morphisms) for all i, j. Show that there exists a unique morphism $f : X \to Y$ such that $f|_{U_i} = f_i$ (the glueing of the f_i).

Solution. *Underlying map.* The hypothesis says in particular that the continuous maps f_i and f_j agree on $U_i \cap U_j$. So there is a unique map $f : X \to Y$ with $f|_{U_i} = f_i$. It is continuous: for $V \subseteq Y$ open, $f^{-1}V = \bigcup_i f_i^{-1}(V)$ with each $f_i^{-1}(V)$ open in U_i, hence open in X. Note $f_i^{-1}(V) = f^{-1}(V) \cap U_i$.

The sheaf map. Let $V \subseteq Y$ be open and $a \in \mathcal{O}_Y(V)$. Put $b_i = f_i^{\#}(V)(a) \in \mathcal{O}_X(f^{-1}V \cap U_i)$ (recall $\mathcal{O}_{U_i} = \mathcal{O}_X|_{U_i}$). The equality $(f_i|_{U_i \cap U_j})^{\#} = (f_j|_{U_i \cap U_j})^{\#}$ at V, applied to a, says exactly that b_i and b_j have the same restriction to $f^{-1}V \cap U_i \cap U_j$. Since the $f^{-1}V \cap U_i$ cover $f^{-1}V$, there is a unique $f^{\#}(V)(a) \in \mathcal{O}_X(f^{-1}V)$ with restriction b_i to each $f^{-1}V \cap U_i$. The map $f^{\#}(V)$ is a ring homomorphism, since sums, products and 1 are checked after restricting to the $f^{-1}V \cap U_i$, where $f^{\#}(V)$ equals $f_i^{\#}(V)$. It is compatible with restriction in V for the same reason. So $f^{\#} : \mathcal{O}_Y \to f_*\mathcal{O}_X$ is a morphism of sheaves of rings.

Locality. Let $x \in X$, say $x \in U_i$. For a germ $\alpha = a_{f(x)}$ ($a \in \mathcal{O}_Y(V)$, $f(x) \in V$) the germ $f_x^{\#}(\alpha)$ is the germ at x of $f^{\#}(V)(a)$, which coincides near x with $f_i^{\#}(V)(a)$. So $f_x^{\#} = (f_i)_x^{\#}$, which is a local homomorphism. Thus $(f, f^{\#})$ is a morphism of ringed spaces.

Properties. By construction $f|_{U_i}$ has the underlying map f_i, and $(f|_{U_i})^{\#}(V)(a) = f^{\#}(V)(a)|_{f^{-1}V \cap U_i} = f_i^{\#}(V)(a)$. So $f|_{U_i} = f_i$.

Uniqueness. If $g : X \to Y$ is a morphism with $g|_{U_i} = f_i$ for all i, then $g = f$ as maps, and for $a \in \mathcal{O}_Y(V)$ the section $g^{\#}(V)(a)$ restricts on $f^{-1}V \cap U_i$ to $(g|_{U_i})^{\#}(V)(a) = f_i^{\#}(V)(a)$. By the uniqueness in the sheaf property, $g^{\#}(V)(a) = f^{\#}(V)(a)$. □

Exercise 2.12

> **Problem.** Let $f : X \to Y$ be a morphism of ringed topological spaces. Let V be an open subset of Y containing $f(X)$. Show that there exists a unique morphism $g : X \to V$ whose composition with the open immersion $V \to Y$ is f.

Solution. Let $j : V \to Y$ be the inclusion; as a morphism of ringed spaces $(V, \mathcal{O}_Y|_V) \to (Y, \mathcal{O}_Y)$, $j^{\#}(W) : \mathcal{O}_Y(W) \to \mathcal{O}_Y(W \cap V)$ is the restriction.

Existence. Let $g : X \to V$ be the map $x \mapsto f(x)$; it is continuous because the topology of V is induced from Y and $g^{-1}(W \cap V) = f^{-1}(W)$. Since $f(X) \subseteq V$ we have $f^{-1}(W) = f^{-1}(W \cap V) =$

$g^{-1}(W \cap V)$ for every open $W \subseteq Y$. For $W \subseteq V$ open (it is open in V and in Y) define

$$g^{\#}(W) = f^{\#}(W) : \mathcal{O}_V(W) = \mathcal{O}_Y(W) \to \mathcal{O}_X(f^{-1}W) = \mathcal{O}_X(g^{-1}W).$$

This is a morphism of sheaves of rings on V (compatible with restrictions between open subsets of V since $f^{\#}$ is). For $x \in X$, $g^{\#}_x = f^{\#}_x$ since $\mathcal{O}_{V,g(x)} = \mathcal{O}_{Y,f(x)}$; this is local. So g is a morphism of ringed spaces. For $W \subseteq Y$ open and $a \in \mathcal{O}_Y(W)$,

$$(j\circ g)^{\#}(W)(a) = g^{\#}(W\cap V)(a|_{W\cap V}) = f^{\#}(W\cap V)(a|_{W\cap V}) = f^{\#}(W)(a)|_{f^{-1}(W\cap V)} = f^{\#}(W)(a),$$

using that $f^{\#}$ commutes with restriction and $f^{-1}(W \cap V) = f^{-1}(W)$. Hence $j \circ g = f$.

Uniqueness. Let g' satisfy $j \circ g' = f$. As j is injective, $g' = g$ as maps. For $W \subseteq V$ open, $j^{\#}(W)$ is the identity of $\mathcal{O}_Y(W)$, so $f^{\#}(W) = (j \circ g')^{\#}(W) = g'^{\#}(W) \circ j^{\#}(W) = g'^{\#}(W)$. Thus $g'^{\#} = f^{\#}|_V = g^{\#}$. □

Exercise 2.13

Problem. Let $f : X \to Y$ be a continuous map of topological spaces. Let $\mathcal{F}$ be a sheaf on X and $\mathcal{G}$ a sheaf on Y.

(a) Show that there exist canonical morphisms of sheaves $\mathcal{G} \to f_* f^{-1}\mathcal{G}$ and $f^{-1} f_* \mathcal{F} \to \mathcal{F}$. Moreover, if f is a closed immersion, then the first morphism is surjective. If f is an open immersion, then the second morphism is an isomorphism.

(b) Using these maps, show that there exists a canonical bijection $\mathrm{Mor}_Y(\mathcal{G}, f_*\mathcal{F}) \simeq \mathrm{Mor}_X(f^{-1}\mathcal{G}, \mathcal{F})$.

Solution. *Remark on hypotheses.* Here "closed immersion" and "open immersion" are meant topologically (a homeomorphism onto a closed, resp. open, subset). In fact the proofs below only use that f is a homeomorphism onto its image (an embedding), plus closedness of the image for the surjectivity claim.

Notation. For $U \subseteq X$ open let $P_{\mathcal{G}}(U) = \varinjlim_{W \supseteq f(U)} \mathcal{G}(W)$ (W open in Y), and write $[s]_W \in P_{\mathcal{G}}(U)$ for the class of $s \in \mathcal{G}(W)$ when $f(U) \subseteq W$. So $f^{-1}\mathcal{G} = P^{\dagger}_{\mathcal{G}}$ with $\theta = \theta_{\mathcal{G}} : P_{\mathcal{G}} \to f^{-1}\mathcal{G}$ as in Prop. 2.15.

Step 0: stalks. For $x \in X$ and $y = f(x)$ the map $P_{\mathcal{G},x} \to \mathcal{G}_y$, sending the germ at x of $[s]_W$ to s_y, is well defined and bijective. Surjective: a germ s_y with $s \in \mathcal{G}(W)$ is the image of the germ of $[s]_W \in P_{\mathcal{G}}(f^{-1}W)$. Injective: if $s_y = 0$ there is an open $W' \ni y$, $W' \subseteq W$ with $s|_{W'} = 0$; then, for a representative $[s]_W \in P_{\mathcal{G}}(U)$ with $x \in U$, its restriction to $U \cap f^{-1}W'$ is $[s]_W = [s|_{W'}]_{W'} = 0$ in $P_{\mathcal{G}}(U \cap f^{-1}W')$ (note $f(U \cap f^{-1}W') \subseteq W'$), so the germ at x is 0. Composing with θ_x (an isomorphism by Prop. 2.15) we get the isomorphism $\kappa : (f^{-1}\mathcal{G})_x \simeq \mathcal{G}_{f(x)}$ of the text, characterized by $\kappa\big(\theta([s]_W)_x\big) = s_{f(x)}$.

Step 1: the morphism $\eta : \mathcal{G} \to f_* f^{-1}\mathcal{G}$. For $V \subseteq Y$ open we have $f(f^{-1}V) \subseteq V$, so $[s]_V \in P_{\mathcal{G}}(f^{-1}V)$ is defined for $s \in \mathcal{G}(V)$. Put $\eta(V)(s) = \theta(f^{-1}V)([s]_V) \in (f^{-1}\mathcal{G})(f^{-1}V) = (f_* f^{-1}\mathcal{G})(V)$. This is a homomorphism, and it is compatible with restrictions in V because the classes $[s]_V$ and θ are.

Step 2: the morphism $\varepsilon : f^{-1} f_* \mathcal{F} \to \mathcal{F}$. Here $P_{f_*\mathcal{F}}(U) = \varinjlim_{W \supseteq f(U)} \mathcal{F}(f^{-1}W)$. Define $\varepsilon'(U) : P_{f_*\mathcal{F}}(U) \to \mathcal{F}(U)$ by $[t]_W \mapsto t|_U$ for $t \in \mathcal{F}(f^{-1}W)$, which makes sense as $U \subseteq f^{-1}W$. It is

compatible with the transition maps of the direct limit (restriction in W followed by restriction to U equals restriction to U) and with restriction in U. So ε' is a morphism of presheaves, and since $\mathcal{F}$ is a sheaf, the universal property of Def. 2.14 gives a unique $\varepsilon : f^{-1}f_*\mathcal{F} \to \mathcal{F}$ with $\varepsilon \circ \theta_{f_*\mathcal{F}} = \varepsilon'$.

Step 3: a lemma. **Lemma.** Let f be an embedding (homeomorphism onto its image with the induced topology), $\mathcal{H}$ a sheaf on X, $x \in X$, $y = f(x)$. Then $c : (f_*\mathcal{H})_y \to \mathcal{H}_x$, $t_y \mapsto t_x$ (for $t \in \mathcal{H}(f^{-1}V)$) is bijective. If moreover $f(X)$ is closed and $y \notin f(X)$ then $(f_*\mathcal{H})_y = 0$.

Proof. Surjective: a germ h_x with $h \in \mathcal{H}(U)$; since f is an embedding, $U = f^{-1}V$ for some open $V \subseteq Y$, so $h \in (f_*\mathcal{H})(V)$ and $c(h_y) = h_x$. Injective: if $t \in \mathcal{H}(f^{-1}V)$ and $t_x = 0$, there is an open $U' \ni x$, $U' \subseteq f^{-1}V$ with $t|_{U'} = 0$. Write $U' = f^{-1}V'$ and replace V' by $V' \cap V$ (which does not change $f^{-1}V'$ because $U' \subseteq f^{-1}V$). Then $V' \subseteq V$ is a neighbourhood of y and $t|_{V'} = t|_{f^{-1}V'} = 0$ in $(f_*\mathcal{H})(V')$, so $t_y = 0$. For the last claim, if $f(X)$ is closed and $y \notin f(X)$, then y has an open neighbourhood V disjoint from $f(X)$, and $(f_*\mathcal{H})(V) = \mathcal{H}(\emptyset) = 0$; so the stalk is 0. $\square$

Step 4: closed immersions. Let f be an embedding with closed image. We show $\eta_y : \mathcal{G}_y \to (f_*f^{-1}\mathcal{G})_y$ is surjective for every $y \in Y$. If $y \notin f(X)$, the target is 0 by the Lemma. If $y = f(x)$, take $s \in \mathcal{G}(V)$ with $y \in V$. Then $\eta_y(s_y)$ is the germ at y of $\theta([s]_V) \in (f^{-1}\mathcal{G})(f^{-1}V)$; applying c gives the germ of this section at x, and then κ gives $s_{f(x)} = s_y$. Hence $\kappa \circ c \circ \eta_y = \mathrm{Id}$ on $\mathcal{G}_y$; since $\kappa \circ c$ is bijective (Step 0 and the Lemma), η_y is bijective, in particular surjective. So η is surjective in the sense of the text (it is even an isomorphism on the stalks at points of $f(X)$).

Step 5: open immersions. Let f be an embedding (e.g. an open immersion). For $x \in X$ and $y = f(x)$, the stalk map $\varepsilon_x : (f^{-1}f_*\mathcal{F})_x \to \mathcal{F}_x$ sends $\theta([t]_W)_x$ (with $t \in \mathcal{F}(f^{-1}W)$) to $(\varepsilon'[t]_W)_x = t_x$. Under the isomorphism $\kappa : (f^{-1}f_*\mathcal{F})_x \simeq (f_*\mathcal{F})_y$ (Step 0 for the sheaf $f_*\mathcal{F}$, which sends that element to t_y) this says $\varepsilon_x \circ \kappa^{-1} = c$, which is bijective by the Lemma. So ε_x is bijective for all x, and ε is an isomorphism by Prop. 2.12. This proves (a).

(b) The bijection. Define $\Phi : \mathrm{Mor}_X(f^{-1}\mathcal{G}, \mathcal{F}) \to \mathrm{Mor}_Y(\mathcal{G}, f_*\mathcal{F})$ by $\Phi(\beta) = f_*(\beta) \circ \eta_{\mathcal{G}}$, i.e. $\Phi(\beta)(V)(s) = \beta(f^{-1}V)\big(\theta([s]_V)\big)$. Define also, for a morphism $\alpha : \mathcal{G} \to f_*\mathcal{F}$, the morphism $f^{-1}\alpha : f^{-1}\mathcal{G} \to f^{-1}f_*\mathcal{F}$ as the unique morphism with $f^{-1}\alpha \circ \theta_{\mathcal{G}} = \theta_{f_*\mathcal{F}} \circ P(\alpha)$, where $P(\alpha)(U)([s]_W) = [\alpha(W)(s)]_W$ (existence and uniqueness by Def. 2.14, since $f^{-1}f_*\mathcal{F}$ is a sheaf), and put $\Psi(\alpha) = \varepsilon_{\mathcal{F}} \circ f^{-1}\alpha$.

We show Φ is bijective with inverse Ψ. By Def. 2.14, $\beta \mapsto \beta \circ \theta$ is a bijection from $\mathrm{Mor}_X(f^{-1}\mathcal{G}, \mathcal{F})$ to the set of presheaf morphisms $\gamma : P_{\mathcal{G}} \to \mathcal{F}$. Next, presheaf morphisms γ correspond bijectively to morphisms $\alpha : \mathcal{G} \to f_*\mathcal{F}$ via

$$\alpha(V)(s) = \gamma(f^{-1}V)([s]_V), \qquad \gamma(U)([s]_W) = \alpha(W)(s)|_U.$$

Indeed, given γ the first formula defines a homomorphism compatible with restrictions (as γ is and $[s|_{V'}]_{V'} = [s]_V$ in $P_{\mathcal{G}}(f^{-1}V')$). Given α the second formula is well defined on the direct limit (if $W' \subseteq W$, $\alpha(W')(s|_{W'})|_U = \alpha(W)(s)|_U$) and compatible with restriction in U. Starting from γ, then α, then recomputing, $\alpha(W)(s)|_U = \gamma(f^{-1}W)([s]_W)|_U = \gamma(U)([s]_W|_U) = \gamma(U)([s]_W)$, and starting from α one gets $\gamma(f^{-1}V)([s]_V) = \alpha(V)(s)$; so the two constructions are inverse. Composing the two bijections, β goes to $\alpha(V)(s) = \beta(f^{-1}V)(\theta([s]_V))$, which is $\Phi(\beta)$. Hence Φ is bijective.

Finally $\Psi = \Phi^{-1}$: let α be given and γ_α the presheaf morphism corresponding to it. Then $\Psi(\alpha) \circ \theta_{\mathcal{G}} = \varepsilon \circ \theta_{f_*\mathcal{F}} \circ P(\alpha) = \varepsilon' \circ P(\alpha)$, and $\varepsilon' P(\alpha)(U)([s]_W) = \alpha(W)(s)|_U = \gamma_\alpha(U)([s]_W)$. So

$\Psi(\alpha) \circ \theta = \gamma_\alpha$, which by the above means $\Phi(\Psi(\alpha)) = \alpha$. As Φ is bijective, $\Psi = \Phi^{-1}$. Thus $\mathrm{Mor}_Y(\mathcal{G}, f_*\mathcal{F}) \simeq \mathrm{Mor}_X(f^{-1}\mathcal{G}, \mathcal{F})$ canonically, both directions being expressed through the maps η and ε of (2.1) (the same argument works for sheaves of rings and ring morphisms). $\square$

Exercise 2.14

Problem. (Quotient space) Let a group G act on a ringed topological space $(X, \mathcal{O}_X)$ (a homomorphism $G \to \mathrm{Aut}(X)$). A quotient of X by G is a ringed space $(Y, \mathcal{O}_Y)$ with a morphism $p : X \to Y$ such that $p = p \circ \sigma$ for every $\sigma \in G$, and every morphism of ringed spaces $f : X \to Z$ with $f = f \circ \sigma$ for all $\sigma \in G$ factors uniquely as $f = \tilde{f} \circ p$. Show the quotient exists: let $Y = X/G$ with the quotient topology and $p : X \to Y$ the projection.

(a) Show that p is an open map.

(b) Show that G acts on the ringed space $p^{-1}(V)$ for $V \subseteq Y$ open. In particular G acts on the ring $\mathcal{O}_X(p^{-1}(V))$.

(c) For $V \subseteq Y$ open, let $\mathcal{O}_Y(V) = \mathcal{O}_X(p^{-1}(V))^G$ (the G-invariant elements). Show that $\mathcal{O}_Y$ is a sheaf of rings on Y.

(d) Let $y \in Y$ and fix $x \in p^{-1}(y)$. Show that $\mathcal{O}_{Y,y} \subseteq \mathcal{O}_{X,x}$. Show that $\mathfrak{m}_y = \mathfrak{m}_x \cap \mathcal{O}_{Y,y}$ is the unique maximal ideal of $\mathcal{O}_{Y,y}$.

(e) Show that $(Y, \mathcal{O}_Y)$ is a ringed space, that p is a morphism of ringed spaces from X to Y, and that p has the universal property of the quotient.

Solution. For $\sigma \in G$ we write $\sigma = (\sigma, \sigma^\#)$ for the corresponding automorphism of $(X, \mathcal{O}_X)$. As σ is invertible as a morphism, σ is a homeomorphism and $\sigma^\# : \mathcal{O}_X \to \sigma_*\mathcal{O}_X$ is an isomorphism of sheaves, i.e. $\sigma^\#(W) : \mathcal{O}_X(W) \to \mathcal{O}_X(\sigma^{-1}W)$ is a ring isomorphism for each open W; and $\sigma^\#_z : \mathcal{O}_{X,\sigma(z)} \to \mathcal{O}_{X,z}$ is an isomorphism of rings for every z (the stalk of $\sigma_*\mathcal{O}_X$ at $\sigma(z)$ is $\mathcal{O}_{X,z}$ because σ is a homeomorphism). The topology on Y is such that p is continuous, and $p \circ \sigma = p$ as maps.

(a) Let $U \subseteq X$ be open. Then $p^{-1}(p(U)) = \bigcup_{\sigma \in G} \sigma(U)$: indeed z lies in the left side iff $p(z) = p(u)$ for some $u \in U$ iff $z = \sigma u$ for some $\sigma \in G, u \in U$. Each $\sigma(U)$ is open since σ is a homeomorphism. So $p^{-1}(p(U))$ is open, i.e. $p(U)$ is open in Y.

(b) Let $W = p^{-1}(V)$. Since $p\sigma = p$, $\sigma(W) \subseteq W$ for every $\sigma \in G$, and applying this to σ^{-1} gives $\sigma(W) = W = \sigma^{-1}(W)$. Thus σ restricts to a homeomorphism of W, and $\sigma^\#$ restricts to an isomorphism $\mathcal{O}_X|_W \to (\sigma|_W)_*(\mathcal{O}_X|_W)$; so $\sigma|_W$ is an automorphism of the ringed space $(W, \mathcal{O}_X|_W)$. In particular each $\sigma^\#(W)$ is a ring automorphism of $\mathcal{O}_X(W)$. By the composition rule, $(\sigma\tau)^\#(W) = \tau^\#(W) \circ \sigma^\#(W)$, so we get an action by ring automorphisms (a right action; the left action $s \mapsto (\sigma^{-1})^\#(W)(s)$ has the same invariants). The only thing we use is the invariant ring $\mathcal{O}_X(W)^G = \{s : \sigma^\#(W)(s) = s \ \forall \sigma \in G\}$, a subring of $\mathcal{O}_X(W)$.

(c) If $V' \subseteq V$ are open and $W' = p^{-1}V' \subseteq W = p^{-1}V$, then $\sigma^\#(W')(s|_{W'}) = \sigma^\#(W)(s)|_{\sigma^{-1}W'} = \sigma^\#(W)(s)|_{W'}$ since $\sigma^\#$ is compatible with restrictions and $\sigma^{-1}W' = W'$. So the restriction of an invariant section is invariant, and $\mathcal{O}_Y$ is a presheaf of rings (with $\mathcal{O}_Y(\emptyset) = \mathcal{O}_X(\emptyset) = 0$). Sheaf property: let $V = \bigcup_a V_a$ and put $W_a = p^{-1}V_a$, so $p^{-1}V = \bigcup W_a$ and $p^{-1}(V_a \cap V_b) = W_a \cap W_b$. If $s \in \mathcal{O}_Y(V)$ restricts to 0 on each V_a, then $s|_{W_a} = 0$ for all a and $s = 0$ in $\mathcal{O}_X(p^{-1}V)$. If $s_a \in \mathcal{O}_Y(V_a)$ agree on the overlaps $V_a \cap V_b$, they agree on $W_a \cap W_b$, so glue to $s \in \mathcal{O}_X(p^{-1}V)$ with $s|_{W_a} = s_a$. For $\sigma \in G$, $\sigma^\#(p^{-1}V)(s)|_{W_a} = \sigma^\#(W_a)(s_a) = s_a$ (as in the displayed formula),

so $\sigma^{\#}(s)$ and s have the same restriction to every W_a and hence are equal. Thus $s \in \mathcal{O}_Y(V)$. So $\mathcal{O}_Y$ is a sheaf of rings.

(d) For $V \ni y$ open and $s \in \mathcal{O}_Y(V) \subseteq \mathcal{O}_X(p^{-1}V)$, since $x \in p^{-1}V$, we can take the germ of s at x. This is compatible with restricting to smaller V, so it defines a ring homomorphism $\iota : \mathcal{O}_{Y,y} \to \mathcal{O}_{X,x}$ (it is $p_x^{\#}$ of part (e)).

ι is injective. Suppose $s \in \mathcal{O}_Y(V)$ has $s_x = 0$ in $\mathcal{O}_{X,x}$. Then there is an open $U \ni x$, $U \subseteq p^{-1}V$, with $s|_U = 0$. For $\sigma \in G$, using $\sigma^{\#}(p^{-1}V)(s) = s$ and compatibility with restriction,

$$\sigma^{\#}(\sigma U)\big(s|_{\sigma U}\big) = \sigma^{\#}(p^{-1}V)(s)|_{\sigma^{-1}\sigma U} = s|_U = 0.$$

As $\sigma^{\#}(\sigma U)$ is injective, $s|_{\sigma U} = 0$. By part (a), $V' = p(U)$ is an open neighbourhood of y contained in V, and $p^{-1}V' = \bigcup_\sigma \sigma U$ (proof of (a)). So $s|_{p^{-1}V'} = 0$, hence $s|_{V'} = 0$ in $\mathcal{O}_Y(V')$ and $s_y = 0$.

So we identify $\mathcal{O}_{Y,y}$ with the subring $A = \iota(\mathcal{O}_{Y,y}) \subseteq \mathcal{O}_{X,x}$ and $\mathfrak{m}_y := \mathfrak{m}_x \cap A$, a proper ideal of A (it does not contain 1). We show every $a \in A \setminus \mathfrak{m}_x$ is a unit of A. Then the non-units of A are exactly the elements of $\mathfrak{m}_x \cap A = \mathfrak{m}_y$ (elements of $\mathfrak{m}_x$ are not units of $\mathcal{O}_{X,x}$, hence not of A), so the non-units form an ideal and A is local with maximal ideal $\mathfrak{m}_y$, as desired.

Let $a = s_x$ with $s \in \mathcal{O}_Y(V)$, $a \notin \mathfrak{m}_x$; thus a is a unit of the local ring $\mathcal{O}_{X,x}$. Let $N = \{z \in p^{-1}V : s_z \in \mathcal{O}_{X,z}^{\times}\}$. If s_z is a unit, there is an open $U_z \ni z$ and $t^z \in \mathcal{O}_X(U_z)$ with $s|_{U_z} t^z = 1$; then every point of U_z is in N, so N is open and $x \in N$. Moreover N is G-stable: from $\sigma^{\#}(p^{-1}V)(s) = s$ we get $\sigma_z^{\#}(s_{\sigma z}) = s_z$, and $\sigma_z^{\#}$ is a ring isomorphism, so $s_{\sigma z}$ is a unit iff s_z is. Hence $N = p^{-1}(V')$ with $V' = p(N)$ open (by (a)) and $y \in V'$. For $z, z' \in N$ the sections $t^z, t^{z'}$ coincide on $U_z \cap U_{z'}$ (both are the inverse of s in the ring $\mathcal{O}_X(U_z \cap U_{z'})$), so they glue to $t \in \mathcal{O}_X(N)$ with $s|_N t = 1$. For $\sigma \in G$, $\sigma^{\#}(N)(t)$ is an inverse of $\sigma^{\#}(N)(s|_N) = s|_N$, so equals t by uniqueness of inverses. Thus $t \in \mathcal{O}_Y(V')$, $s|_{V'} t = 1$ in $\mathcal{O}_Y(V')$, and s_y is a unit of $\mathcal{O}_{Y,y}$; i.e. a is a unit of A.

(e) By (c) and (d), $(Y, \mathcal{O}_Y)$ is a ringed topological space. Let $p^{\#}(V) : \mathcal{O}_Y(V) = \mathcal{O}_X(p^{-1}V)^G \hookrightarrow \mathcal{O}_X(p^{-1}V) = (p_*\mathcal{O}_X)(V)$ be the inclusion; this is a morphism of sheaves of rings. The stalk map $p_x^{\#} : \mathcal{O}_{Y,p(x)} \to \mathcal{O}_{X,x}$ is the map ι of (d), and $(p_x^{\#})^{-1}(\mathfrak{m}_x) = \iota^{-1}(\mathfrak{m}_x \cap A) = \mathfrak{m}_{p(x)}$, so it is local. So $p : X \to Y$ is a morphism of ringed spaces. For $\sigma \in G$, $(p\sigma)^{\#}(V)(a) = \sigma^{\#}(p^{-1}V)(p^{\#}(V)(a)) = a = p^{\#}(V)(a)$ because a is invariant, so $p \circ \sigma = p$ as morphisms.

Universal property. Let $f : X \to Z$ be a morphism with $f \circ \sigma = f$ for all $\sigma \in G$ (equality of morphisms, so the maps are equal and $(f\sigma)^{\#} = f^{\#}$). The map f is constant on G-orbits, so $f = \tilde{f} \circ p$ for a unique map $\tilde{f} : Y \to Z$, and $\tilde{f}$ is continuous because $p^{-1}(\tilde{f}^{-1}W) = f^{-1}W$ is open for each open $W \subseteq Z$ and Y has the quotient topology. For $W \subseteq Z$ open and $u \in \mathcal{O}_Z(W)$ put $b = f^{\#}(W)(u) \in \mathcal{O}_X(f^{-1}W) = \mathcal{O}_X(p^{-1}\tilde{f}^{-1}W)$. From $(f\sigma)^{\#} = f^{\#}$ we get $\sigma^{\#}(f^{-1}W)(b) = (f\sigma)^{\#}(W)(u) = b$, so b is invariant, i.e. $b \in \mathcal{O}_Y(\tilde{f}^{-1}W) = (\tilde{f}_*\mathcal{O}_Y)(W)$. Define $\tilde{f}^{\#}(W)(u) = b$. This is a morphism of sheaves of rings $\mathcal{O}_Z \to \tilde{f}_*\mathcal{O}_Y$, since $f^{\#}$ is one and $\mathcal{O}_Y \subseteq p_*\mathcal{O}_X$ is a subsheaf of rings. For $x \in X$, $y = p(x)$, we have $\iota \circ \tilde{f}_y^{\#} = f_x^{\#}$ as maps $\mathcal{O}_{Z,f(x)} \to \mathcal{O}_{X,x}$ (both send the germ of u to the germ at x of $f^{\#}(W)(u)$). If $c \in \mathfrak{m}_{f(x)}$ then $\iota(\tilde{f}_y^{\#}(c)) = f_x^{\#}(c) \in \mathfrak{m}_x$, and it lies in $A = \iota(\mathcal{O}_{Y,y})$, so $\iota(\tilde{f}_y^{\#}(c)) \in \mathfrak{m}_x \cap A = \iota(\mathfrak{m}_y)$, and by injectivity of ι, $\tilde{f}_y^{\#}(c) \in \mathfrak{m}_y$. So $\tilde{f}_y^{\#}$ is local and $\tilde{f}$ is a morphism of ringed spaces. Finally $(\tilde{f} \circ p)^{\#}(W)(u) = p^{\#}(\tilde{f}^{-1}W)(\tilde{f}^{\#}(W)(u)) = f^{\#}(W)(u)$, so $\tilde{f} \circ p = f$.

Uniqueness. Let $h : Y \to Z$ be a morphism with $h \circ p = f$. Since p is surjective, $h = \tilde{f}$ as maps. For $u \in \mathcal{O}_Z(W)$ we have $f^{\#}(W)(u) = (hp)^{\#}(W)(u) = p^{\#}(h^{-1}W)(h^{\#}(W)(u)) = h^{\#}(W)(u)$,

viewed inside $\mathcal{O}_X(p^{-1}h^{-1}W)$, because $p^\#$ is an inclusion. Hence $h^\#(W)(u) = \tilde{f}^\#(W)(u)$ and $h = \tilde{f}$. Therefore $(Y, \mathcal{O}_Y)$ with p is a quotient of X by G. □

2.3 Schemes

Throughout, rings are commutative with 1, and results of the text are cited by their numbers in Liu ("Prop. 3.25", "Lemma 3.40", ...). For a morphism of schemes $g : Z \to X$ the naturality (3.4) of the map ρ of (3.3) reads

$$\rho(f \circ g) = g^\#(X) \circ \rho(f) \qquad (f : X \to Y).$$

We refer to this identity as (N).

Honest notes. Ex. 3.3(b) is printed with a closure bar that is essential: the image of f itself is in general not equal to $V(\operatorname{Ker}\varphi)$, and the solution proves the statement for $\overline{\operatorname{Im} f}$. Ex. 3.10 ("$X$ is affine iff $n = 0$") is false for $A = 0$ (then $\mathbb{P}^n_A = \emptyset$ is affine), so the second half is proved for $A \neq 0$. In Ex. 3.11(b) the hypothesis "B_e finitely generated over A" is read as "finitely generated A-module" and "B_{ed} is generated by B_e^d" as "B_{ed} is spanned over A by products of d elements of B_e", where B is a graded A-algebra.

Exercise 3.1

> **Problem.** Let X be an open subscheme of an affine scheme Y. Show that the canonical morphism $X \to Y$ corresponds, by the map ρ of Proposition 3.25, to the restriction $\mathcal{O}_Y(Y) \to \mathcal{O}_X(X)$.

Solution. By definition (Def. 3.10) X is an open subset of Y with the structure sheaf $\mathcal{O}_Y|_X$, and the canonical morphism is $i = (i, i^\#)$, where $i : X \to Y$ is the inclusion and $i^\# : \mathcal{O}_Y \to i_*(\mathcal{O}_Y|_X)$ is given, for each open $V \subseteq Y$, by the restriction map $\mathcal{O}_Y(V) \to \mathcal{O}_Y(V \cap X) = (i_*\mathcal{O}_X)(V)$. By definition of ρ in (3.3),

$$\rho(i) = i^\#(Y) : \ \mathcal{O}_Y(Y) \longrightarrow (i_*\mathcal{O}_X)(Y) = \mathcal{O}_X(X),$$

and by the description of $i^\#$ this is the restriction map $\mathcal{O}_Y(Y) \to \mathcal{O}_Y(X) = \mathcal{O}_X(X)$. Since Y is affine, Prop. 3.25 says that i is the unique morphism $X \to Y$ with this image under ρ. □

Exercise 3.2

> **Problem.** Let $U = \operatorname{Spec} B$ be an affine open subscheme of $X = \operatorname{Spec} A$. Show that the restriction $A \to B$ is a flat homomorphism.

Solution. Here "$U = \operatorname{Spec} B$" means that we are given an isomorphism of schemes $\sigma : \operatorname{Spec} B \to (U, \mathcal{O}_X|_U)$; the restriction $A = \mathcal{O}_X(X) \to \mathcal{O}_X(U) \simeq B$ is the ring homomorphism $\varphi := \tau^\#(X) = \rho(\tau)$, where $\tau : \operatorname{Spec} B \to X$ is σ followed by the canonical open immersion $U \to X$ (use Ex. 3.1 and (N)). By Lemma 3.23, $\tau = f_\varphi$ is the morphism of Prop. 3.14 attached to φ. Let $\mathfrak{q} \in \operatorname{Spec} B$

and $\mathfrak{p} = \varphi^{-1}(\mathfrak{q}) = \tau(\mathfrak{q})$. By Prop. 3.14 the stalk map $\tau^{\#}_{\mathfrak{q}} : \mathcal{O}_{X,\mathfrak{p}} = A_{\mathfrak{p}} \to \mathcal{O}_{\operatorname{Spec} B,\mathfrak{q}} = B_{\mathfrak{q}}$ is the localization of φ. Since τ is an isomorphism onto the open subscheme U, the stalk maps of τ are isomorphisms. Hence

$$A_{\mathfrak{p}} \xrightarrow{\sim} B_{\mathfrak{q}} \quad \text{for every } \mathfrak{q} \in \operatorname{Spec} B,\ \mathfrak{p} = \varphi^{-1}(\mathfrak{q}).$$

Call this (1).

Now let $u : N' \to N$ be an injective homomorphism of A-modules, and let K be the kernel of $u \otimes_A B : N' \otimes_A B \to N \otimes_A B$, a B-module. Fix $\mathfrak{q}$ and $\mathfrak{p}$ as above. Localization at $\mathfrak{q}$ is exact, so $K_{\mathfrak{q}}$ is the kernel of $N' \otimes_A B_{\mathfrak{q}} \to N \otimes_A B_{\mathfrak{q}}$. Elements of $A \setminus \mathfrak{p}$ are invertible in $B_{\mathfrak{q}}$, so $N' \otimes_A B_{\mathfrak{q}} = N'_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} B_{\mathfrak{q}}$ and likewise for N. By (1), $B_{\mathfrak{q}} \simeq A_{\mathfrak{p}}$ as $A_{\mathfrak{p}}$-modules, so the map in question is isomorphic to $N'_{\mathfrak{p}} \to N_{\mathfrak{p}}$, which is injective (localization is exact). Thus $K_{\mathfrak{q}} = 0$ for all $\mathfrak{q}$. A module all of whose localizations at primes vanish is zero (if $0 \neq x \in K$, the ideal $\operatorname{Ann}(x)$ lies in a maximal ideal $\mathfrak{q}$ and $x/1 \neq 0$ in $K_{\mathfrak{q}}$). Hence $K = 0$ and B is flat over A. $\square$

Exercise 3.3

Problem. Closure of a subset.

(a) Let F be a subset of an affine scheme $\operatorname{Spec} A$. Show that the closure $\overline{F}$ of F in (the underlying topological space of) $\operatorname{Spec} A$ is equal to $V(I)$, where $I = \cap_{\mathfrak{p} \in F} \mathfrak{p}$.

(b) Let $\varphi : A \to B$ be a ring homomorphism. Let $f : \operatorname{Spec} B \to \operatorname{Spec} A$ be the morphism of schemes associated to φ. Show that $\overline{\operatorname{Im} f} = V(\operatorname{Ker} \varphi)$. Study the situation where B is the localization of A at a prime ideal.

Solution. (a) (For $F = \emptyset$ we have $I = A$ and $V(I) = \emptyset = \overline{F}$.) The set $V(I)$ is closed, and it contains F because $I \subseteq \mathfrak{p}$ for every $\mathfrak{p} \in F$. Conversely let $V(J)$ be any closed set containing F. Then $J \subseteq \mathfrak{p}$ for all $\mathfrak{p} \in F$, so $J \subseteq I$ and $V(I) \subseteq V(J)$. Thus $V(I)$ is the smallest closed set containing F, i.e. $V(I) = \overline{F}$.

(b) We have $f(\mathfrak{q}) = \varphi^{-1}(\mathfrak{q})$. By (a), $\overline{\operatorname{Im} f} = V(I)$ with

$$I = \bigcap_{\mathfrak{q} \in \operatorname{Spec} B} \varphi^{-1}(\mathfrak{q}) = \varphi^{-1}\Big(\bigcap_{\mathfrak{q}} \mathfrak{q}\Big) = \varphi^{-1}(\operatorname{Nil} B) = \sqrt{\operatorname{Ker} \varphi},$$

because $\varphi(a)$ is nilpotent iff $\varphi(a^m) = 0$ for some $m \geq 1$ (Lemma 1.6: the intersection of all primes is the nilradical). Since $V(\sqrt{J}) = V(J)$ for every ideal J, we get $\overline{\operatorname{Im} f} = V(\operatorname{Ker} \varphi)$.

Remarks. (i) The inclusion $\operatorname{Im} f \subseteq V(\operatorname{Ker} \varphi)$ is obvious, but equality can fail without the closure: for $\varphi : \mathbb{Z} \to \mathbb{Q}$ we have $\operatorname{Im} f = \{(0)\}$ while $V(\operatorname{Ker} \varphi) = \operatorname{Spec} \mathbb{Z}$. If φ is surjective, then $\operatorname{Im} f = V(\operatorname{Ker} \varphi)$: this is Lemma 3.17, since $B \simeq A/\operatorname{Ker} \varphi$.

(ii) Let $\mathfrak{p} \in \operatorname{Spec} A$, $B = A_{\mathfrak{p}}$ and $\varphi : A \to A_{\mathfrak{p}}$ the localization map. A prime $\mathfrak{q}$ of A is of the form $\varphi^{-1}(\mathfrak{Q})$ for a prime $\mathfrak{Q}$ of $A_{\mathfrak{p}}$ iff $\mathfrak{q} \cap (A \setminus \mathfrak{p}) = \emptyset$, i.e. iff $\mathfrak{q} \subseteq \mathfrak{p}$ (the primes of $A_{\mathfrak{p}}$ correspond bijectively to these $\mathfrak{q}$, via $\mathfrak{q}A_{\mathfrak{p}}$). Hence

$$\operatorname{Im} f = \{\mathfrak{q} \in \operatorname{Spec} A : \mathfrak{q} \subseteq \mathfrak{p}\} = \bigcap_{s \in A \setminus \mathfrak{p}} D(s),$$

the set of generizations of $\mathfrak{p}$. Moreover $\operatorname{Ker}\varphi = \{a \in A : sa = 0 \text{ for some } s \notin \mathfrak{p}\}$, and $\overline{\operatorname{Im} f} = V(\operatorname{Ker}\varphi)$ by (b). This set contains $\mathfrak{p}$ and hence the closure $V(\mathfrak{p})$ of $\{\mathfrak{p}\}$. Therefore $\operatorname{Im} f$ can be closed only if $V(\mathfrak{p}) \subseteq \{\mathfrak{q} : \mathfrak{q} \subseteq \mathfrak{p}\}$, i.e. only if $\mathfrak{p}$ is a maximal ideal. In general $\operatorname{Im} f$ is not closed: for $A = \mathbb{Z}$ and $\mathfrak{p} = (p)$ the image is $\{(0), (p)\}$, which is not closed, with closure all of $\operatorname{Spec}\mathbb{Z}$ (here $\operatorname{Ker}\varphi = 0$). If A is a domain then $\operatorname{Ker}\varphi = 0$ and $\operatorname{Im} f$ is dense in $\operatorname{Spec} A$ for every $\mathfrak{p}$. If A is local with maximal ideal $\mathfrak{p}$, then φ is an isomorphism and $\operatorname{Im} f = \operatorname{Spec} A$. □

Exercise 3.4

> **Problem.** Let X be a scheme and $f \in \mathcal{O}_X(X)$. Show that $U \mapsto f|_U\mathcal{O}_X(U)$ for every affine open subset U defines a sheaf of ideals on X. We denote this sheaf by $f\mathcal{O}_X$. Show that $\operatorname{Supp} f\mathcal{O}_X$ is closed (see Exercise 2.5).

Solution. Define, for every open $W \subseteq X$,

$$\mathcal{I}(W) = \{s \in \mathcal{O}_X(W) : \ s_x \in f_x\mathcal{O}_{X,x} \text{ for every } x \in W\}.$$

This is an $\mathcal{O}_X(W)$-submodule of $\mathcal{O}_X(W)$ (an ideal), stable under restriction. It is a sheaf: the condition on s is a condition on the germs s_x, so if $s \in \mathcal{O}_X(W)$ restricts to elements of $\mathcal{I}(W_j)$ on an open covering $\{W_j\}$ of W, then $s \in \mathcal{I}(W)$; and the sections of $\mathcal{I}$ are sections of the sheaf $\mathcal{O}_X$, so the uniqueness and glueing conditions follow from those of $\mathcal{O}_X$ (a glued section of $\mathcal{O}_X$ has germs in $f_x\mathcal{O}_{X,x}$ by the same local property). Thus $\mathcal{I}$ is a sheaf of ideals of $\mathcal{O}_X$.

Claim: for every affine open U, $\mathcal{I}(U) = f|_U\mathcal{O}_X(U)$. Write $U = \operatorname{Spec} A$ (we identify U with $\operatorname{Spec} A$) and $a = f|_U \in A$. The inclusion $\supseteq$ is clear. Conversely let $s \in \mathcal{I}(U)$. For every $\mathfrak{p} \in U$ we have $s_\mathfrak{p} \in aA_\mathfrak{p}$ (Prop. 3.1(b)), i.e. $s/1 = (b/t)a$ in $A_\mathfrak{p}$ for some $b \in A$, $t \in A \setminus \mathfrak{p}$, i.e. $t'(ts - ba) = 0$ for some $t' \notin \mathfrak{p}$, so $t't\,s \in aA$ with $t't \notin \mathfrak{p}$. So the ideal $J = \{t \in A : ts \in aA\}$ is not contained in any prime $\mathfrak{p}$, hence $J = A$, $s \in aA$. This proves the claim.

Since the affine open subsets form a base of the topology and $\mathcal{I}$ is a sheaf, $\mathcal{I}$ is the unique sheaf of ideals whose sections over each affine open U are $f|_U\mathcal{O}_X(U)$ (with the restriction maps of $\mathcal{O}_X$). This is the sheaf $f\mathcal{O}_X$ of the statement; its stalk at x is $(f\mathcal{O}_X)_x = f_x\mathcal{O}_{X,x}$: indeed $f \in \mathcal{I}(X)$, so $f_x\mathcal{O}_{X,x} \subseteq \mathcal{I}_x$, and $\subseteq$ in the other direction holds by definition of $\mathcal{I}$.

Support. $\operatorname{Supp} f\mathcal{O}_X = \{x \in X : f_x\mathcal{O}_{X,x} \neq 0\} = \{x : f_x \neq 0\}$. Its complement $\{x : f_x = 0\}$ is open: if $f_x = 0$, there is an open neighbourhood V of x with $f|_V = 0$, hence $f_y = 0$ for all $y \in V$. So the support is closed. (Contrast with Ex. 2.5: for a general sheaf the support need not be closed.) □

Exercise 3.5

> **Problem.** Let Y be a scheme that satisfies the conclusion of Proposition 3.25 for every affine scheme X. Show that Y is affine.

Solution. The hypothesis is that for every affine scheme X the map $\rho_X : \operatorname{Mor}(X, Y) \to \operatorname{Hom}(\mathcal{O}_Y(Y), \mathcal{O}_X(X))$ is bijective. Put $A = \mathcal{O}_Y(Y)$ and $Z = \operatorname{Spec} A$, so $\mathcal{O}_Z(Z) = A$ (Prop. 3.1). We use (N) freely.

By Prop. 3.25 applied to the affine target Z and the source Y, there is a unique morphism $u : Y \to Z$ with $\rho_Z(u) = u^\#(Z) = \mathrm{id}_A$. By the hypothesis applied to the affine scheme Z, there is a unique morphism $v : Z \to Y$ with $\rho_Y(v) = v^\#(Y) = \mathrm{id}_A \in \mathrm{Hom}(A, \mathcal{O}_Z(Z))$.

$u \circ v = \mathrm{id}_Z$. By (N), $\rho_Z(u \circ v) = v^\#(Y) \circ \rho_Z(u) = \mathrm{id}_A = \rho_Z(\mathrm{id}_Z)$. As $\rho_Z : \mathrm{Mor}(Z, Z) \to \mathrm{Hom}(A, A)$ is injective (Lemma 3.23), $u \circ v = \mathrm{id}_Z$.

$v \circ u = \mathrm{id}_Y$. Let $w : W \to Y$ be the inclusion of an affine open subset W of Y. By (N),

$$\rho_Y(v \circ u \circ w) = (u \circ w)^\#(Z) \circ \rho_Y(v) = \rho_Z(u \circ w) = w^\#(Y) \circ \rho_Z(u) = w^\#(Y) = \rho_Y(w),$$

where $w^\#(Y)$ is the restriction $A \to \mathcal{O}_Y(W)$ (Ex. 3.1). Since W is affine, $\rho_Y : \mathrm{Mor}(W, Y) \to \mathrm{Hom}(A, \mathcal{O}_Y(W))$ is injective by hypothesis, so $v \circ u \circ w = w$. The affine open subsets W cover Y, and two morphisms of schemes $Y \to Y$ that agree after composition with the inclusion of every member of an open covering are equal (they agree as maps, and the sheaf maps agree on a base; cf. Ex. 2.11). Hence $v \circ u = \mathrm{id}_Y$.

Therefore $u : Y \to \operatorname{Spec} A$ is an isomorphism and Y is affine. □

Exercise 3.6

Problem. Generalize Example 3.32 to the case when k is an arbitrary ring.

Solution. Let k be a ring, $I \subseteq k[T_1, \ldots, T_n]$ an ideal, $C = k[T_1, \ldots, T_n]/I$ and $X = \operatorname{Spec} C$, a k-scheme via $\iota : k \to C$ (Prop. 3.14). Let $X(k)$ be the set of sections of $X \to \operatorname{Spec} k$, i.e. of morphisms of k-schemes $\sigma : \operatorname{Spec} k \to X$ (Def. 3.28). Put

$$Z = \{\alpha = (\alpha_1, \ldots, \alpha_n) \in k^n : \ P(\alpha) = 0 \text{ for all } P \in I\}.$$

Claim: there is a canonical bijection $\lambda : Z \to X(k)$. (For a field k this is Example 3.32; the point $\lambda(\alpha) \in X$ is the image of the section, namely the prime $\mathfrak{m}_\alpha = (T_1 - \alpha_1, \ldots, T_n - \alpha_n)/I$.)

Let $\mathrm{Hom}_{k\text{-alg}}(C, k)$ be the set of k-algebra homomorphisms. By Prop. 3.25 (target X affine, source $\operatorname{Spec} k$), ρ is a bijection $\mathrm{Mor}(\operatorname{Spec} k, X) \to \mathrm{Hom}_{\mathrm{rings}}(C, k)$. A morphism σ is a morphism of k-schemes iff $\pi \circ \sigma = \mathrm{id}_{\operatorname{Spec} k}$, where $\pi : X \to \operatorname{Spec} k$ is the structural morphism with $\rho(\pi) = \iota$. By (N), $\rho(\pi \circ \sigma) = \sigma^\#(X) \circ \rho(\pi) = \rho(\sigma) \circ \iota$, while $\rho(\mathrm{id}) = \mathrm{id}_k$; as ρ is bijective (Lemma 3.23), σ is a section iff $\rho(\sigma) \circ \iota = \mathrm{id}_k$, i.e. iff $\rho(\sigma)$ is a k-algebra homomorphism $C \to k$. So

$$X(k) \ \xrightarrow[\rho]{\sim} \ \mathrm{Hom}_{k\text{-alg}}(C, k).$$

Finally, a k-algebra homomorphism $\psi : C \to k$ is the same as a k-algebra homomorphism $\tilde{\psi} : k[T_1, \ldots, T_n] \to k$ vanishing on I. Such a $\tilde{\psi}$ is determined by $\alpha_i = \tilde{\psi}(T_i)$ and is the evaluation $P \mapsto P(\alpha)$, and any $\alpha \in k^n$ gives one. It vanishes on I iff $P(\alpha) = 0$ for all $P \in I$, i.e. iff $\alpha \in Z$. Hence $\alpha \mapsto$ (evaluation at α) is a bijection from Z onto $\mathrm{Hom}_{k\text{-alg}}(C, k)$, and composing with ρ^{-1} we obtain the bijection $\lambda : Z \to X(k)$. □

Exercise 3.7

Problem. Let X be a scheme over a field k. Let $\varphi : k[T_1, \dots, T_n] \to \mathcal{O}_X(X)$ be a homomorphism of k-algebras and $f : X \to \mathbb{A}^n_k$ the morphism induced by φ. Show that for any rational point $x \in X(k)$, via the identification $\mathbb{A}^n_k(k) = k^n$, we have $f(x) = (f_1(x), \dots, f_n(x))$, where $f_i = \varphi(T_i)$ and $f_i(x)$ is the image of f_i in $k(x) = k$.

Solution. Here f is the morphism with $\rho(f) = f^\#(\mathbb{A}^n_k) = \varphi$ (Prop. 3.25). It is a morphism of k-schemes: with π_X, π the structural morphisms of X and $\mathbb{A}^n_k$, by (N) $\rho(\pi \circ f) = \varphi|_k = \rho(\pi_X)$ since φ is a k-algebra map, so $\pi \circ f = \pi_X$ by injectivity of ρ (Prop. 3.25).

Let $x \in X$ have $k(x) = k$ (Example 3.29) and $y = f(x) \in \mathbb{A}^n_k$. The local homomorphism $f^\#_x : \mathcal{O}_{\mathbb{A}^n_k, y} \to \mathcal{O}_{X,x}$ induces a field homomorphism $\bar{f}_x : k(y) \to k(x) = k$ which is a homomorphism of k-algebras (because f is a k-morphism). Since $k(y)$ is a field containing k (as the image of k is k itself, $k \to k(y)$ is injective), and $\bar{f}_x$ is a k-linear field embedding into k, we get $k(y) = k$ and $\bar{f}_x = \mathrm{id}_k$. So y is a rational point of $\mathbb{A}^n_k$, and under Example 3.32 (with $I = 0$) it corresponds to $\alpha = (\alpha_1, \dots, \alpha_n)$, where α_i is the image of T_i in $k(y) = k$ (i.e. $y = \mathfrak{m}_\alpha$).

Now, for the global section T_i of $\mathbb{A}^n_k$ we have $f^\#_x((T_i)_y) = (f^\#(\mathbb{A}^n_k)(T_i))_x = (\varphi(T_i))_x = (f_i)_x$. Passing to residue fields, $\bar{f}_x$ sends the class α_i of $(T_i)_y$ in $k(y)$ to the class $f_i(x)$ of $(f_i)_x$ in $k(x)$. Since $\bar{f}_x = \mathrm{id}_k$, we get $\alpha_i = f_i(x)$, i.e. $f(x) = (f_1(x), \dots, f_n(x))$. □

Exercise 3.8

Problem. Let X be a scheme over a ring A. Let $f_0, \dots, f_n \in \mathcal{O}_X(X)$ be such that the $f_{i,x}$ generate the unit ideal of $\mathcal{O}_{X,x}$ for every $x \in X$. Show that X is the union of the X_{f_i} (see Definition 3.11), and that we have a morphism $f : X \to \mathbb{P}^n_A$ such that $f^{-1}(D_+(T_i)) = X_{f_i}$ and that $f|_{X_{f_i}}$ is induced by the homomorphism $A[T_i^{-1}T_j]_j \to \mathcal{O}_X(X_{f_i})$ given by $T_i^{-1}T_j \mapsto f_i^{-1}f_j$. If $A = k$ is a field and $x \in X(k)$, determine $f(x)$ as in the preceding exercise.

Solution. Write $B = A[T_0, \dots, T_n]$, $\mathbb{P}^n_A = \operatorname{Proj} B$ (Ex. 3.39), $U_i = X_{f_i}$ (open by Prop. 3.12), and $D_i = D_+(T_i) = \operatorname{Spec} B_{(T_i)}$ with $B_{(T_i)} = A[T_i^{-1}T_j]_j$.

A general fact. (*) Let $g : U \to \operatorname{Spec} C$ be a morphism of schemes and $h \in C$. Then $g^{-1}(D(h)) = U_{\rho(g)(h)}$. Indeed, for $x \in U$ and $\mathfrak{p} = g(x)$: $x \in g^{-1}(D(h))$ iff $h \notin \mathfrak{p}$ iff the image of h in $C_\mathfrak{p}$ is a unit. As $g^\#_x : C_\mathfrak{p} \to \mathcal{O}_{U,x}$ is a local homomorphism, an element of $C_\mathfrak{p}$ is a unit iff its image in $\mathcal{O}_{U,x}$ is a unit (a non-unit lies in the maximal ideal, which maps into the maximal ideal). The image of h in $\mathcal{O}_{U,x}$ is $(\rho(g)(h))_x$, whence (*).

$X = \bigcup U_i$. Let $x \in X$. The ideal of the local ring $\mathcal{O}_{X,x}$ generated by the $f_{i,x}$ is the unit ideal, so not all $f_{i,x}$ lie in the maximal ideal. Hence some $f_{i,x}$ is a unit, i.e. $x \in U_i$. Moreover, as in the proof of Prop. 3.12, $f_i|_{U_i}$ is invertible in $\mathcal{O}_X(U_i)$ (glue local inverses); put f_i^{-1} for its inverse.

Construction of f. Note that $B_{(T_i)}$ is the polynomial ring over A in the n variables $S_j = T_i^{-1}T_j$, $j \neq i$ (the elements of degree 0 in $A[T_0, \dots, T_n, T_i^{-1}]$ are the Laurent polynomials in T_j with total degree 0 and non-negative exponents for $j \neq i$; these are exactly the polynomials in the S_j). Since $\mathcal{O}_X(U_i)$ is an A-algebra (Cor. 3.26), there is a unique A-algebra homomorphism

$$\psi_i : B_{(T_i)} \to \mathcal{O}_X(U_i), \qquad T_i^{-1}T_j \mapsto f_i^{-1}f_j \quad (j \neq i),$$

and then also $T_i^{-1}T_i = 1 \mapsto f_i^{-1}f_i = 1$. By Prop. 3.25 (the target D_i is affine) there is a unique morphism $g_i : U_i \to D_i \subseteq \mathbb{P}^n_A$ with $\rho(g_i) = \psi_i$; it is a morphism of A-schemes, by the argument of Ex. 3.7 (use (N) and injectivity of ρ into $\operatorname{Spec} A$).

Compatibility. Put $U_{ij} = U_i \cap U_j$ and $R_{ij} = B_{(T_iT_j)} = \mathcal{O}_{\mathbb{P}^n_A}(D_i \cap D_j)$. By Lemma 3.36(b),(c) with $g = T_iT_j$, $f = T_i$ (so $\alpha = T_j/T_i$), $D_i \cap D_j = D(T_i^{-1}T_j) \subseteq D_i$, and $R_{ij} = (B_{(T_i)})_{T_i^{-1}T_j}$, and similarly with i, j exchanged. By (*), $g_i^{-1}(D_i \cap D_j) = (U_i)_{\psi_i(T_i^{-1}T_j)} = (U_i)_{f_i^{-1}f_j} = U_i \cap U_j$ (as f_i is a unit on U_i, $f_i^{-1}f_j$ is a unit at $x \in U_i$ iff $f_{j,x}$ is). So g_i and g_j restrict to morphisms $U_{ij} \to \operatorname{Spec} R_{ij}$. Consider the ring homomorphism $\Theta : A[T_0, \dots, T_n] \to \mathcal{O}_X(U_{ij})[u, u^{-1}]$ given by $T_k \mapsto f_k u$. It sends T_iT_j to the unit $f_if_ju^2$ (units on U_{ij}), so it extends to $B_{T_iT_j}$, and it maps the degree-0 part R_{ij} into $\mathcal{O}_X(U_{ij})$ (the elements of degree 0 go to elements of u-degree 0). We obtain an A-algebra homomorphism $\theta_{ij} : R_{ij} \to \mathcal{O}_X(U_{ij})$ with $\theta_{ij}(T_k/T_i) = f_k/f_i$ and $\theta_{ij}(T_k/T_j) = f_k/f_j$ for all k. The restriction of θ_{ij} to $B_{(T_i)}$ is ψ_i (followed by restriction to U_{ij}), and to $B_{(T_j)}$ it is ψ_j. Now $\rho(g_i|_{U_{ij}})$ also restricts to ψ_i on $B_{(T_i)}$, and a homomorphism out of the localization $R_{ij} = (B_{(T_i)})_{T_i^{-1}T_j}$ is determined by its restriction to $B_{(T_i)}$; so $\rho(g_i|_{U_{ij}}) = \theta_{ij}$, and likewise $\rho(g_j|_{U_{ij}}) = \theta_{ij}$. By the injectivity in Prop. 3.25 (the target $\operatorname{Spec} R_{ij}$ is affine), $g_i|_{U_{ij}} = g_j|_{U_{ij}}$.

By glueing (Ex. 2.11) there is a morphism $f : X \to \mathbb{P}^n_A$ with $f|_{U_i} = g_i$, a morphism of A-schemes. Finally $f^{-1}(D_i) \cap U_j = g_j^{-1}(D_j \cap D_i) = U_i \cap U_j$ for all j (computed above, also for $j = i$), so $f^{-1}(D_i) = \bigcup_j (U_i \cap U_j) = U_i = X_{f_i}$. By construction $f|_{X_{f_i}}$ is the morphism induced by $T_i^{-1}T_j \mapsto f_i^{-1}f_j$.

The case $A = k$ a field, $x \in X(k)$. Then $f_j(x) \in k(x) = k$ is the image of $f_{j,x}$. Because $f_{i,x}$ is a unit iff its image $f_i(x)$ in the residue field is $\neq 0$, the hypothesis says that $f_0(x), \dots, f_n(x)$ are not all 0. Choose i with $f_i(x) \neq 0$; then $x \in U_i$ and $x \in U_i(k)$ (Remark 3.31). Apply Ex. 3.7 to U_i, the k-algebra homomorphism ψ_i, and the identification $D_i \simeq \mathbb{A}^n_k$: the point $f(x) = g_i(x) \in D_i$ is the rational point with coordinates $T_i^{-1}T_j \mapsto \psi_i(T_i^{-1}T_j)(x) = f_j(x)/f_i(x)$. Comparing with Lemma 3.43 (whose proof shows that the point of $\mathbb{P}^n_k(k)$ with homogeneous coordinates $(\beta_0, \dots, \beta_n)$, $\beta_i \neq 0$, lies in $D_+(T_i)$ and corresponds there to the ideal generated by the $T_i^{-1}T_j - \beta_i^{-1}\beta_j$), we conclude

$$f(x) = (f_0(x) : f_1(x) : \cdots : f_n(x)) \in \mathbb{P}(k^{n+1}) = \mathbb{P}^n_k(k).$$

This does not depend on the choice of i since proportional coordinates give the same point. □

Exercise 3.9

> **Problem.** Let B be a graded ring. Let $f \in B$ be non-nilpotent, homogeneous of degree 0. Show that B_f possesses a natural grading, and that $D_+(f) \simeq \operatorname{Proj} B_f$.

Solution. *Grading.* Since $f \in B_0$, localization commutes with direct sums: $B_f = \bigoplus_{d \geq 0} (B_d)_f$, where $(B_d)_f = \{b/f^m : b \in B_d, m \geq 0\}$ is the localization of the B_0-module B_d. Put $(B_f)_d = (B_d)_f$ for $d \geq 0$. Then $(B_f)_d(B_f)_e \subseteq (B_f)_{d+e}$ and $1 \in (B_f)_0$, so B_f is a graded ring and $B \to B_f$ is a homomorphism of graded rings; we have $(B_f)_0 = (B_0)_f$. The hypothesis that f is not nilpotent just says that $B_f \neq 0$; the result below holds without it (if f is nilpotent, both sides are empty).

The isomorphism. Let $\iota : B \to B_f$. It is a graded homomorphism with $r = 1$ in Lemma 3.40, and $M = \iota(B_+)B_f = (B_f)_+$, so $V_+(M) = \emptyset$. By Lemma 3.40 we have a morphism $\tilde{f} : \operatorname{Proj} B_f \to$

$\operatorname{Proj} B$, $\mathfrak{q} \mapsto \iota^{-1}(\mathfrak{q})$, such that for every homogeneous $h \in B_+$, $\tilde{f}^{-1}(D_+(h)) = D_+(h/1)$ and $\tilde{f}$ restricted to it is the morphism of affine schemes

$$\operatorname{Spec}(B_f)_{(h/1)} \to \operatorname{Spec} B_{(h)}$$

induced by $B_{(h)} \to (B_f)_{(h)}$ (we write h for $h/1$). Since $f/1$ is a unit of B_f, $f \notin \iota^{-1}(\mathfrak{q})$, so $\tilde{f}$ maps into $D_+(f)$.

Let $h \in B_+$ be homogeneous of degree $d > 0$. Because $f \in B_0$ we have $f/1 \in B_{(h)}$. The natural map $(B_{(h)})_f \to (B_f)_{(h)}$, $(a/h^k)/f^m \mapsto (a/f^m)/h^k$ ($a \in B_{kd}$), is an isomorphism: it is surjective because an element of $(B_f)_{(h)}$ has the form $(a/f^m)/h^k$ with $a \in B_{kd}$; it is injective because $a/(f^m h^k) = 0$ in B_{fh} means $(fh)^s a = 0$ in B for some s, i.e. a/h^k is killed by a power of f in $B_{(h)}$. Therefore $B_{(h)} \to (B_f)_{(h)}$ is the localization at f, and by Lemma 3.7 and Example 3.15 $\operatorname{Spec}(B_f)_{(h)} \to \operatorname{Spec} B_{(h)}$ is an isomorphism onto the open subscheme $D(f) \subseteq \operatorname{Spec} B_{(h)}$. By Lemma 3.36(b) (applied to $D_+(fh) \subseteq D_+(h)$: here $r = d = \deg(fh)$, so $\alpha = (fh)^d h^{-d} = f^d \in B_{(h)}$ and $D(\alpha) = D(f)$), $D(f)$ corresponds to $D_+(fh) = D_+(f) \cap D_+(h)$. Thus $\tilde{f}$ induces an isomorphism of schemes

$$\tilde{f}^{-1}(D_+(h)) \xrightarrow{\sim} D_+(f) \cap D_+(h).$$

Since $\tilde{f}$ lands in $D_+(f)$, we have $\tilde{f}^{-1}(D_+(f) \cap D_+(h)) = \tilde{f}^{-1}(D_+(h))$. The open sets $D_+(f) \cap D_+(h)$, $h \in B_+$ homogeneous, cover $D_+(f)$ (the $D_+(h)$ cover $\operatorname{Proj} B$). So $\tilde{f} : \operatorname{Proj} B_f \to D_+(f)$ is an isomorphism over each member of an open covering of the target; hence it is an isomorphism (the local inverses glue). Thus $D_+(f) \simeq \operatorname{Proj} B_f$. □

Exercise 3.10

> **Problem.** Let A be a ring, $X = \mathbb{P}^n_A$. Show that $\mathcal{O}_X(X) = A$ (use the covering $\mathcal{U} = \{D_+(T_i)\}_i$ of X and the complex associated to $\mathcal{U}$ and $\mathcal{O}_X$ as in Lemma 2.7). Deduce from this that X is affine if and only if $n = 0$.

Solution. Let $B = A[T_0, \dots, T_n]$, $U_i = D_+(T_i)$, $U_{ij} = U_i \cap U_j = D_+(T_iT_j)$. By Prop. 3.38, $\mathcal{O}_X(U_i) = B_{(T_i)}$ and $\mathcal{O}_X(U_{ij}) = B_{(T_iT_j)}$, and the restriction $B_{(T_i)} \to B_{(T_iT_j)}$ is the map of Lemma 3.36(c); with $g = T_iT_j = T_i \cdot T_j$ it sends aT_i^{-N} to $(aT_j^N)(T_iT_j)^{-N} = aT_i^{-N}$, i.e. it is the inclusion of subrings of $B_{T_iT_j}$.

By Lemma 2.7, since $\mathcal{O}_X$ is a sheaf the sequence

$$0 \to \mathcal{O}_X(X) \to \bigoplus_i \mathcal{O}_X(U_i) \to \bigoplus_{i,j} \mathcal{O}_X(U_{ij}), \qquad (s_i) \mapsto (s_i|_{U_{ij}} - s_j|_{U_{ij}}),$$

is exact. So $\mathcal{O}_X(X) = \{(s_i)_i : s_i \in B_{(T_i)},\ s_i = s_j \text{ in } B_{(T_iT_j)} \text{ for all } i, j\}$.

All rings involved are free A-modules with monomial bases. Let $M_0 = \{a \in \mathbb{Z}^{n+1} : a_0 + \cdots + a_n = 0\}$ and $T^a = \prod T_k^{a_k}$. Then $B_{(T_i)}$ has A-basis $\{T^a : a \in S_i\}$ with $S_i = \{a \in M_0 : a_k \geq 0 \text{ for } k \neq i\}$ (the degree 0 part of B_{T_i}), and $B_{(T_iT_j)}$ has basis $\{T^a : a \in S_{ij}\}$, $S_{ij} = \{a \in M_0 : a_k \geq 0 \text{ for } k \neq i, j\} \supseteq S_i \cup S_j$. Everything lies in the free A-module $A[M_0]$ with basis $\{T^a\}_{a \in M_0}$ (inside the Laurent polynomial ring $A[T_0^{\pm 1}, \dots, T_n^{\pm 1}]$), and the restriction maps are inclusions. Write $s_i = \sum_{a \in S_i} c_i(a) T^a$. The condition $s_i = s_j$ in $A[M_0]$ says $c_i(a) = c_j(a)$ for all $a \in M_0$ (with $c_i(a) = 0$ for $a \notin S_i$). So there is one family $c : M_0 \to A$ with $c(a) = 0$ unless $a \in \bigcap_i S_i$,

and $s_i = \sum c(a)T^a$ for all i. Now $a \in \bigcap_i S_i$ means $a_k \geq 0$ for all k (for each k there is an $i \neq k$ if $n \geq 1$, and for $n = 0$, $M_0 = \{0\}$), and together with $\sum a_k = 0$ this forces $a = 0$. Hence $s_i = c(0) \in A$ for all i, and $\mathcal{O}_X(X) = A \cdot 1$. Since $A \to B_{(T_0)} = A[T_0^{-1}T_j]$ is injective, the structure homomorphism $A \to \mathcal{O}_X(X)$ is an isomorphism.

Affineness. If $n = 0$, $\mathbb{P}^0_A = \operatorname{Spec} A[T_0]_{(T_0)} = \operatorname{Spec} A$ is affine. For $A = 0$ we have $\mathbb{P}^n_A = \emptyset$, which is affine for every n, so the "only if" part requires $A \neq 0$. Assume $A \neq 0$, $n \geq 1$ and, by contradiction, X affine. Then $X \simeq \operatorname{Spec} C$ with $C = \mathcal{O}_X(X) \simeq A$, and the structural morphism $\pi : X \to \operatorname{Spec} A$ corresponds by Lemma 3.23 to the structure homomorphism $\iota : A \to \mathcal{O}_X(X)$, which is an isomorphism by what we just showed. Thus π is the composition of an isomorphism $X \simeq \operatorname{Spec} \mathcal{O}_X(X)$ and $\operatorname{Spec} \iota$ (a homeomorphism); in particular π is injective on points.

Restrict to $U_0 = \operatorname{Spec} A[S_1, \ldots, S_n]$, $S_j = T_0^{-1}T_j$. By definition of the A-scheme structure (Prop. 3.38), $\pi|_{U_0}$ is the morphism attached to the inclusion $A \to A[S_1, \ldots, S_n]$, so on points $\mathfrak{q} \mapsto \mathfrak{q} \cap A$. Take a prime $\mathfrak{p}$ of A ($A \neq 0$) and put

$$\mathfrak{q}_c = \mathfrak{p}A[S_1, \ldots, S_n] + (S_1 - c, S_2, \ldots, S_n), \qquad c = 0, 1.$$

Then $A[S]/\mathfrak{q}_c \simeq A/\mathfrak{p}$ is a domain, so $\mathfrak{q}_c$ is prime, and $\mathfrak{q}_c \cap A = \mathfrak{p}$ for $c = 0, 1$. But $\mathfrak{q}_0 \neq \mathfrak{q}_1$, since $S_1 \in \mathfrak{q}_0$ whereas $S_1 \notin \mathfrak{q}_1$ (otherwise $1 = S_1 - (S_1 - 1) \in \mathfrak{q}_1$). So π is not injective, a contradiction. Hence $\mathbb{P}^n_A$ is not affine for $n \geq 1$ and $A \neq 0$. □

Exercise 3.11

Problem. Let $B = \bigoplus_{d \geq 0} B_d$ be a graded ring. Let $e \geq 1$ be an integer.

(a) Let us denote the graded ring $\bigoplus_{d \geq 0} B_{de}$ by C (so $C_d := B_{de}$). Show that $\operatorname{Proj} B \simeq \operatorname{Proj} C$.

(b) Let us suppose that for every $d \geq 1$, B_{ed} is generated by B_e^d, and that B_e is finitely generated over a ring A. Show that $\operatorname{Proj} B$ is a projective scheme over A.

Solution. (a) The inclusion $\iota : C \hookrightarrow B$ is a homomorphism of graded rings in the sense of Lemma 3.40, with $r = e$ (it maps $C_d = B_{de}$ into B_{ed}). Let $M = \iota(C_+)B$. We claim $V_+(M) = \emptyset$. If $\mathfrak{p} \in \operatorname{Proj} B$ contains M, it contains every B_{de}, $d \geq 1$. For a homogeneous $b \in B_k$, $k \geq 1$, we have $b^e \in B_{ke} \subseteq \mathfrak{p}$, so $b \in \mathfrak{p}$, whence $B_+ \subseteq \mathfrak{p}$, contradicting $\mathfrak{p} \in \operatorname{Proj} B$. By Lemma 3.40 we get a morphism $\phi : \operatorname{Proj} B \to \operatorname{Proj} C$, $\mathfrak{p} \mapsto \mathfrak{p} \cap C$, such that for every homogeneous $h \in C_+$ (say $h \in C_d = B_{de}$, $d \geq 1$), $\phi^{-1}(D_+(h)) = D_+(h)$ (here the right side is taken in $\operatorname{Proj} B$, h viewed as an element of $B_{de} \subseteq B_+$), and ϕ restricted to it is induced by the homomorphism $\gamma : C_{(h)} \to B_{(h)}$, $c/h^N \mapsto c/h^N$.

We claim γ is bijective. Elements of $B_{(h)}$ are b/h^N with $b \in B_{Nde} = C_{Nd}$, so $b \in C$ and γ is surjective. If $c/h^N \in C_{(h)}$ maps to 0 in B_h, then $h^s c = 0$ in B for some s; this is an equation in C, so $c/h^N = 0$ already in C_h. Hence γ is an isomorphism, and ϕ restricts to an isomorphism of schemes $D_+(h) \to D_+(h)$ (the first in $\operatorname{Proj} B \simeq \operatorname{Spec} B_{(h)}$, the second in $\operatorname{Proj} C \simeq \operatorname{Spec} C_{(h)}$, by Prop. 3.38). The sets $D_+(h)$, $h \in C_+$ homogeneous, cover $\operatorname{Proj} C$. So ϕ is an isomorphism over each member of an open covering of the target, hence an isomorphism, and $\operatorname{Proj} B \simeq \operatorname{Proj} C$. This is an isomorphism of A-schemes for any ring homomorphism $A \to B_0 = C_0$.

(b) Here B is a graded A-algebra (so $A \to B_0$) and $\operatorname{Proj} B$ is an A-scheme. By (a), $\operatorname{Proj} B \simeq \operatorname{Proj} C$ as A-schemes, with $C_d = B_{de}$. Let $x_0, \ldots, x_n \in B_e = C_1$ generate B_e as an A-module (if

$B_e = 0$ take $n = 0$, $x_0 = 0$). Let $R = A[T_0, \dots, T_n]$ with its usual grading and $\varphi : R \to C$ the A-algebra homomorphism with $T_i \mapsto x_i \in C_1$; it is a graded homomorphism ($r = 1$). For $d \geq 1$, $\varphi(R_d)$ is the A-span of the monomials of degree d in the x_i, which is the A-span of all products of d elements of B_e (expand each factor in the x_i), and this is $B_{ed} = C_d$ by hypothesis. Thus

$$\varphi(R_d) = C_d \quad \text{for all } d \geq 1.$$

Hence $\varphi(R_+)C = C_+$, and Lemma 3.40 gives a morphism of A-schemes $g : \operatorname{Proj} C \to \operatorname{Proj} R = \mathbb{P}^n_A$ (the domain is everything because $V_+(\varphi(R_+)C) = V_+(C_+) = \emptyset$), with $g^{-1}(D_+(T_i)) = D_+(x_i)$ and $g|_{D_+(x_i)}$ the morphism of affine schemes $\operatorname{Spec} C_{(x_i)} \to \operatorname{Spec} R_{(T_i)}$ induced by $\varphi_i : R_{(T_i)} \to C_{(x_i)}$, $P/T_i^N \mapsto \varphi(P)/x_i^N$. (If $x_i = 0$ then $C_{(x_i)} = 0$ and $D_+(x_i) = \emptyset$.)

φ_i is surjective. An element of $C_{(x_i)}$ has the form c/x_i^N with $c \in C_N$. If $N \geq 1$, then $c = \varphi(P)$ with $P \in R_N$ and $c/x_i^N = \varphi_i(P/T_i^N)$. If $N = 0$, then $c \in C_0$, and $c/1 = (cx_i)/x_i$ with $cx_i \in C_1 = \varphi(R_1)$, say $cx_i = \varphi(P)$, so $c/1 = \varphi_i(P/T_i)$. Hence φ_i is surjective and, by Lemma 3.17, $g|_{D_+(x_i)} : D_+(x_i) \to D_+(T_i)$ is a closed immersion.

Since the $D_+(T_i)$ cover $\mathbb{P}^n_A$ and $g^{-1}(D_+(T_i)) \to D_+(T_i)$ is a closed immersion for each i, g is a closed immersion. (Being a closed immersion is local on the target: g is injective, $g(\operatorname{Proj} C) \cap D_+(T_i)$ is closed in $D_+(T_i)$ for all i, so $g(\operatorname{Proj} C)$ is closed, g is a homeomorphism onto its image because this holds over each $D_+(T_i)$, and surjectivity of $g^\#$ is checked on stalks, i.e. locally on the target (Ex. 2.7).) Therefore $\operatorname{Proj} B \simeq \operatorname{Proj} C$ is isomorphic to a closed subscheme of $\mathbb{P}^n_A$, i.e. it is a projective scheme over A (Def. 3.42). □

Notes for this part. Results of the text are cited by number ("Prop. 3.25", ...). For a morphism $f : X \to Y$ of schemes and $V \subseteq Y$ open, $f^\#(V) : \mathcal{O}_Y(V) \to \mathcal{O}_X(f^{-1}V)$ denotes the induced ring homomorphism, and $\rho : \operatorname{Mor}(X, Y) \to \operatorname{Hom}(\mathcal{O}_Y(Y), \mathcal{O}_X(X))$ is the bijection of Prop. 3.25 when Y is affine. For $g : Z \to X$ the diagram (3.4) says that $\rho(f \circ g) = g^\#(X) \circ \rho(f)$. Ringed spaces are locally ringed and morphisms of ringed spaces have local stalk maps (Def. 2.19, 2.20). Ex. 3.13 uses the Veronese statement of Ex. 3.11(a), and its argument is included. Ex. 3.16 is used in Ex. 3.17(b). Ex. 3.19 uses, as black boxes, that $\mathcal{O}_K$ is a Dedekind domain and that the class group is finite. Ex. 3.21 uses Ex. 3.20.

Exercise 3.12

> **Problem.** Let B be a Noetherian graded ring. (a) Show that for any homogeneous $f \in B_+$, $B_{(f)}$ is Noetherian. (b) Show that $\operatorname{Proj} B$ is Noetherian.

Solution. Here $B = \bigoplus_{d \geq 0} B_d$.

(a) Let $r = \deg f \geq 1$. The localization B_f is a $\mathbb{Z}$-graded ring: $(B_f)_n$ consists of the fractions b/f^N with b homogeneous of degree $Nr + n$. Then $B_{(f)} = (B_f)_0$. Let I be an ideal of $B_{(f)}$. The ideal IB_f of the Noetherian ring B_f is finitely generated. Each generator is a finite B_f-combination of elements of I, so we can choose generators $a_1, \dots, a_m \in I$ of IB_f. We claim $I = (a_1, \dots, a_m)B_{(f)}$. Let $x \in I$. Then $x = \sum_i c_i a_i$ with $c_i \in B_f$. Taking degree-0 components and using $\deg a_i = 0$ gives $x = \sum_i (c_i)_0 a_i$ with $(c_i)_0 \in B_{(f)}$. So I is finitely generated, and $B_{(f)}$ is Noetherian.

(b) The ideal B_+ is homogeneous, and B is Noetherian. So B_+ is generated by finitely many homogeneous elements $f_1, \dots, f_n$ of positive degree (it is generated by its homogeneous elements,

and finitely many of them suffice). No point of $\operatorname{Proj} B$ contains B_+, so $\emptyset = V_+(B_+) = \bigcap_i V_+(f_i)$, that is $\operatorname{Proj} B = \bigcup_i D_+(f_i)$ (as in the text before Lemma 3.36). By Prop. 3.38, each $D_+(f_i)$ is an affine open subscheme isomorphic to $\operatorname{Spec} B_{(f_i)}$, and $B_{(f_i)}$ is Noetherian by (a). Hence $\operatorname{Proj} B$ is a finite union of affine opens with Noetherian rings, i.e. Noetherian (Def. 3.45). If $B_+ = 0$ the union is empty and $\operatorname{Proj} B = \emptyset$ is trivially Noetherian. □

Exercise 3.13

> **Problem.** Let $B = A[X, Y, Z]$ be a polynomial ring. Let B_d be the sub-A-module of B generated by the elements of the form $X^aY^bZ^c$ with $a + 2b + 3c = d$. (a) Show that the B_d induce a grading on B. (b) Determine B_6 and show that B_{6d} is generated by the elements of B_6^d. Deduce from this that $\operatorname{Proj} B$ is isomorphic to a closed subscheme of $\mathbb{P}^6_A$.

Solution. (a) The monomials $X^aY^bZ^c$ form an A-basis of B, and each monomial has exactly one weight $a+2b+3c$. So $B = \bigoplus_{d\geq 0} B_d$ as A-modules, with B_d free on the monomials of weight d. Weights add under multiplication of monomials, so $B_dB_e \subseteq B_{d+e}$. Also $1 \in B_0 = A$. Hence this is a grading of the A-algebra B (with $B_0 = A$).

(b) The monomials of weight 6 are: for $c = 0$, $X^6,\ X^4Y,\ X^2Y^2,\ Y^3$; for $c = 1$ (so $a + 2b = 3$), $X^3Z,\ XYZ$; for $c = 2$ (so $a = b = 0$), Z^2. Thus B_6 is free over A with basis

$$m_0 = X^6,\ m_1 = X^4Y,\ m_2 = X^2Y^2,\ m_3 = Y^3,\ m_4 = X^3Z,\ m_5 = XYZ,\ m_6 = Z^2.$$

Claim: for $d \geq 1$, B_{6d} is generated by the products of d elements of B_6. The products B_6^d lie in B_{6d}. It suffices to show that every monomial $\mu = X^aY^bZ^c$ of weight $N = 6d \geq 6$ is divisible by one of $m_0, \ldots, m_6$; then $\mu = m_i\mu'$ with μ' of weight $6(d-1)$, and induction on d (the case $d = 1$ being trivial) proves the claim. Note that N is even.

- If $c \geq 2$, then $Z^2 = m_6$ divides μ.
- If $c = 1$, then $a + 2b = N - 3$ is odd, so a is odd and $a \geq 1$. If $a \geq 3$, $m_4 = X^3Z$ divides μ. If $a = 1$, then $b = (N - 4)/2 \geq 1$ and $m_5 = XYZ$ divides μ.
- If $c = 0$, then $a + 2b = N$. If $b \geq 3$, $m_3 = Y^3$ divides μ. If $b = 2$, then $a = N - 4 \geq 2$ and m_2 divides μ. If $b = 1$, then $a = N - 2 \geq 4$ and m_1 divides μ. If $b = 0$, then $a = N \geq 6$ and m_0 divides μ.

This proves the claim.

Veronese subring. Let $C = \bigoplus_{d\geq 0} C_d$ with $C_d = B_{6d}$. This is a graded subring of B (as $B_{6d}B_{6d'} \subseteq B_{6(d+d')}$), and $C_0 = A$. We show that $\operatorname{Proj} B \simeq \operatorname{Proj} C$ (this is Ex. 3.11(a) for $e = 6$; the argument works for any $e \geq 1$). Let $\varphi : C \to B$ be the inclusion; it satisfies $\varphi(C_d) \subseteq B_{6d}$. Let $M = \varphi(C_+)B$ be the ideal of Lemma 3.40. Let $\mathfrak{p} \in \operatorname{Proj} B$ be a prime containing M. For every homogeneous $g \in B_+$ we have $g^6 \in B_{6\deg g} \subseteq C_+ \subseteq M \subseteq \mathfrak{p}$, so $g \in \mathfrak{p}$. Hence $B_+ \subseteq \mathfrak{p}$, which is excluded in $\operatorname{Proj} B$. Therefore $V_+(M) = \emptyset$, and Lemma 3.40 gives a morphism of A-schemes $\pi : \operatorname{Proj} B \to \operatorname{Proj} C$, $\mathfrak{p} \mapsto \mathfrak{p} \cap C$, such that for every homogeneous $h \in C_+$ we have $\pi^{-1}(D_+(h)) = D_+(h)$ and $\pi|_{D_+(h)}$ is the morphism of affine schemes induced by $C_{(h)} \to B_{(h)}$. If $h \in C_\delta = B_{6\delta}$, then

$$C_{(h)} = \{a/h^N :\ a \in C_{N\delta} = B_{6N\delta}\} = \{a/h^N :\ a \in B_{6N\delta}\} = B_{(h)}.$$

More precisely, $C_h \to B_h$ is injective (localization is exact), and the map $C_{(h)} \to B_{(h)}$ is injective and surjective by the displayed equalities. So π restricts to an isomorphism $D_+(h) \to D_+(h)$ over each member of the open cover $\{D_+(h)\}_h$ of $\operatorname{Proj} C$. Being an isomorphism is local on the target, so π is an isomorphism.

Conclusion. By the claim, $C_d = C_1^d$ for $d \geq 1$, so C is generated as an A-algebra by $C_1 = B_6$. Hence the graded A-algebra homomorphism $A[T_0, \ldots, T_6] \to C$, $T_i \mapsto m_i$, is surjective. Its kernel I is a homogeneous ideal, and $C \simeq A[T_0, \ldots, T_6]/I$. By Lemma 3.41, $\operatorname{Proj} C$ is isomorphic to a closed subscheme of $\mathbb{P}^6_A$. So is $\operatorname{Proj} B \simeq \operatorname{Proj} C$. □

Exercise 3.14

> **Problem.** We say that a scheme X is quasi-compact if it is quasi-compact as a topological space (Exercise 1.6). Show that a scheme is quasi-compact if and only if it is a finite union of affine schemes.

Solution. ($\Rightarrow$) Cover X by affine open subsets. By quasi-compactness a finite subfamily already covers X.

($\Leftarrow$) Suppose $X = U_1 \cup \cdots \cup U_n$ with $U_i = \operatorname{Spec} A_i$ open. Each $\operatorname{Spec} A_i$ is quasi-compact (Ex. 1.6; in short, if $\operatorname{Spec} A = \bigcup_j D(g_j)$ then $1 \in (g_j)_j$, so $1 = \sum_{j \in F} c_j g_j$ for a finite set F, and $\operatorname{Spec} A = \bigcup_{j \in F} D(g_j)$; since the $D(g)$ form a base, every open cover has a finite subcover). Now let $X = \bigcup_\lambda W_\lambda$ be an open cover. For each i, finitely many W_λ cover U_i. Taking all of them for $i = 1, \ldots, n$ gives a finite subcover of X. □

Exercise 3.15

> **Problem.** Let X be a quasi-compact scheme, $A = \mathcal{O}_X(X)$. Consider the morphism $f : X \to \operatorname{Spec} A$ induced by the identity on A (Prop. 3.25). Show that $f(X)$ is dense in $\operatorname{Spec} A$.

Solution. We prove a slightly more general statement, which is also used in Ex. 3.17(e).

Lemma A. Let X be a quasi-compact scheme, R a ring and $\psi : R \to \mathcal{O}_X(X)$ a ring homomorphism whose kernel consists of nilpotent elements. Let $h : X \to \operatorname{Spec} R$ be the morphism with $\rho(h) = \psi$ (Prop. 3.25). Then $h(X)$ is dense in $\operatorname{Spec} R$.

Proof. Step 1: for $r \in R$, $h^{-1}(D(r)) = X_{\psi(r)}$ (Def. 3.11). Let $x \in X$ and let $\mathfrak{p}$ be the prime $h(x)$. The stalk map $h^\#_x : R_\mathfrak{p} \to \mathcal{O}_{X,x}$ is local, and it sends $r/1$ to the germ $\psi(r)_x$ (because $h^\#(\operatorname{Spec} R) = \psi$). If $r \notin \mathfrak{p}$, then $r/1$ is a unit and so is $\psi(r)_x$. If $r \in \mathfrak{p}$, then $r/1$ lies in the maximal ideal, so $\psi(r)_x$ lies in $\mathfrak{m}_x$ and is not a unit. This proves Step 1.

Step 2: if $D(r) \cap h(X) = \emptyset$, then r is nilpotent. Indeed, by Step 1, $X_{\psi(r)} = \emptyset$. Write $X = U_1 \cup \cdots \cup U_m$ with $U_i = \operatorname{Spec} B_i$ affine open (Ex. 3.14), and let $a_i \in B_i$ be the restriction of $\psi(r)$. Then $D(a_i) = (U_i)_{a_i} = X_{\psi(r)} \cap U_i = \emptyset$. So a_i lies in every prime of B_i, hence is nilpotent. Choose N with $a_i^N = 0$ for all i. Then $\psi(r)^N$ restricts to 0 on each U_i, so $\psi(r^N) = \psi(r)^N = 0$ by the sheaf axiom. Hence $r^N \in \operatorname{Ker} \psi$ is nilpotent, so r is nilpotent.

Step 3: a nonempty open subset of $\operatorname{Spec} R$ contains a nonempty $D(r)$, and $D(r) \neq \emptyset$ means r is not nilpotent. By Step 2, $D(r)$ meets $h(X)$. So $h(X)$ is dense. □

The exercise is the case $R = A$, $\psi = \mathrm{id}_A$, whose kernel is 0. □

Exercise 3.16

> **Problem.** Let X be a scheme. Show that X is locally Noetherian if and only if any affine open subscheme of X is Noetherian.

Solution. Here an affine open subscheme $U = \operatorname{Spec} A$ is called Noetherian when A is a Noetherian ring (equivalently, U is a Noetherian scheme, by Prop. 3.46(b) and Def. 3.45).

($\Leftarrow$) Every point has an affine open neighborhood $U = \operatorname{Spec} A$. By hypothesis A is Noetherian, so U is a Noetherian scheme (a one-term union in Def. 3.45). Hence X is locally Noetherian.

($\Rightarrow$) Let $U = \operatorname{Spec} A$ be an affine open subscheme of X. Let $x \in U$. By assumption x has an open neighborhood W which is a Noetherian scheme. Choose $g \in A$ with $x \in D(g) \subseteq U \cap W$ (the $D(g)$ form a base of U). Then $D(g)$ is an affine open subset of W (it is isomorphic to $\operatorname{Spec} A_g$ by Lemma 3.7). By Prop. 3.46(b) applied to W, the ring $\mathcal{O}_X(D(g)) = A_g$ is Noetherian. So U is covered by principal opens $D(g)$ with A_g Noetherian. Since U is quasi-compact (Ex. 1.6), finitely many of them, say $D(g_1), \ldots, D(g_n)$, cover U. Thus U is a finite union of affine opens with Noetherian rings, i.e. a Noetherian scheme. Applying Prop. 3.46(b) to the Noetherian scheme U and its affine open subset U itself, $A = \mathcal{O}_X(U)$ is Noetherian. □

Exercise 3.17

> **Problem.** Let $f : X \to Y$ be a morphism of schemes. We say that f is quasi-compact if the inverse image of any affine open subset is quasi-compact. (a) Show that every closed immersion is quasi-compact. (b) Show that an open immersion $f : X \to Y$ is quasi-compact if Y is locally Noetherian. Let us suppose in what follows that f is quasi-compact. (c) Let $\mathcal{J} = \operatorname{Ker}(f^{\#} : \mathcal{O}_Y \to f_*\mathcal{O}_X)$. Show that the ringed topological subspace $Z := V(\mathcal{J})$ of Y (see Lemma 2.23) is a scheme (reduce to the case when Y is affine and draw inspiration from the proofs of Propositions 3.12 and 3.20). (d) Let $j : Z \to Y$ be the closed immersion. Show that we have a morphism $g : X \to Z$ such that $f = j \circ g$, and that if f decomposes into a morphism $g' : X \to Z'$ followed by a closed immersion $j' : Z' \to Y$, then Z is a closed subscheme of Z'. (e) Show that $f(X)$ is dense in Z (reduce to the case when Z is affine). We call Z the scheme-theoretic closure of $f(X)$ in Y. If X is reduced, show that Z is reduced.

Solution. (a) Let $j : Z \to Y$ be a closed immersion and $V \subseteq Y$ an affine open subset. The restriction $j^{-1}(V) \to V$ is again a closed immersion (the topological map is a closed embedding and the stalk maps are unchanged). By Prop. 3.20, $j^{-1}(V)$ is affine, hence quasi-compact (Ex. 3.14).

(b) Let $V = \operatorname{Spec} A \subseteq Y$ be affine open. By Ex. 3.16, A is Noetherian. Since f is an open immersion, $f^{-1}(V)$ is homeomorphic to the open subset $U = f(X) \cap V$ of $\operatorname{Spec} A$. Write $U = V \setminus V(I)$ with $I = (g_1, \ldots, g_r)$ finitely generated. Then $U = \bigcup_s D(g_s)$ is a finite union of affine opens (Lemma 3.7), hence quasi-compact (Ex. 3.14).

For the rest, f is quasi-compact. Recall that $\mathcal{J}$ is a sheaf of ideals of $\mathcal{O}_Y$ with $\mathcal{J}(V) = \operatorname{Ker} f^{\#}(V)$, and that $Z = V(\mathcal{J}) = \{y : \mathcal{J}_y \neq \mathcal{O}_{Y,y}\}$ carries the sheaf $j^{-1}(\mathcal{O}_Y/\mathcal{J})$ (Lemma 2.23).

A computation in the affine case. Let $V = \operatorname{Spec} A$ be an affine open subset of Y and let $X' = f^{-1}(V)$, which is quasi-compact. Write $X' = U_1 \cup \cdots \cup U_n$ with $U_i = \operatorname{Spec} B_i$ affine open, and let $\varphi_i = (f|_{U_i})^{\#}(V) : A \to B_i$. Let $\psi = f^{\#}(V) : A \to \mathcal{O}_X(X')$ and $I = \operatorname{Ker} \psi$. Since $\mathcal{O}_X(X') \to \prod_i B_i$ is injective (sheaf axiom), $I = \bigcap_i \operatorname{Ker} \varphi_i$. We claim: $\mathcal{J}(D(g)) = IA_g$ for every $g \in A$. By Prop. 3.14, $f^{-1}(D(g)) \cap U_i = D(\varphi_i(g))$, and $f^{\#}(D(g)) : A_g \to \mathcal{O}_X(f^{-1}D(g)) \hookrightarrow \prod_i (B_i)_{\varphi_i(g)}$ is, componentwise, the localization of φ_i. So $a/g^k \in A_g$ is killed by $f^{\#}(D(g))$ iff for each i there is m_i with $\varphi_i(g^{m_i}a) = 0$. Taking $m = \max m_i$ (finitely many i), this holds iff $g^m a \in I$ for some m, iff $a/g^k \in IA_g$. This proves the claim. Let $i : \operatorname{Spec} A/I \to V$ be the closed immersion of Lemma 3.17. It gives $(\operatorname{Ker} i^{\#})(D(g)) = I \otimes_A A_g = IA_g = \mathcal{J}(D(g))$. Two subsheaves of $\mathcal{O}_V$ which agree on all $D(g)$ are equal (a section over U is determined by its restrictions to the $D(g) \subseteq U$). So $\operatorname{Ker} i^{\#} = \mathcal{J}|_V$. By Prop. 2.24 applied to i, $\operatorname{Spec} A/I \simeq V(\operatorname{Ker} i^{\#}) = Z \cap V$ as ringed spaces. $(*)$

(c) Cover Y by affine opens V. Each $f^{-1}(V)$ is quasi-compact. By $(*)$, each $Z \cap V$ is isomorphic to an affine scheme, and the $Z \cap V$ form an open cover of Z by affine schemes. So Z is a scheme, and j is a closed immersion of schemes by Lemma 2.23.

(d) *Factorization.* Let $x \in X$, $y = f(x)$. If $s \in \mathcal{J}_y$ is represented by $s \in \mathcal{J}(V)$, then $f^{\#}(V)(s) = 0$, so $f_x^{\#}(s) = 0$. Hence $\mathcal{J}_y \subseteq \operatorname{Ker} f_x^{\#}$. This kernel is a proper ideal, since $f_x^{\#}$ is a ring homomorphism to a nonzero ring. So $\mathcal{J}_y \neq \mathcal{O}_{Y,y}$, i.e. $y \in Z$. Thus $f(X) \subseteq Z$, and $f = j \circ g$ for a continuous $g : X \to Z$ (the same map). The morphism $f^{\#} : \mathcal{O}_Y \to f_*\mathcal{O}_X$ kills $\mathcal{J}$, so it factors through $\mathcal{O}_Y/\mathcal{J} \to f_*\mathcal{O}_X = j_*g_*\mathcal{O}_X$. Applying j^{-1} and using $j^{-1}j_* = \mathrm{id}$ for the closed subset Z (Ex. 2.13, as in the proof of Prop. 2.24), we obtain $g^{\#} : \mathcal{O}_Z = j^{-1}(\mathcal{O}_Y/\mathcal{J}) \to g_*\mathcal{O}_X$. Its stalk at x is $\mathcal{O}_{Y,y}/\mathcal{J}_y \to \mathcal{O}_{X,x}$, induced by $f_x^{\#}$; it is local because $f_x^{\#}$ is and the maximal ideal of the quotient is the image of $\mathfrak{m}_y$. So g is a morphism of schemes, and $j \circ g = f$ by construction.

Minimality. Suppose $f = j' \circ g'$ with $j' : Z' \to Y$ a closed immersion. By Prop. 2.24, we may identify Z' with $V(\mathcal{J}')$, $\mathcal{J}' = \operatorname{Ker} j'^{\#}$, with its structure sheaf $\mathcal{O}_Y/\mathcal{J}'$ and the canonical j'. We have $f^{\#} = j'_*(g'^{\#}) \circ j'^{\#}$, hence $\mathcal{J}' \subseteq \mathcal{J}$. Consequently $\mathcal{J}'_y \neq \mathcal{O}_{Y,y}$ whenever $\mathcal{J}_y \neq \mathcal{O}_{Y,y}$, i.e. $Z \subseteq Z'$, and Z is closed in Z'. The canonical surjection $\mathcal{O}_Y/\mathcal{J}' \to \mathcal{O}_Y/\mathcal{J}$, restricted to Z' and using that the stalks of $\mathcal{O}_Y/\mathcal{J}$ vanish off Z (so that its restriction to Z' equals $h_*\mathcal{O}_Z$, $h : Z \to Z'$ the inclusion), gives $h^{\#} : \mathcal{O}_{Z'} \to h_*\mathcal{O}_Z$. Its stalks $\mathcal{O}_{Y,z}/\mathcal{J}'_z \to \mathcal{O}_{Y,z}/\mathcal{J}_z$ are surjective and local. So h is a closed immersion with $j' \circ h = j$, which says that Z is a closed subscheme of Z'.

(e) *Density.* It suffices to show that $f(X) \cap Z \cap V$ is dense in $Z \cap V$ for every V in an affine open cover of Y (a subset which is dense in each member of an open cover is dense). Fix $V = \operatorname{Spec} A$ and use the notation of $(*)$. Then $Z \cap V \simeq \operatorname{Spec} A/I$, and $f(X) \cap V = f(X') = g(X')$. The morphism $X' \to Z \cap V \simeq \operatorname{Spec}(A/I)$ corresponds under ρ to the map $\bar{\psi} : A/I \to \mathcal{O}_X(X')$ induced by ψ (compare with (3.4): composing with $\operatorname{Spec} A/I \to \operatorname{Spec} A$ gives ψ). The map $\bar{\psi}$ is injective by the definition of I. By Lemma A (Ex. 3.15) applied to the quasi-compact scheme X', $g(X')$ is dense in $\operatorname{Spec}(A/I)$.

Reducedness. Assume X is reduced. Then $\mathcal{O}_X(U)$ is reduced for every open U: if $s^n = 0$, then s_x is nilpotent in the reduced ring $\mathcal{O}_{X,x}$, so $s_x = 0$ for all x and $s = 0$. In particular $\mathcal{O}_X(X')$ is reduced. The subring $A/I \subseteq \mathcal{O}_X(X')$ is reduced. Thus $Z \cap V \simeq \operatorname{Spec} A/I$ has reduced local rings (localizations of a reduced ring). As V varies, these cover Z, so Z is reduced. $\square$

Exercise 3.18

Problem. Let X be an affine algebraic variety over a field k. Show that there exists a projective variety $\overline{X}$ over k such that X is isomorphic to a dense open subscheme of $\overline{X}$.

Solution. Write $X = \operatorname{Spec} A$ with $A = k[t_1, \dots, t_n]/I$ a finitely generated k-algebra (Def. 3.47). Let $S = k[T_0, \dots, T_n]$ with the usual grading. For $P \in k[t_1, \dots, t_n]$ nonzero of degree δ let $P^h = T_0^\delta P(T_1/T_0, \dots, T_n/T_0) \in S_\delta$ be its homogenization, and set $0^h = 0$. Let $\bar{I} \subseteq S$ be the (homogeneous) ideal generated by all P^h, $P \in I$. Put $B = S/\bar{I}$ and $\overline{X} = \operatorname{Proj} B$. This is a closed subscheme of $\mathbb{P}^n_k$ (Lemma 3.41), hence a projective variety over k. Let $\epsilon : S \to k[t_1, \dots, t_n]$ be the k-algebra homomorphism $T_0 \mapsto 1$, $T_j \mapsto t_j$, and write $b(1, t) = \epsilon(b)$.

Step 1: for $b \in S$ homogeneous, $b \in \bar{I} \iff \epsilon(b) \in I$. If $b = \sum_j c_j P_j^h$ with $c_j \in S$ and $P_j \in I$, then $\epsilon(b) = \sum_j \epsilon(c_j) P_j \in I$ because $\epsilon(P_j^h) = P_j$. Conversely let $\epsilon(b) = P \in I$, $b \neq 0$. Write $b = T_0^m b'$ with b' homogeneous and not divisible by T_0. Then b' has a monomial free of T_0, so $\deg P = \deg b'$, and ϵ is injective on homogeneous polynomials of a fixed degree. Hence $b' = P^h$ and $b = T_0^m P^h \in \bar{I}$.

Step 2: T_0 is a non-zero-divisor on B. Since $\bar{I}$ is homogeneous, it suffices to treat a homogeneous b with $T_0 b \in \bar{I}$. Then $\epsilon(b) = \epsilon(T_0 b) \in I$ by Step 1, so $b \in \bar{I}$ by Step 1.

Step 3: $D_+(T_0) \simeq X$. By Prop. 3.38, $D_+(T_0) \simeq \operatorname{Spec} B_{(T_0)}$. We have $S_{(T_0)} = k[T_j/T_0]_j \simeq k[t_1, \dots, t_n]$ via $b/T_0^N \mapsto \epsilon(b)$ for $b \in S_N$. Since $B_{T_0} = S_{T_0}/\bar{I}S_{T_0}$ is $\mathbb{Z}$-graded and $\bar{I}S_{T_0}$ is a homogeneous ideal, $B_{(T_0)} = S_{(T_0)}/(\bar{I}S_{T_0})_0$. The degree-0 part $(\bar{I}S_{T_0})_0$ consists of the sums of elements b/T_0^N with $b \in \bar{I}$ homogeneous of degree N; under the isomorphism above these become the elements $\epsilon(b)$, which lie in I by Step 1, and every $P \in I$ occurs (as $\epsilon(P^h)$). So the image of $(\bar{I}S_{T_0})_0$ is exactly I, and $B_{(T_0)} \simeq k[t_1, \dots, t_n]/I = A$. Hence $X = \operatorname{Spec} A$ is isomorphic, as a k-scheme, to the open subscheme $D_+(T_0)$ of $\overline{X}$.

Step 4: $D_+(T_0)$ is dense in $\overline{X}$. A nonempty open subset of $\overline{X}$ contains a nonempty $D_+(g)$ with $g \in B_+$ homogeneous. Suppose $D_+(g) \cap D_+(T_0) = D_+(gT_0) = \emptyset$. Then $B_{(gT_0)} = 0$ (Prop. 3.38), so $1 = 0$ in B_{gT_0}, i.e. $(gT_0)^m = 0$ in B for some m. By Step 2, $g^m = 0$. Then $B_g = 0$, so $B_{(g)} = 0$ and $D_+(g) = \emptyset$, a contradiction. Thus every nonempty open set meets $D_+(T_0)$. □

Exercise 3.19

Problem. Let K be a number field. Let $\mathcal{O}_K$ be the ring of integers of K (i.e., the set of elements of K which are integral over $\mathbb{Z}$). Using the finiteness theorem of the class group $\operatorname{cl}(K)$, show that every open subset of $\operatorname{Spec} \mathcal{O}_K$ is principal. Deduce from this that every open subscheme of $\operatorname{Spec} \mathcal{O}_K$ is affine.

Solution. We use the standard facts: $\mathcal{O}_K$ is a Dedekind domain (so its nonzero prime ideals are maximal, and every nonzero ideal is a product of prime ideals, uniquely), and the class group $\operatorname{cl}(K)$ of invertible ideals modulo principal ones is finite. Let $h = |\operatorname{cl}(K)|$.

Let U be an open subset of $X = \operatorname{Spec} \mathcal{O}_K$, $U = X \setminus V(J)$ for an ideal J. If $U = \emptyset$, then $U = D(0)$. Otherwise $J \neq 0$ (as $V(0) = X$), and we factor $J = \mathfrak{p}_1^{e_1} \cdots \mathfrak{p}_r^{e_r}$ with distinct nonzero primes ($r \geq 0$). A prime contains J iff it contains some $\mathfrak{p}_i$, iff it equals $\mathfrak{p}_i$ (the $\mathfrak{p}_i$ are maximal and a prime containing a product contains a factor). So $V(J) = \{\mathfrak{p}_1, \dots, \mathfrak{p}_r\}$.

By Lagrange, the class of $\mathfrak{p}_i$ in the finite group $\operatorname{cl}(K)$ satisfies $[\mathfrak{p}_i]^h = 1$, i.e. $\mathfrak{p}_i^h = \pi_i \mathcal{O}_K$ is principal. Put $g = \pi_1 \cdots \pi_r$ ($g = 1$ if $r = 0$). Then $(g) = (\mathfrak{p}_1 \cdots \mathfrak{p}_r)^h$, and the primes containing

g are exactly $\mathfrak{p}_1, \ldots, \mathfrak{p}_r$. So $V(g) = V(J)$ and $U = D(g)$. Hence every open subset of X is principal.

Consequently every open subscheme $U = D(g)$ of $\operatorname{Spec} \mathcal{O}_K$ is isomorphic to $\operatorname{Spec}(\mathcal{O}_K)_g$ by Lemma 3.7, hence is affine. $\square$

Exercise 3.20

Problem. Let A be a ring, G a finite group of automorphisms of A, and A^G the subring of elements of A which are invariant under G. Let $p : \operatorname{Spec} A \to \operatorname{Spec} A^G$ denote the morphism induced by the inclusion $A^G \to A$. (a) Show that G acts naturally on $\operatorname{Spec} A$. Show that $p(x_1) = p(x_2)$ if and only if there exists $\sigma \in G$ such that $\sigma(x_1) = x_2$. (b) Show that A is integral over A^G (let $a \in A$, and consider the polynomial $P(T) := \prod_{\sigma \in G}(T - \sigma a) \in A^G[T]$). Deduce from this that p is surjective (Exercise 1.8). (c) Let $a \in A$. Let $P(T)$ be as above and write $P(T) = T^d + \sum_{i \le d-1} b_i T^i$, $b_i \in A^G$. Show that $p(D(a)) = \bigcup_i D(b_i)$. Show that p is open (in the topological sense). (d) Show that for any $b \in A^G$, we have $p^{-1}(D(b)) = D(bA)$ and $(A^G)_b = (A_b)^G$. Let V be an open subset of $\operatorname{Spec} A^G$. Show that G acts on the scheme $p^{-1}(V)$ and that $\mathcal{O}_{\operatorname{Spec} A^G}(V) = \mathcal{O}_{\operatorname{Spec} A}(p^{-1}(V))^G$.

Solution. Write $X = \operatorname{Spec} A$, $S = \operatorname{Spec} A^G$, $d = |G|$.

(a) For $\sigma \in G$ let $\varphi_\sigma = \operatorname{Spec}(\sigma^{-1}) : X \to X$. It sends a prime $\mathfrak{q}$ to $(\sigma^{-1})^{-1}(\mathfrak{q}) = \sigma(\mathfrak{q})$. Since Spec is contravariant, $\varphi_\sigma \varphi_\tau = \operatorname{Spec}(\tau^{-1}\sigma^{-1}) = \varphi_{\sigma\tau}$. So $\sigma \mapsto \varphi_\sigma$ is a homomorphism $G \to \operatorname{Aut}(X)$ (automorphisms of schemes, Prop. 3.14), and we write $\sigma(x) = \varphi_\sigma(x)$. On global sections, $\varphi_\sigma^{\#}(X) = \sigma^{-1}$ (Prop. 3.14), so the invariants of $\mathcal{O}_X(X)$ are A^G. Since σ^{-1} fixes A^G, we have $p \circ \varphi_\sigma = p$. So $p(\mathfrak{q}) = \mathfrak{q} \cap A^G$ is constant on G-orbits: $\sigma(\mathfrak{q}) \cap A^G = \sigma(\mathfrak{q} \cap A^G) = \mathfrak{q} \cap A^G$.

Conversely, let $\mathfrak{q}_1, \mathfrak{q}_2$ be primes with $\mathfrak{q}_1 \cap A^G = \mathfrak{q}_2 \cap A^G =: P$. Let $b \in \mathfrak{q}_2$. Then $N(b) = \prod_{\sigma \in G} \sigma(b)$ is G-invariant and lies in $\mathfrak{q}_2$ (the factor for $\sigma = 1$), so $N(b) \in \mathfrak{q}_2 \cap A^G = P \subseteq \mathfrak{q}_1$. As $\mathfrak{q}_1$ is prime, $\sigma(b) \in \mathfrak{q}_1$ for some σ, i.e. $b \in \sigma^{-1}(\mathfrak{q}_1)$. Therefore $\mathfrak{q}_2 \subseteq \bigcup_\sigma \sigma(\mathfrak{q}_1)$, a finite union of primes. By prime avoidance, $\mathfrak{q}_2 \subseteq \tau(\mathfrak{q}_1)$ for some τ. By symmetry, $\mathfrak{q}_1 \subseteq \rho(\mathfrak{q}_2)$ for some ρ. Then $\mathfrak{q}_2 \subseteq \tau(\mathfrak{q}_1) \subseteq \tau\rho(\mathfrak{q}_2)$. Put $\sigma_0 = \tau\rho$ and let n be its order. Then $\mathfrak{q}_2 \subseteq \sigma_0(\mathfrak{q}_2) \subseteq \sigma_0^2(\mathfrak{q}_2) \subseteq \cdots \subseteq \sigma_0^n(\mathfrak{q}_2) = \mathfrak{q}_2$, so all these are equal and $\tau(\mathfrak{q}_1) = \mathfrak{q}_2$. This proves (a).

(b) For $a \in A$, the polynomial $P(T) = \prod_{\sigma \in G}(T - \sigma a) \in A[T]$ is monic of degree d and $P(a) = 0$ (factor $\sigma = 1$). For $\tau \in G$, applying τ to the coefficients permutes the factors, so $\tau P = P$. Hence $P \in A^G[T]$ and a is integral over A^G.

Surjectivity of p (this is Ex. 1.8(c) for the injective integral map $A^G \to A$; here is the argument). Let P_0 be a prime of $R = A^G$ and $T = R \setminus P_0$. Since $R \subseteq A$ and $0 \notin T$, the ring $T^{-1}A$ is nonzero. Let M be a maximal ideal of $T^{-1}A$. The ring $T^{-1}A$ is integral over $T^{-1}R = R_{P_0}$. The domain $D = R_{P_0}/(M \cap R_{P_0})$ is contained in the field $F = T^{-1}A/M$ and F is integral over D. Then D is a field: if $0 \neq x \in D$ and x^{-1} satisfies $x^{-m} + c_1 x^{-m+1} + \cdots + c_m = 0$ with $c_i \in D$, multiplying by x^{m-1} gives $x^{-1} \in D$. So $M \cap R_{P_0} = P_0 R_{P_0}$, and the prime $\mathfrak{q} = \{a \in A : a/1 \in M\}$ satisfies $\mathfrak{q} \cap R = P_0$. Hence p is surjective.

(c) Write $P(T) = T^d + \sum_{i<d} b_i T^i$. Here $b_i = \pm e_{d-i}(\sigma a)_{\sigma \in G}$ is, up to sign, the elementary symmetric polynomial of degree $d - i \geq 1$ in the σa.

($\subseteq$) Let $\mathfrak{q} \in D(a)$ and $P_0 = \mathfrak{q} \cap A^G$. If all $b_i \in P_0 \subseteq \mathfrak{q}$, then $a^d = -\sum_{i<d} b_i a^i \in \mathfrak{q}$, so $a \in \mathfrak{q}$, a contradiction. So some $b_i \notin P_0$, i.e. $p(\mathfrak{q}) \in D(b_i)$.

($\supseteq$) Let $P_0 \in D(b_i)$ for some i. By (b) there is $\mathfrak{q}$ with $\mathfrak{q} \cap A^G = P_0$. If $\sigma(a) \in \mathfrak{q}$ for all $\sigma \in G$, then b_i, being a sum of products of $d - i \geq 1$ of the σa, lies in $\mathfrak{q} \cap A^G = P_0$, a contradiction. So $\sigma(a) \notin \mathfrak{q}$ for some σ, i.e. $a \notin \sigma^{-1}(\mathfrak{q})$. The prime $\sigma^{-1}(\mathfrak{q})$ lies in $D(a)$ and over P_0 by (a). Hence $P_0 \in p(D(a))$.

Since the sets $D(a)$ form a base of X and p commutes with unions, p maps open sets to open sets.

(d) For $b \in A^G$, $p^{-1}(D(b)) = \{\mathfrak{q} : b \notin \mathfrak{q} \cap A^G\} = D(b) \subseteq X$, which is $D(bA)$ in the notation of the statement. G acts on A_b by $\sigma(a/b^n) = \sigma(a)/b^n$, and $(A^G)_b \to A_b$ is injective (localization is exact) with image in $(A_b)^G$. Conversely, let $a/b^n \in (A_b)^G$. For each σ, $\sigma(a)/b^n = a/b^n$ in A_b, so $b^{m_\sigma}(\sigma a - a) = 0$ for some m_σ. With $m = \max_\sigma m_\sigma$, the element $b^m a$ is G-invariant, and $a/b^n = b^m a/b^{n+m} \in (A^G)_b$. So $(A^G)_b = (A_b)^G$.

Let $V \subseteq S$ be open and $W = p^{-1}(V)$. Since $p\varphi_\sigma = p$, W is stable under every φ_σ, and φ_σ restricts to an automorphism of the open subscheme W. So G acts on W, and on the ring $\mathcal{O}_X(W)$ by the $\varphi_\sigma^\#(W)$; the invariants are the elements fixed by all of them. Moreover, the maps $\varphi_\sigma^\#$ commute with restriction to stable opens. Write $V = \bigcup_j D(b_j)$ with $b_j \in A^G$. Then $W = \bigcup_j D(b_j)$ (in X). By the first part, $\mathcal{O}_S(D(b_j)) = (A^G)_{b_j} = (A_{b_j})^G = \mathcal{O}_X(D(b_j))^G$, and likewise for $D(b_j b_l)$. We show that $p^\#(V) : \mathcal{O}_S(V) \to \mathcal{O}_X(W)^G$ is bijective. Injectivity: if $s \in \mathcal{O}_S(V)$ maps to 0, then each $s|_{D(b_j)} \in (A^G)_{b_j} \subseteq A_{b_j}$ is 0, so $s = 0$. Surjectivity: let $t \in \mathcal{O}_X(W)^G$. Then each $t|_{D(b_j)}$ is G-invariant, hence lies in $\mathcal{O}_S(D(b_j))$. The sections $t_j = t|_{D(b_j)}$ agree on $D(b_j b_l)$ in $A_{b_j b_l}$, hence in the subring $(A^G)_{b_j b_l}$. By the sheaf property on S they come from a unique $s \in \mathcal{O}_S(V)$, and $p^\#(V)(s) = t$ by the sheaf property on X. So $\mathcal{O}_S(V) = \mathcal{O}_X(p^{-1}V)^G$, via $p^\#(V)$. □

Exercise 3.21

> **Problem.** Let X be a scheme and G a finite group acting on X (i.e. G is endowed with a group homomorphism $G \to \mathrm{Aut}(X)$). We define the quotient scheme X/G by the universal property of Exercise 2.14, where we replace the ringed topological spaces by schemes. It does not always exist. (a) Let A be a ring and G a finite group of automorphisms of A, which we identify with a group of automorphisms of the scheme $\mathrm{Spec}\, A$. Show that the quotient scheme $(\mathrm{Spec}\, A)/G$ exists and is isomorphic to $\mathrm{Spec}(A^G)$. Show that this is also the quotient as a ringed topological space (Exercise 2.14). (b) Let U be an open subscheme of $\mathrm{Spec}\, A$ that is stable under G. Preserving the notation of the preceding exercise, show that the quotient scheme U/G exists and is isomorphic to $p(U)$. (c) Let G be a finite group acting on a scheme X. We suppose that every point $x \in X$ has an affine open neighborhood that is stable under G (see Exercise 3.3.23 for examples of such X). Show that the quotient scheme X/G exists.

Solution. *Setting.* A quotient scheme of X by G is a scheme Y with a morphism $q : X \to Y$ such that $q \circ \sigma = q$ for all $\sigma \in G$, and such that every morphism of schemes $f : X \to Z$ with $f \circ \sigma = f$ for all σ factors as $f = \tilde{f} \circ q$ for a unique morphism $\tilde{f} : Y \to Z$. It is unique up to unique isomorphism. Let $X = \mathrm{Spec}\, A$, $S = \mathrm{Spec}\, A^G$, $p : X \to S$, and use the notation φ_σ of Ex. 3.20. The group G is a finite group of automorphisms of A. (If G is given as acting on A through a homomorphism $G \to \mathrm{Aut}(A)$ with a kernel, replace G by its image H: invariants, orbits and the condition $f\sigma = f$ only involve the image.)

Lemma U. Let $V \subseteq S$ be open and $W = p^{-1}(V)$. If $g, g' : V \to Z$ are morphisms of schemes with $g \circ p|_W = g' \circ p|_W$, then $g = g'$.

Proof. The map $p : W \to V$ is surjective (Ex. 3.20(b)), so $g = g'$ as maps of sets. Now $(g \circ p)^\# = g_*(p^\#) \circ g^\#$ and the same for g'. By Ex. 3.20(d), $p^\#(V'')$ is injective for every open $V'' \subseteq V$ (it is the inclusion of G-invariants), so $g_*(p^\#)$ is injective. Hence $g^\# = g'^\#$. $\square$

Lemma Q. Let $V' \subseteq S$ be open and $W = p^{-1}(V')$, with G acting on W as in Ex. 3.20(d). Then $p|_W : W \to V'$ is a quotient scheme of W by G.

Proof. We have $p\varphi_\sigma = p$. Let $f : W \to Z$ be a morphism of schemes with $f\varphi_\sigma = f$ for all σ. Cover Z by affine opens $Z_i = \operatorname{Spec} C_i$. Then $W_i = f^{-1}(Z_i)$ is open and G-stable (since $f\varphi_\sigma = f$). Let $V_i = p(W_i)$; it is open by Ex. 3.20(c). By Ex. 3.20(a), the fibres of p are the G-orbits, so $p^{-1}(V_i) = \bigcup_\sigma \sigma(W_i) = W_i$. The V_i cover V' (p is surjective and the W_i cover W).

Let $\psi_i = \rho(f|_{W_i}) : C_i \to \mathcal{O}_X(W_i)$ (Prop. 3.25). By (3.4), $\rho(f\varphi_\sigma|_{W_i}) = \varphi_\sigma^\#(W_i) \circ \psi_i$, which equals ψ_i. So ψ_i takes values in $\mathcal{O}_X(W_i)^G = \mathcal{O}_S(V_i)$ (Ex. 3.20(d)). Let $g_i : V_i \to Z_i \subseteq Z$ be the morphism with $\rho(g_i) = \psi_i$ (viewed as a map $C_i \to \mathcal{O}_S(V_i)$). Then $\rho(g_i \circ p|_{W_i}) = p^\#(V_i) \circ \psi_i = \psi_i = \rho(f|_{W_i})$, so $g_i \circ p = f$ on W_i by injectivity of ρ. On $V_i \cap V_j$, whose preimage is $W_i \cap W_j$, both g_i and g_j satisfy $g \circ p = f$. By Lemma U they coincide. Gluing morphisms (Ex. 2.11) we get $\tilde{f} : V' \to Z$ with $\tilde{f} \circ p = f$. Uniqueness follows from Lemma U. $\square$

(a) and (b). Taking $V' = S$ in Lemma Q shows that (S, p) is a quotient scheme of X, so $(\operatorname{Spec} A)/G$ exists and is isomorphic to $\operatorname{Spec} A^G$. For (b), let $U \subseteq X$ be open and G-stable. Then $p(U)$ is open (Ex. 3.20(c)) and $p^{-1}(p(U)) = \bigcup_\sigma \sigma(U) = U$. Lemma Q with $V' = p(U)$ shows that $p|_U : U \to p(U)$ is the quotient scheme of U by G.

Ringed space quotient. Let $Y = X/G$ be the set of orbits with the quotient topology and $q : X \to Y$, and let $\mathcal{O}_Y(V) = \mathcal{O}_X(q^{-1}V)^G$, as in Ex. 2.14. By Ex. 3.20(a), p induces a bijection $\theta : Y \to S$ with $\theta q = p$. It is a homeomorphism: p is continuous, open (Ex. 3.20(c)) and surjective, so a subset $V \subseteq S$ is open iff $p^{-1}(V)$ is open, which is the definition of the topology on Y. By Ex. 3.20(d), $p^\#$ identifies $\mathcal{O}_S(V)$ with $\mathcal{O}_X(p^{-1}V)^G = \mathcal{O}_Y(\theta^{-1}V)$. So $(\theta, \cdot) : (Y, \mathcal{O}_Y) \to (S, \mathcal{O}_S)$ is an isomorphism of ringed spaces compatible with the maps from X. By Ex. 2.14, $(Y, \mathcal{O}_Y, q)$ is the quotient in the category of ringed topological spaces, and hence so is (S, p).

(c) Let $\{U_i\}$ be a cover of X by G-stable affine opens, $U_i = \operatorname{Spec} A_i$. The group G acts on U_i by scheme automorphisms, i.e. by ring automorphisms of A_i (Prop. 3.25), through a finite group of automorphisms (as in the Setting). Let $S_i = \operatorname{Spec} A_i^G$ and $p_i : U_i \to S_i$.

Let $Y = X/G$ be the set of orbits with the quotient topology and $q : X \to Y$. It is open: $q^{-1}(q(W)) = \bigcup_\sigma \sigma(W)$ is open for W open. Define $\mathcal{O}_Y(V) = \mathcal{O}_X(q^{-1}V)^G$. This is a sheaf of rings: restriction preserves invariants, and if $s_a \in \mathcal{O}_Y(V_a)$ are compatible on overlaps for an open cover $V = \bigcup V_a$, then the glued section of $\mathcal{O}_X(q^{-1}V)$ is invariant, because $\sigma(s)$ and s agree on every $q^{-1}V_a$. Let $V_i = q(U_i)$, which is open, with $q^{-1}(V_i) = U_i$ because U_i is stable. The bijection $\theta_i : V_i \to S_i$ induced by p_i (both are the set of orbits in U_i) is a homeomorphism: $V \subseteq V_i$ is open iff $q^{-1}V$ is open iff $p_i^{-1}(\theta_i V)$ is open iff $\theta_i V$ is open, as p_i is open, continuous and surjective. By Ex. 3.20(d), $p_i^\#$ identifies $\mathcal{O}_{S_i}(\theta_i V)$ with $\mathcal{O}_X(q^{-1}V)^G = \mathcal{O}_Y(V)$. Hence $(V_i, \mathcal{O}_Y|_{V_i}) \simeq S_i$ is an affine scheme. The V_i cover Y, so Y is a scheme. Moreover $q|_{U_i}$ corresponds to p_i under θ_i, so q is a morphism of schemes (locality of stalk maps is checked on the cover U_i), and $q\varphi_\sigma = q$.

Universal property. Let $f : X \to Z$ be a morphism of schemes with $f\varphi_\sigma = f$. Since $q^{-1}(V_i) = U_i \simeq p_i^{-1}(S_i)$, Lemma Q (for S_i in place of S and $V' = S_i$) yields a morphism $f_i : V_i \to Z$ with $f_i \circ q = f$ on U_i. On $V_i \cap V_j$, whose preimage is $U_i \cap U_j$, the morphisms f_i and f_j satisfy $g \circ q = f$; since $V_i \cap V_j$ is an open subset of $V_i \simeq S_i$, Lemma U (for S_i) shows $f_i = f_j$ there. Gluing gives

$\tilde{f} : Y \to Z$ with $\tilde{f} \circ q = f$. If $\tilde{f}'$ is another one, then $\tilde{f}|_{V_i}$ and $\tilde{f}'|_{V_i}$ agree by Lemma U, so $\tilde{f} = \tilde{f}'$. Thus (Y, q) is the quotient scheme X/G. □

Exercise 3.22

> **Problem.** Let k be a field of characteristic 0. We let $G := \mathbb{Z}$ act on the polynomial ring $k[T]$ by $n : T \mapsto T + n$ if $n \in \mathbb{Z}$. (a) Show that G is a subgroup of the group of automorphisms (of k-algebras) of $k[T]$. We identify it with a group of automorphisms of $\mathbb{A}^1_k$. Show that the only open subschemes of $\mathbb{A}^1_k$ that are stable under G are $\emptyset$ and $\mathbb{A}^1_k$ itself. (b) Show that the quotient scheme $\mathbb{A}^1_k/G$ exists and is equal to $\operatorname{Spec} k$. (c) Show that $\mathbb{A}^1_k/G$ is not the quotient as a ringed topological space (Exercise 2.14).

Solution. Here $G = \mathbb{Z}$ is infinite, so the finite-group results of Ex. 3.20–3.21 are not used. The definition of quotient scheme is the one in Ex. 3.21 (universal property, G acting through $n \mapsto \varphi_n = \operatorname{Spec}(\tau_{-n})$).

(a) Let $\tau_n : k[T] \to k[T]$ be the k-algebra homomorphism with $T \mapsto T + n$ (where n means $n \cdot 1 \in k$). Then $\tau_n\tau_m = \tau_{n+m}$ and $\tau_0 = \mathrm{id}$, so each τ_n is an automorphism and $n \mapsto \tau_n$ is a homomorphism $\mathbb{Z} \to \operatorname{Aut}_k(k[T])$. It is injective since $\operatorname{char} k = 0$: $\tau_n = \mathrm{id}$ means $T + n = T$, i.e. $n = 0$ in k. So G is identified with a subgroup. It acts on $X = \mathbb{A}^1_k = \operatorname{Spec} k[T]$ by $\varphi_n = \operatorname{Spec}(\tau_{-n})$, which sends a prime $\mathfrak{q}$ to $\tau_n(\mathfrak{q})$.

Let $U \neq \emptyset$ be an open G-stable subset and $F = X \setminus U$, which is closed and G-stable. Since $k[T]$ is a principal ideal domain, a proper closed subset is a finite set of maximal ideals, so F is either X or a finite set of maximal ideals. Suppose $F \neq \emptyset$ is finite and contains (π) with $\pi = T^m + c_{m-1}T^{m-1} + \dots$ monic irreducible, $m \geq 1$. Then $\tau_n((\pi)) = (\pi(T + n))$, and $\pi(T+n) = T^m + (c_{m-1} + nm)T^{m-1} + \dots$ is monic of degree m. Two monic polynomials generating the same ideal are equal, and for $n \neq n'$ the coefficients of T^{m-1} differ, since $m(n - n') \neq 0$ in characteristic 0. So the orbit of (π) is infinite, contradicting finiteness of F and stability. Hence $F = \emptyset$ or $F = X$ (which is impossible for $U \neq \emptyset$). So $U = X$. The only stable open subsets are $\emptyset$ and X.

(b) Let $\pi : X \to \operatorname{Spec} k$ be the structural morphism. Since each τ_n is a k-algebra map, $\pi\varphi_n = \pi$.

Existence. Let $f : X \to Z$ be a morphism of schemes with $f\varphi_n = f$ for all n. Pick $x \in X$ and an affine open $Z_0 = \operatorname{Spec} C$ of Z containing $f(x)$. Then $f^{-1}(Z_0)$ is a nonempty open set, stable under G because $f\varphi_n = f$. By (a), $f^{-1}(Z_0) = X$, so f is a morphism $X \to Z_0$. Let $\psi = \rho(f) : C \to k[T]$ (Prop. 3.25). By (3.4), $\rho(f\varphi_n) = \tau_{-n} \circ \psi = \psi$ for all n. So the image of ψ lies in $k[T]^G = \{P : P(T+n) = P(T)\ \forall n \in \mathbb{Z}\}$. For such P, the polynomial $P(T) - P(0)$ vanishes at $T = 0, 1, 2, \dots$, which are pairwise distinct elements of k as $\operatorname{char} k = 0$. So $P = P(0) \in k$ is constant, and $k[T]^G = k$. Hence ψ factors as $C \xrightarrow{\psi_0} k \hookrightarrow k[T]$. Let $\tilde{f} : \operatorname{Spec} k \to Z_0 \subseteq Z$ be the morphism with $\rho(\tilde{f}) = \psi_0$. Then $\rho(\tilde{f} \circ \pi) = \psi = \rho(f)$, so $\tilde{f} \circ \pi = f$ by injectivity of ρ.

Uniqueness. Let $\tilde{f}, \tilde{f}' : \operatorname{Spec} k \to Z$ satisfy $\tilde{f}\pi = \tilde{f}'\pi$. Since π is surjective, the images of the single point of $\operatorname{Spec} k$ coincide; choose an affine open $Z_0 = \operatorname{Spec} C$ containing it. Both morphisms land in Z_0, and $\rho(\tilde{f}\pi) = \rho(\tilde{f})$ followed by $k \hookrightarrow k[T]$ by (3.4). Since $k \to k[T]$ is injective, $\rho(\tilde{f}) = \rho(\tilde{f}')$, so $\tilde{f} = \tilde{f}'$ by Prop. 3.25.

So $(\operatorname{Spec} k, \pi)$ is the quotient scheme $\mathbb{A}^1_k/G$.

(c) We show that $\pi : X \to \operatorname{Spec} k$ does not have the universal property among ringed topological spaces. Let ξ be the generic point of X and let $Z = \{\eta, c\}$ be a two-point space with the indiscrete topology (the only open sets are $\emptyset$ and Z), with sheaf $\mathcal{O}_Z(Z) = k$, $\mathcal{O}_Z(\emptyset) = 0$. This is a sheaf, and both stalks equal k, a local ring, so Z is a ringed topological space. Define $f : X \to Z$ by $f(\xi) = \eta$ and $f(x) = c$ for every closed point x, and $f^{\#}(Z) : k \to \mathcal{O}_X(X) = k[T]$ the inclusion. Then f is continuous (the inverse images of $\emptyset$ and Z are open), $f^{\#}$ is a morphism of sheaves of rings, and every stalk map $k \to \mathcal{O}_{X,x}$ is local (the maximal ideal of k is 0). Also $f \circ \varphi_n = f$: the maps φ_n fix ξ and permute closed points, and $\varphi_n^{\#} \circ f^{\#} = f^{\#}$ because τ_n is the identity on k. If $f = \tilde{f} \circ \pi$ for some map $\tilde{f}$ from the one-point space $\operatorname{Spec} k$, then f would be constant, but it takes the two values η and c. So no factorization exists, and $\operatorname{Spec} k$ is not the quotient in the category of ringed topological spaces. (Any other morphism $p' : X \to \operatorname{Spec} k$ with $p'\varphi_n = p'$ is π composed with an automorphism of $\operatorname{Spec} k$, since $\rho(p') : k \to k[T]$ lands in $k[T]^G = k$; so the same argument applies to it.)

Remark. By Ex. 2.14 the quotient ringed space exists: it is the set X/G with the quotient topology and $\mathcal{O}(V) = \mathcal{O}_X(q^{-1}V)^G$. By (a) its only open sets are $\emptyset$ and the whole space, and its ring of global sections is $k[T]^G = k$. So it is the indiscrete space on the infinite set X/G (it contains the point ξ and the G-orbits of closed points) with global sections k. This has many points and is not $\operatorname{Spec} k$, and f above factors through it by $X \to X/G \to Z$. □

2.4 Reduced schemes and integral schemes

Conventions for this section. Schemes denoted X, Y are non-empty whenever we speak of connectedness, irreducibility or integrality. For a scheme X, a point $x \in X$ and $s \in \mathcal{O}_X(U)$ with $x \in U$, s_x denotes the germ; $\mathfrak{m}_x$ is the maximal ideal of $\mathcal{O}_{X,x}$. For an irreducible space, a *generic point* is a point ξ with $\overline{\{\xi\}} = X$; it is unique when it exists on a scheme, because the underlying space of a scheme is T_0 (Ex. 4.7 below). We say that y *specializes to* x (or x is a specialization of y) if $x \in \overline{\{y\}}$.

Exercise 4.1

> **Problem.** Let k be a field and $P \in k[T_1, \ldots, T_n]$. Show that $\operatorname{Spec}(k[T_1, \ldots, T_n]/(P))$ is reduced (resp. irreducible; resp. integral) if and only if P has no square factor (resp. admits only one irreducible factor; resp. is irreducible).

Solution. *Remark on the hypotheses.* The statement is meant for P non-zero and non-constant. If $P = 0$ the scheme is $\mathbb{A}^n_k$, which is integral although 0 is not irreducible, and if P is a unit the scheme is empty. So we assume $P \neq 0$ and P not a unit. "Only one irreducible factor" means: $P = uQ^e$ with u a unit, Q irreducible, $e \geq 1$.

Let $R = k[T_1, \ldots, T_n]$ (a UFD), $A = R/(P)$ and $X = \operatorname{Spec} A$. Write $P = u\, P_1^{e_1} \cdots P_r^{e_r}$ with $r \geq 1$, P_i pairwise non-associate irreducible (hence prime) elements, $e_i \geq 1$.

Radical of (P). We claim $\sqrt{(P)} = (P_1 \cdots P_r)$. If $Q^m \in (P)$ then each P_i divides Q^m, hence Q (P_i prime); since the P_i are pairwise non-associate primes in a UFD, $P_1 \cdots P_r \mid Q$. Conversely $(P_1 \cdots P_r)^m$ is divisible by P for $m = \max e_i$. Thus the nilradical of A is $\sqrt{(P)}/(P) = (P_1 \cdots P_r)/(P)$.

Reduced. By Prop. 4.2(a),(b), the affine scheme X is reduced iff A is reduced, iff $\sqrt{(P)} = (P)$, iff $P \mid P_1 \cdots P_r$, iff all $e_i = 1$, i.e. P has no square factor.

Irreducible. By Prop. 4.7(a),(c) applied to $\operatorname{Spec} A$ and the ideal 0, X is irreducible iff the nilradical of A is prime. Under the correspondence between ideals of A and ideals of R containing (P), this holds iff $\sqrt{(P)} = (P_1 \cdots P_r)$ is a prime ideal of R. If $r = 1$ it is (P_1), which is prime. If $r \geq 2$, put $a = P_1$ and $b = P_2 \cdots P_r$. Then $ab \in (P_1 \cdots P_r)$, but $P_1 \cdots P_r \nmid a$ (as $P_2 \nmid P_1$) and $P_1 \cdots P_r \nmid b$ (as $P_1 \nmid b$), so the ideal is not prime. Hence X is irreducible iff $r = 1$, i.e. P has only one irreducible factor.

Integral. X is integral iff A is an integral domain (remark after Def. 4.16), iff (P) is a prime ideal of R, iff P is a prime element, iff P is irreducible (R is a UFD). This is consistent with the two previous cases: $r = 1$ and $e_1 = 1$. □

Exercise 4.2

> **Problem.** Let X be a scheme and $x \in X$. Show that the image of the morphism $\operatorname{Spec} \mathcal{O}_{X,x} \to X$ is the set of points of X that specialize to x.

Solution. Recall (Example 3.16) that the morphism is the composition $\operatorname{Spec} \mathcal{O}_{X,x} \to U = \operatorname{Spec} A \hookrightarrow X$ for any affine open $U \ni x$, where $A = \mathcal{O}_X(U)$, the first map being induced by $A \to \mathcal{O}_{X,x} = A_{\mathfrak{p}}$ and $\mathfrak{p}$ the prime ideal corresponding to x. We compute the image in U.

Image of $\operatorname{Spec} A_{\mathfrak{p}} \to \operatorname{Spec} A$. The primes of $A_{\mathfrak{p}}$ are the $\mathfrak{q}A_{\mathfrak{p}}$ with $\mathfrak{q}$ a prime of A contained in $\mathfrak{p}$, and the contraction of $\mathfrak{q}A_{\mathfrak{p}}$ is $\mathfrak{q}$. So the image is $\{\mathfrak{q} \in \operatorname{Spec} A : \mathfrak{q} \subseteq \mathfrak{p}\}$.

Points specializing to x. Let $y \in X$ with $x \in \overline{\{y\}}$. Since U is an open neighbourhood of x and x lies in the closure of $\{y\}$, U meets $\{y\}$, so $y \in U$. Let y correspond to the prime $\mathfrak{q}$ of A. The closure of $\{y\}$ in U is $\overline{\{y\}} \cap U = V(\mathfrak{q})$ (Lemma 4.11(a)), so $x \in \overline{\{y\}}$ iff $\mathfrak{p} \in V(\mathfrak{q})$ iff $\mathfrak{q} \subseteq \mathfrak{p}$ (Lemma 4.11(b)). Conversely a point in the image lies in U and corresponds to some $\mathfrak{q} \subseteq \mathfrak{p}$, hence specializes to x. Therefore the image of $\operatorname{Spec} \mathcal{O}_{X,x} \to X$ is exactly $\{y \in X : x \in \overline{\{y\}}\}$. □

Exercise 4.3

> **Problem.** Let $\mathcal{O}_K$ be a discrete valuation ring with field of fractions K and uniformizing parameter t (a generator of the maximal ideal). Show that $\operatorname{Spec} K[T]$ can be identified with an open subscheme of $\operatorname{Spec} \mathcal{O}_K[T]$. Determine the set of closed points of $\operatorname{Spec} K[T]$ which specialize to the point corresponding to the maximal ideal (T, t) of $\mathcal{O}_K[T]$.

Solution. Let v be the normalized valuation of $\mathcal{O}_K$, $k = \mathcal{O}_K/t\mathcal{O}_K$ the residue field, $B = \mathcal{O}_K[T]$.

Open subscheme. Every non-zero element of $\mathcal{O}_K$ is ut^m with u a unit and $m \geq 0$, hence $K = \mathcal{O}_K[1/t] = (\mathcal{O}_K)_t$ and $K[T] = B_t$. By Example 3.15 (and Lemma 3.7), $\operatorname{Spec} K[T] = \operatorname{Spec} B_t \to \operatorname{Spec} B$ is an isomorphism onto the principal open subscheme $D(t)$. Its points are the primes of B not containing t, corresponding to the primes of $K[T]$ by $\mathfrak{q} \mapsto \mathfrak{q} \cap B$.

Gauss lemma. Call $G \in B$ *primitive* if its image in $k[T]$ is non-zero, i.e. some coefficient is a unit. Since $k[T]$ is a domain, a product of primitive polynomials is primitive. Consequently, if $F' \in B$ is primitive, $H \in K[T]$ and $F'H \in B$, then $H \in B$: write $H = t^{-r}H_0$ with $r \in \mathbf{Z}$ and

$H_0 \in B$ primitive; then $F'H_0$ is primitive and $t^{-r}F'H_0 \in B$ forces $r \leq 0$. Hence for primitive F':

$$F'K[T] \cap B = F'B.$$

Closed points. The closed points of $\operatorname{Spec} K[T]$ are the maximal ideals (F) with F monic irreducible in $K[T]$ ($K[T]$ is a PID; closed points of an affine scheme are the maximal ideals). Write $F = T^d + a_{d-1}T^{d-1} + \cdots + a_0$, $a_d := 1$. Let $s = -\min_{0\leq i\leq d} v(a_i) \geq 0$ (the minimum is ≤ 0 because $a_d = 1$), and $F' = t^s F \in B$. Then F' is primitive (some coefficient has valuation 0 by choice of s). The prime of B corresponding to (F) is $\mathfrak{q}_F = FK[T] \cap B = F'K[T] \cap B = F'B$.

Specialization. A point y of $D(t)$ with prime $\mathfrak{q}$ specializes to the point $\mathfrak{m} = (T, t)$ of $\operatorname{Spec} B$ iff $\mathfrak{m} \in \overline{\{y\}} = V(\mathfrak{q})$, i.e. iff $\mathfrak{q} \subseteq \mathfrak{m}$ (Lemma 4.11). Now $\mathfrak{m} = \{G \in B : G(0) \in t\mathcal{O}_K\}$ (the kernel of $B \to k$, $T \mapsto 0$). So $\mathfrak{q}_F \subseteq \mathfrak{m}$ iff $F' \in \mathfrak{m}$ iff $v(t^s a_0) \geq 1$ iff $v(a_0) > -s = \min_i v(a_i)$.

Answer. The closed points of $\operatorname{Spec} K[T]$ specializing to (T, t) are exactly the maximal ideals (F) with $F = T^d + a_{d-1}T^{d-1} + \cdots + a_0 \in K[T]$ monic irreducible such that the constant term does not realize the minimal valuation of the coefficients:

$$v(a_0) > \min_{0\leq i\leq d} v(a_i) \qquad (\text{with } v(0) = +\infty,\ a_d = 1).$$

Equivalently, the primitive multiple F' of F in $\mathcal{O}_K[T]$ has constant term in $t\mathcal{O}_K$. Examples: $F = T$; $F = T - a$ with $a \in t\mathcal{O}_K$ (here $\min(v(a), 0) = 0 < v(a)$ for $v(a) \geq 1$ and the condition says exactly $v(a) \geq 1$); every monic irreducible $F \in \mathcal{O}_K[T]$ with $F(0) \in t\mathcal{O}_K$; and also some F with non-integral coefficients, e.g. for $\mathcal{O}_K = \mathbf{Z}_{(p)}$ and $F = T^2 + p^{-1}T + 1$ (irreducible over $\mathbf{Q}$ since its discriminant $p^{-2} - 4 = (1 - 4p^2)/p^2$ is negative, hence not a square), $F' = pT^2 + T + p \in (T, p)$. □

Exercise 4.4

> **Problem.** We say that a scheme X is connected if the underlying topological space of X is connected. For example, an irreducible scheme is connected. Let X be a scheme having only a finite number of irreducible components $\{X_i\}_i$. Show that X is connected if and only if for any pair i, j, there exist indices $i_0 = i, i_1, \ldots, i_r = j$ such that $X_{i_l} \cap X_{i_{l+1}} \neq \emptyset$ for every $l < r$. Also show that X is integral if and only if X is connected, and integral at every point $x \in X$.

Solution. Throughout $X \neq \emptyset$. Every irreducible space is connected: if $Z = C_1 \sqcup C_2$ with C_i closed then $Z = C_1$ or $Z = C_2$. The X_i are closed and cover X (they are the irreducible components).

First equivalence. Say $i \sim j$ if a chain as in the statement exists; this is an equivalence relation. Let $\mathcal{C}$ be the set of classes, and for $C \in \mathcal{C}$ put $Y_C = \bigcup_{i\in C} X_i$, a finite union of closed sets, hence closed. If $C \neq C'$ then $Y_C \cap Y_{C'} = \emptyset$ (a point in $X_i \cap X_j$ with $i \in C$, $j \in C'$ would give $i \sim j$). So $X = \bigsqcup_C Y_C$ is a finite disjoint union of closed sets, and each Y_C is also open (its complement is the finite union of the other $Y_{C'}$). Moreover each Y_C is connected: it is the union of connected sets X_i ($i \in C$), any two of which are linked by a chain of pairwise meeting ones, and a union of connected sets, two of which meet, is connected. In summary, *the Y_C are exactly the connected components of X; they are open and closed, and finite in number* (we call this fact $(*)$). Thus

X is connected iff there is only one class, i.e. iff any i, j are linked by a chain. (If X is connected and $X = \bigsqcup Y_C$ with clopen non-empty pieces, there is a single piece. Conversely, if there is one class, $X = Y_C$ is connected.)

Second equivalence. ($\Rightarrow$) An integral scheme is irreducible, hence connected. It is integral at each point: pick an affine open $U \ni x$; $\mathcal{O}_X(U)$ is an integral domain by Prop. 4.17, and $\mathcal{O}_{X,x}$ is a localization of it, hence a domain.

($\Leftarrow$) Assume X is connected and every $\mathcal{O}_{X,x}$ is a domain. Then X is reduced. For $x \in X$, $\operatorname{Spec} \mathcal{O}_{X,x}$ is irreducible (Prop. 4.7(c), a domain has a unique minimal prime (0)), so by Prop. 4.12(b) exactly one irreducible component of X passes through x. Hence distinct irreducible components of X are disjoint. Now let i, j be arbitrary; by the first part there is a chain $i = i_0, \ldots, i_r = j$ with $X_{i_l} \cap X_{i_{l+1}} \neq \emptyset$; disjointness forces $X_{i_l} = X_{i_{l+1}}$ for all l, so $X_i = X_j$. Thus X has a single irreducible component, namely X itself, so X is irreducible. Being reduced and irreducible, X is integral. □

Exercise 4.5

> **Problem.** Let X be a scheme. Show that every irreducible component of X is contained in a connected component (in the topological sense). Show that if X is locally Noetherian, then the connected components are open. If X is Noetherian, then there are only finitely many connected components.

Solution. *Irreducible components.* Let Z be an irreducible component, $x \in Z$, and C the connected component of x, the union of all connected subsets containing x. Z is irreducible, hence connected (Ex. 4.4), and contains x, so $Z \subseteq C$.

Locally Noetherian case. Let C be a connected component of X and $x \in C$. Choose an affine open $U = \operatorname{Spec} A \ni x$ with A Noetherian (definition of locally Noetherian). The ring A is Noetherian, so U has finitely many irreducible components (Prop. 4.9, for the Noetherian scheme U). By the argument of ($*$) in Ex. 4.4 (the classes of components under the chain relation give a partition of U into finitely many clopen connected sets), U is a finite disjoint union of connected clopen subsets of U. Let W be the one containing x; W is connected and open in U, hence open in X. As W is a connected set containing x, $W \subseteq C$. So every point of C has an open neighbourhood contained in C, and C is open.

Noetherian case. A Noetherian scheme is quasi-compact and locally Noetherian. The connected components are open by the above, pairwise disjoint and cover X. By quasi-compactness finitely many of them cover X; since they are disjoint and non-empty, no other component can exist. So there are finitely many connected components. □

Exercise 4.6

> **Problem.** Let A be a ring. We say that $e \in A$ is idempotent if $e \neq 0$ and $e^2 = e$. We will say that e is indecomposable if it cannot be written as a sum of two idempotent elements. Let X be a scheme.
>
> (a) Show the equivalence of the following properties: (i) X is connected; (ii) $\mathcal{O}_X(X)$ has no other idempotent elements than 1; (iii) $\operatorname{Spec} \mathcal{O}_X(X)$ is connected.

(b) Show that any local scheme (i.e., the spectrum of a local ring) is connected.

(c) Let us suppose that the connected components of X are open (e.g., X locally Noetherian). Let U be a connected component of X. Show that there exists a unique idempotent element e of $\mathcal{O}_X(X)$ such that $e|_U = 1$ and $e|_{X\setminus U} = 0$. Show that this establishes a canonical bijection from the set of connected components of X onto the set of indecomposable idempotent elements of $\mathcal{O}_X(X)$, the converse map being given by $e \mapsto V(1-e)$.

Solution. *Correction about "indecomposable".* Read literally (any two idempotents e_1, e_2 with $e = e_1 + e_2$), no element is indecomposable in a ring of characteristic 2: in $\mathbf{F}_2 \times \mathbf{F}_2$ we have $(1,0) = (1,1) + (0,1)$. The intended notion, which we use, is: e is *indecomposable* if it is not a sum $e = e_1 + e_2$ of two idempotents with $e_1 e_2 = 0$ (orthogonal idempotents). When 2 is invertible the orthogonality is automatic: $e_1 + e_2$ idempotent gives $2e_1e_2 = 0$. Also $V(f)$ for a global section f means $\{x \in X : f_x \in \mathfrak{m}_x\}$. We take $X \neq \emptyset$ in (a), and note that the zero ring has no idempotent in the sense above.

Preliminary: idempotents are locally 0 *or* 1. If R is a local ring and $\varepsilon^2 = \varepsilon$, then $\varepsilon \in \{0,1\}$: if $\varepsilon \notin \mathfrak{m}$ it is a unit and $\varepsilon(\varepsilon - 1) = 0$ gives $\varepsilon = 1$; if $\varepsilon \in \mathfrak{m}$ then $1 - \varepsilon$ is a unit and $\varepsilon(1-\varepsilon) = 0$ gives $\varepsilon = 0$. Now let $e \in \mathcal{O}_X(X)$ with $e^2 = e$ (we also allow $e = 0$ here). Then $e_x \in \{0,1\}$ for all x. Put $U_e = \{x : e_x = 1\} = V(1-e)$ (as $(1-e)_x \in \mathfrak{m}_x$ iff $e_x = 1$) and $V_e = \{x : e_x = 0\}$. Then $X = U_e \sqcup V_e$. If $e_x = 1$ then $e = 1$ on a neighbourhood of x (a germ equality means equality of sections on a neighbourhood), so U_e is open; likewise V_e is open. So U_e is open and closed. Moreover $U_e = \emptyset$ iff $e = 0$, and $U_e = X$ iff $e = 1$ (the sheaf property). Conversely for a clopen $W \subseteq X$ the section 1_W equal to 1 on W and 0 on $X \setminus W$ (glued on the open cover $\{W, X \setminus W\}$) is an idempotent or zero, with $U_{1_W} = W$. So

$$e \longmapsto U_e = V(1-e)$$

is a bijection from $\{e \in \mathcal{O}_X(X) : e^2 = e\}$ onto the set of clopen subsets of X, with inverse $W \mapsto 1_W$. Under it $e_1e_2 = 0$ and $e = e_1 + e_2$ correspond to $U_{e_1} \cap U_{e_2} = \emptyset$ and $U_e = U_{e_1} \sqcup U_{e_2}$ (stalkwise, $e_{1,x}e_{2,x} = 0$ means not both are 1, and then the sum is the indicator of the union; conversely the indicator functions of a disjoint decomposition are orthogonal idempotents summing to 1_W).

(a) (i)$\Leftrightarrow$(ii): By the bijection, $\mathcal{O}_X(X)$ has an idempotent $e \neq 0, 1$ iff X has a clopen subset different from $\emptyset$ and X, iff X is disconnected. For a non-empty X the idempotent 1 always exists. (ii)$\Leftrightarrow$(iii): apply the equivalence (i)$\Leftrightarrow$(ii), which holds for every non-empty scheme, to the affine scheme $\operatorname{Spec} \mathcal{O}_X(X)$, whose ring of global sections is $\mathcal{O}_X(X)$ itself.

(b) Let R be local. By the preliminary, R (with $X = \operatorname{Spec} R$, $\mathcal{O}_X(X) = R$) has no idempotents other than 1 (its only idempotent in the sense of the exercise), so $\operatorname{Spec} R$ is connected by (a). (Alternatively: the closed point $\mathfrak{m}$ lies in every non-empty closed set $V(I)$, so two disjoint non-empty closed sets cannot exist.)

(c) Let U be a connected component; it is closed, and by assumption open, so $X \setminus U$ is open too. Put $e = 1_U$. Then $e^2 = e$ and $e \neq 0$, and the section e is unique with $e|_U = 1$, $e|_{X\setminus U} = 0$ because $\mathcal{O}_X$ is a sheaf and $\{U, X \setminus U\}$ covers X. We have $U_e = U = V(1-e)$.

Bijection. By the bijection above, indecomposable idempotents correspond to non-empty clopen subsets $W \subseteq X$ admitting no decomposition $W = W_1 \sqcup W_2$ into two non-empty clopen subsets of X, i.e. to non-empty clopen subsets W that are connected (a subset of W is clopen in X iff it is clopen in W, since W is clopen). So we must show: the connected components of X are

exactly the non-empty connected clopen subsets. Each component U is clopen (by assumption) and connected. Conversely, if W is a non-empty connected clopen subset, W lies in a connected component C and W is clopen in the connected C, so $W = C$. Therefore $U \mapsto e_U = 1_U$ is a bijection from connected components onto indecomposable idempotents, whose inverse is $e \mapsto U_e = V(1-e)$. □

Exercise 4.7

> **Problem.** We say that a topological space X verifies the separation axiom T_0 if for every pair of points $x \neq y$, there exists an open subset which contains one of the points and not the other one. Show that the underlying topological space of a scheme verifies T_0.

Solution. Let X be a scheme and $x \neq y$ in X. Take an affine open $U = \operatorname{Spec} A$ containing x. If $y \notin U$, then U is open, contains x and not y. If $y \in U$, let x, y correspond to distinct primes $\mathfrak{p} \neq \mathfrak{q}$ of A. Then $\mathfrak{p} \not\subseteq \mathfrak{q}$ or $\mathfrak{q} \not\subseteq \mathfrak{p}$; say $\mathfrak{p} \not\subseteq \mathfrak{q}$, and pick $a \in \mathfrak{p} \setminus \mathfrak{q}$. The principal open set $D(a)$ contains $\mathfrak{q}$ (i.e. y) and not $\mathfrak{p}$ (i.e. x). It is open in U, hence in X. In the other case, symmetrically, there is an open set containing x and not y. □

Exercise 4.8

> **Problem.** Let X be a quasi-compact scheme (Exercise 3.14). Show that X contains a closed point. See also Exercise 3.3.26 for a counterexample when X is not quasi-compact.

Solution. Assume $X \neq \emptyset$ (otherwise there is nothing to prove, or the statement is meant for non-empty X).

Step 1: a minimal non-empty closed subset exists. Order the non-empty closed subsets of X by inclusion. Let (Z_α) be a chain. If $\bigcap Z_\alpha = \emptyset$, the complements $X \setminus Z_\alpha$ form an open cover of X; by quasi-compactness, finitely many cover X, and since the chain is totally ordered the largest of these complements is X, so the corresponding $Z_\alpha = \emptyset$, a contradiction. So $\bigcap Z_\alpha$ is a non-empty closed lower bound. By Zorn's lemma there is a minimal non-empty closed subset Z.

Step 2: $\overline{\{z\}} = Z$ for every $z \in Z$. Let $z \in Z$ and choose an affine open $V = \operatorname{Spec} B \ni z$. Then $Z \cap V$ is a non-empty closed subset of V, so it is $V(I)$ with $I \neq B$, and $V(I)$ contains a maximal ideal $\mathfrak{m}$ of B; call the corresponding point w. The point w is closed in V, and $w \in Z$. Then $\overline{\{w\}} \subseteq Z$ is closed and non-empty, hence equals Z by minimality. Therefore $Z \cap V = \overline{\{w\}} \cap V$ is the closure of $\{w\}$ in V, which is $\{w\}$. Since $z \in Z \cap V$, $z = w$, and so $\overline{\{z\}} = Z$.

Step 3: Z is one point. For $z, z' \in Z$, Step 2 gives $\overline{\{z\}} = Z = \overline{\{z'\}}$. An open set containing z must meet $\{z'\}$ (because $z \in \overline{\{z'\}}$), so contains z', and symmetrically. So z, z' have the same open neighbourhoods, and $z = z'$ by Ex. 4.7. Thus $Z = \{z\}$ is a closed point of X. □

Exercise 4.9

> **Problem.** Let X be a Noetherian scheme. Show that the set of points $x \in X$ such that $\mathcal{O}_{X,x}$ is reduced (resp. is an integral domain) is open.

Solution. Both properties of x depend only on $\mathcal{O}_{X,x} = \mathcal{O}_{U,x}$ for any open $U \ni x$, and openness of a subset is local. X is covered by affine opens $\operatorname{Spec} A$ with A Noetherian, so it suffices to treat $X = \operatorname{Spec} A$, A Noetherian.

Reduced locus. Let N be the nilradical of A. For a prime $\mathfrak{p}$, the nilradical of $A_\mathfrak{p}$ is $N_\mathfrak{p}$: if a/s is nilpotent then $ta^m = 0$ for some m and $t \notin \mathfrak{p}$, so $(ta)^m = 0$, $ta \in N$ and $a/s = ta/ts \in N_\mathfrak{p}$; the other inclusion is clear. As A is Noetherian, $N = (n_1, \dots, n_m)$. Then $A_\mathfrak{p}$ is reduced iff $n_i/1 = 0$ in $A_\mathfrak{p}$ for all i iff $\operatorname{Ann}(n_i) \not\subseteq \mathfrak{p}$ for all i. So the non-reduced locus is $\bigcup_{i=1}^m V(\operatorname{Ann}(n_i))$, a finite union of closed sets, so closed. The reduced locus is open.

Domain locus. Since A is Noetherian it has finitely many minimal primes $\mathfrak{p}_1, \dots, \mathfrak{p}_s$ (Prop. 4.9 and Prop. 4.7(b)). The minimal primes of $A_\mathfrak{p}$ are the $\mathfrak{q}A_\mathfrak{p}$ with $\mathfrak{q} \subseteq \mathfrak{p}$ minimal among primes of A contained in $\mathfrak{p}$, that is (any prime inside $\mathfrak{q}$ is inside $\mathfrak{p}$) exactly the minimal primes $\mathfrak{p}_i$ of A contained in $\mathfrak{p}$. A local ring is a domain iff it is reduced and has exactly one minimal prime (if reduced, 0 is the intersection of its minimal primes). Hence $A_\mathfrak{p}$ is a domain iff $\mathfrak{p} \notin$ non-reduced locus and $\mathfrak{p}$ contains exactly one $\mathfrak{p}_i$, i.e. $\mathfrak{p} \notin \bigcup_{i \neq j} V(\mathfrak{p}_i) \cap V(\mathfrak{p}_j)$. The set of domain points is thus the complement of the closed set

$$\bigcup_{i=1}^m V(\operatorname{Ann}(n_i)) \;\cup\; \bigcup_{i \neq j} \big(V(\mathfrak{p}_i) \cap V(\mathfrak{p}_j)\big),$$

so it is open. □

Exercise 4.10

> **Problem.** Let $f : X \to \operatorname{Spec} A$ be a quasi-compact morphism (Exercise 3.17). Let I be an ideal of A. Show that $f(X) \subseteq V(I)$ if and only if $f^\#(\operatorname{Spec} A)(I) \subseteq \mathcal{O}_X(X)$ is nilpotent.

Solution. *Interpretation.* We write $\varphi = f^\#(\operatorname{Spec} A) : A \to \mathcal{O}_X(X)$ and read "$\varphi(I)$ is nilpotent" as: every element of $\varphi(I)$ is nilpotent. If I is finitely generated this is the same as nilpotence of the ideal $\varphi(I)\mathcal{O}_X(X)$ (a finitely generated ideal of nilpotent elements is nilpotent). For general I the stronger reading (the ideal is nilpotent) is false: let $A = \mathcal{O}_X(X) = k[x_1, x_2, \dots]/(x_i^i)$, $X = \operatorname{Spec} A$, $f = \mathrm{id}$, $I = (x_1, x_2, \dots)$. Then $f(X) = V(I)$ since I is in the nilradical, but $I^N \ni x_{N+1}^N \neq 0$ for each N.

Key formula. Let $x \in X$, $f(x) = \mathfrak{p}$. The local homomorphism $f_x^\# : A_\mathfrak{p} = \mathcal{O}_{\operatorname{Spec} A, \mathfrak{p}} \to \mathcal{O}_{X,x}$ satisfies $(f_x^\#)^{-1}(\mathfrak{m}_x) = \mathfrak{p}A_\mathfrak{p}$. For $a \in A$, the image of a in $\mathcal{O}_{X,x}$ is $\varphi(a)_x$. Hence

$$a \in \mathfrak{p} \iff a/1 \in \mathfrak{p}A_\mathfrak{p} \iff \varphi(a)_x \in \mathfrak{m}_x.$$

($\Leftarrow$) Suppose each $\varphi(a)$, $a \in I$, is nilpotent. For $x \in X$, $\varphi(a)_x$ is nilpotent, hence in $\mathfrak{m}_x$, so $a \in f(x)$ for all $a \in I$, i.e. $f(x) \in V(I)$. (Quasi-compactness is not needed here.)

($\Rightarrow$) Suppose $f(X) \subseteq V(I)$ and let $a \in I$. For every $x \in X$, $a \in f(x)$, so $\varphi(a)_x \in \mathfrak{m}_x$. Since $\operatorname{Spec} A$ is affine and f is quasi-compact, $X = f^{-1}(\operatorname{Spec} A)$ is quasi-compact (Ex. 3.14), hence a finite union of affine opens $U_j = \operatorname{Spec} B_j$, $1 \le j \le m$. Let $b_j = \varphi(a)|_{U_j} \in B_j$. For every prime $\mathfrak{q}$ of B_j, corresponding to a point $x \in U_j$, we have $b_j/1 \in \mathfrak{q}(B_j)_\mathfrak{q} = \mathfrak{m}_x$, hence $b_j \in \mathfrak{q}$. So b_j lies in all primes of B_j, hence is nilpotent (the nilradical is the intersection of all primes): $b_j^{n_j} = 0$.

With $n = \max n_j$, $\varphi(a)^n$ restricts to 0 on each U_j, so $\varphi(a)^n = 0$. This holds for each $a \in I$ (with n depending on a), proving the claim. If $I = (a_1, \ldots, a_r)$ is finitely generated, the ideal generated by the nilpotent elements $\varphi(a_i)$ is nilpotent. □

Exercise 4.11

> **Problem.** Let $f : X \to Y$ be a morphism of irreducible schemes with respective generic points ξ_X, ξ_Y. We say that f is dominant if $f(X)$ is dense in Y. Let us suppose that X, Y are integral. Show that the following properties are equivalent: (i) f is dominant; (ii) $f^\# : \mathcal{O}_Y \to f_*\mathcal{O}_X$ is injective; (iii) for every open subset V of Y and every open subset $U \subseteq f^{-1}(V)$, the map $\mathcal{O}_Y(V) \to \mathcal{O}_X(U)$ is injective; (iv) $f(\xi_X) = \xi_Y$; (v) $\xi_Y \in f(X)$.

Solution. *Correction in (iii).* For $U = \emptyset$ we have $\mathcal{O}_X(U) = 0$, so (iii) as printed fails trivially for $V \neq \emptyset$. We therefore read (iii) with U and V non-empty (then $V \neq \emptyset$ is automatic). Note that for non-empty open V of Y we have $\xi_Y \in V$ (since $\overline{\{\xi_Y\}} = Y$), and $\mathcal{O}_Y(V)$ is an integral domain, in particular non-zero.

We prove (i)⇔(iv)⇔(v), (iv)⇒(ii)⇒(i) and (iv)⇒(iii)⇒(v).

(iv)⇒(v): trivial. *(v)⇒(i)*: $f(X) \ni \xi_Y$ and $\overline{\{\xi_Y\}} = Y$.

(i)⇒(iv): By continuity, $f(X) = f(\overline{\{\xi_X\}}) \subseteq \overline{\{f(\xi_X)\}}$. Since $f(X)$ is dense, $Y \subseteq \overline{\{f(\xi_X)\}}$, so $f(\xi_X)$ is a generic point of Y. Two generic points η, η' of Y have the same open neighbourhoods (as above), hence are equal by Ex. 4.7. So $f(\xi_X) = \xi_Y$.

(iv)⇒(iii) and (ii). Let $V \subseteq Y$ be a non-empty open subset and $U \subseteq f^{-1}(V)$ a non-empty open subset. Since $U \neq \emptyset$ we have $\xi_X \in U$, and $f(\xi_X) = \xi_Y$. Consider the maps

$$\mathcal{O}_Y(V) \xrightarrow{f^\#(V)} \mathcal{O}_X(f^{-1}(V)) \to \mathcal{O}_X(U) \to \mathcal{O}_{X,\xi_X},$$

whose composition equals $\mathcal{O}_Y(V) \to \mathcal{O}_{Y,\xi_Y} \xrightarrow{f^\#_{\xi_X}} \mathcal{O}_{X,\xi_X}$ by the definition of the stalk map. The first map $\mathcal{O}_Y(V) \to \mathcal{O}_{Y,\xi_Y} = K(Y)$ is injective by Prop. 4.18(b), and $f^\#_{\xi_X} : K(Y) \to K(X)$ is a ring homomorphism between fields, hence injective. So the composition is injective, and so is its first factor $\mathcal{O}_Y(V) \to \mathcal{O}_X(U)$. This is (iii). For (ii): let V be any open subset of Y. If $V = \emptyset$ there is nothing to show. If $V \neq \emptyset$ then $\xi_Y \in V$, so $\xi_X \in f^{-1}(V)$ and $U = f^{-1}(V)$ is non-empty; by the above $\mathcal{O}_Y(V) \to \mathcal{O}_X(f^{-1}(V)) = (f_*\mathcal{O}_X)(V)$ is injective. Hence $f^\#$ is injective.

(ii)⇒(i): Suppose $f(X)$ is not dense. Then $V = Y \setminus \overline{f(X)}$ is a non-empty open set with $f^{-1}(V) = \emptyset$, so $(f_*\mathcal{O}_X)(V) = \mathcal{O}_X(\emptyset) = 0$ while $\mathcal{O}_Y(V) \neq 0$. Hence $f^\#(V)$ is not injective, contradicting (ii).

(iii)⇒(v): Let $x \in X$, and choose an affine open $V = \operatorname{Spec} B \ni f(x)$ in Y and an affine open $U = \operatorname{Spec} C \ni x$ of X contained in $f^{-1}(V)$. As X, Y are integral, B and C are integral domains (Prop. 4.17). The morphism $f|_U : U \to V$ is $\operatorname{Spec}\psi$ where $\psi : B \to C$ is the map $\mathcal{O}_Y(V) \to \mathcal{O}_X(U)$ of (iii) (Prop. 3.25), hence injective by assumption. The point (0) of $\operatorname{Spec} C$ maps to $\psi^{-1}(0) = \operatorname{Ker}\psi = (0) \in \operatorname{Spec} B$. The point (0) of $\operatorname{Spec} B$ is the generic point of V; its closure in Y contains V, which is dense in the irreducible Y (Prop. 4.5(a)); hence its closure is Y and the point is ξ_Y (uniqueness of the generic point, as in (i)⇒(iv)). Hence $\xi_Y \in f(U) \subseteq f(X)$. □

Exercise 4.12

> **Problem.** Let B be a graded ring. Let Y be a reduced closed subscheme of $\operatorname{Proj} B$. Show that there exists a homogeneous ideal I of B such that $Y \simeq \operatorname{Proj} B/I$.

Solution. Here $B = \bigoplus_{d\geq 0} B_d$ (as in Subsection 2.3.3) and Y is a closed subscheme with the closed immersion $j : Y \to \operatorname{Proj} B$. We prove more precisely that I can be chosen so that Y and $\operatorname{Proj} B/I$ are isomorphic *as closed subschemes* of $\operatorname{Proj} B$.

Choice of I. Let $Z = j(Y)$, a closed subset of $\operatorname{Proj} B$, so $Z = V_+(J)$ for a homogeneous ideal J. Let $I = \bigcap_{\mathfrak{p}\in Z} \mathfrak{p}$ ($I = B$ if $Z = \emptyset$). It is an intersection of homogeneous ideals, so it is homogeneous, and radical. Moreover $V_+(I) = Z$: every $\mathfrak{p} \in Z$ contains I, so $Z \subseteq V_+(I)$; conversely $I \supseteq J$ (each $\mathfrak{p} \in Z$ contains J), so $V_+(I) \subseteq V_+(J) = Z$.

The closed immersion $\operatorname{Proj} B/I \to \operatorname{Proj} B$. Let $\phi : B \to B/I$ be the quotient map; it is a homomorphism of graded rings with $\phi(B_+)(B/I) = (B/I)_+$. By Lemma 3.40 (with $r = 1$, $M = (B/I)_+$) we get a morphism $g : \operatorname{Proj} B/I \to \operatorname{Proj} B$ with $g^{-1}(D_+(h)) = D_+(\phi(h))$ for homogeneous $h \in B_+$, and on this open set g is Spec of $B_{(h)} \to (B/I)_{(\phi(h))}$. This ring map is surjective: an element of $(B/I)_{(\phi(h))}$ is $\phi(a)/\phi(h)^n$ with a homogeneous of degree $n \deg h$, which is the image of a/h^n. Its kernel is $\{a/h^n : \ h^m a \in I \text{ for some } m\} = IB_h \cap B_{(h)} = I_{(h)}$. The open sets $D_+(h)$ cover $\operatorname{Proj} B$ (Subsection 2.3.3). By Lemma 3.17, on each $D_+(h)$ the morphism is a closed immersion with image $V(I_{(h)})$, where $I_{(h)} = IB_h \cap B_{(h)}$; by Lemma 3.36(d) this is $V_+(I) \cap D_+(h)$ under $D_+(h) \simeq \operatorname{Spec} B_{(h)}$. Being a homeomorphism onto a closed subset, and surjectivity of the map of sheaves, are local on the target; hence g is a closed immersion with image $V_+(I) = Z$.

$\operatorname{Proj} B/I$ *is reduced.* For homogeneous $h \in B_+$ the ring $(B/I)_{(\phi(h))}$ is the degree-0 part of the localization $(B/I)_{\phi(h)}$. The ring B/I is reduced (as I is radical), hence so is its localization, hence so is its subring $(B/I)_{(\phi(h))}$. Since these rings are the rings of sections of the affine opens $D_+(\phi(h))$ covering $\operatorname{Proj} B/I$, the scheme $\operatorname{Proj} B/I$ is reduced (Prop. 4.2(b)).

Conclusion. Y (via j) and $\operatorname{Proj} B/I$ (via g) are two reduced closed subschemes of $\operatorname{Proj} B$ with the same underlying closed subset Z. By the uniqueness in Prop. 4.2(e), they coincide: there is an isomorphism $Y \to \operatorname{Proj} B/I$ compatible with the immersions into $\operatorname{Proj} B$. In particular $Y \simeq \operatorname{Proj} B/I$. □

2.5 Dimension

Throughout, k is a field, rings are commutative with 1, and ht is the height. "Algebraic variety" is as in Def. 3.47, and a "graded ring" is $\mathbb{N}$-graded as in Section 2.3.2. Results of the text are cited by number. Honest notes: Ex. 5.5 is false as printed when the ideal contains non-zero homogeneous elements of degree 0 (a counterexample is given), and it is proved for $I \subseteq A_+$, which is the case needed in practice; in Ex. 5.12(d) the set X must be non-empty (and $n \geq 1$), and the proof uses the separatedness of $\mathbb{P}^n_k$ from Chapter 3 (Prop. 3.3.6, Example 3.3.7); in Ex. 5.15 the ring B is a homogeneous algebra over k (so $B_0 = k$, generated by B_1).

Exercise 5.1

Problem. Let X be a topological space and $\{X_i\}_i$ a covering of X by closed subsets which is locally finite, that is, every $x \in X$ has an open neighborhood U meeting only finitely many X_i. Show that $\dim X = \sup_i \dim X_i$.

Solution. By Prop. 5.5(a), $\dim X_i \leq \dim X$ for all i, so $\sup_i \dim X_i \leq \dim X$.

Conversely, let $Z_0 \subsetneq \cdots \subsetneq Z_n$ be a chain of irreducible closed subsets of X. Pick $x \in Z_0$ (irreducible sets are non-empty) and an open neighborhood U of x meeting only $X_1, \ldots, X_m$ (after renumbering). The set $Z_n \cap U$ is a non-empty open subset of the irreducible space Z_n (it contains x), hence irreducible and dense in Z_n (Prop. 4.5(a)). Since the X_i cover X and only $X_1, \ldots, X_m$ meet U,

$$Z_n \cap U = \bigcup_{i=1}^{m} \big((Z_n \cap U) \cap X_i\big),$$

a finite union of closed subsets of $Z_n \cap U$. By irreducibility $Z_n \cap U \subseteq X_i$ for some i. As X_i is closed and $Z_n \cap U$ is dense in Z_n, we get $Z_n \subseteq X_i$. Then $Z_0 \subsetneq \cdots \subsetneq Z_n$ is a chain of irreducible closed subsets of X_i, so $n \leq \dim X_i$. Hence $n \leq \sup_i \dim X_i$ and, taking the supremum over chains, $\dim X \leq \sup_i \dim X_i$. □

Exercise 5.2

Problem. Let X be a scheme and Z a closed subset of X. Show that for all $x \in X$ we have $\operatorname{codim}(\overline{\{x\}}, X) = \dim \mathcal{O}_{X,x}$, and (for $Z \neq \emptyset$) $\operatorname{codim}(Z, X) = \min_{z \in Z} \dim \mathcal{O}_{X,z}$.

Solution. (The codimension of the point x means that of the irreducible closed set $\overline{\{x\}}$.)

First formula. Choose an affine open $U = \operatorname{Spec} A \ni x$ and let x correspond to the prime $\mathfrak{p}$, so $\mathcal{O}_{X,x} = A_{\mathfrak{p}}$. An irreducible closed subset Z' of X contains $\overline{\{x\}}$ iff $x \in Z'$ (as Z' is closed). For such Z' the set $Z' \cap U$ is a non-empty open subset of Z', hence irreducible, dense in Z', and closed in U. Conversely if W is an irreducible closed subset of U with $x \in W$, its closure $\overline{W}$ in X is irreducible, closed, contains x, and $\overline{W} \cap U = W$. These two operations are mutually inverse and preserve inclusions: $Z' \mapsto Z' \cap U$ and $W \mapsto \overline{W}$ (for the first, $\overline{Z' \cap U} = Z'$ by density). So chains of irreducible closed subsets of X containing $\overline{\{x\}}$ correspond to chains of irreducible closed subsets of U containing x. By Prop. 4.7 (irreducible closed subsets of $\operatorname{Spec} A$ are the $V(\mathfrak{q})$ with $\mathfrak{q}$ prime, and $x \in V(\mathfrak{q})$ iff $\mathfrak{q} \subseteq \mathfrak{p}$) these are the chains of primes contained in $\mathfrak{p}$. Therefore $\operatorname{codim}(\overline{\{x\}}, X) = \operatorname{ht}(\mathfrak{p}) = \dim A_{\mathfrak{p}} = \dim \mathcal{O}_{X,x}$ (Prop. 5.8(b)).

Second formula. Let Z_j $(j \in J)$ be the irreducible components of Z, with generic points z_j (Prop. 4.12(a)), so $Z_j = \overline{\{z_j\}}$. By definition and the first part, $\operatorname{codim}(Z, X) = \inf_j \operatorname{codim}(Z_j, X) = \inf_j \dim \mathcal{O}_{X,z_j}$. In particular $\operatorname{codim}(Z, X) \geq \inf_{z \in Z} \dim \mathcal{O}_{X,z}$. Conversely, let $z \in Z$. The irreducible closed set $\overline{\{z\}} \subseteq Z$ lies in some irreducible component Z_j of Z (every irreducible subset lies in a maximal one, by Zorn's lemma). Every chain of irreducible closed subsets of X starting with a set containing Z_j also starts with a set containing $\overline{\{z\}}$, so $\operatorname{codim}(Z_j, X) \leq \operatorname{codim}(\overline{\{z\}}, X) = \dim \mathcal{O}_{X,z}$. Hence $\operatorname{codim}(Z, X) \leq \dim \mathcal{O}_{X,z}$ for every $z \in Z$. Thus $\operatorname{codim}(Z, X) = \inf_{z \in Z} \dim \mathcal{O}_{X,z}$, and the infimum is a minimum because the values lie in $\mathbb{N} \cup \{\infty\}$. □

Exercise 5.3

Problem. (a) Let Z be a closed subset of a topological space X. Show that $\operatorname{codim}(Z, X) = 0$ if and only if Z contains an irreducible component of X. Give an example (with X not irreducible) where $\operatorname{codim}(Z, X) = 0$ and $\dim Z < \dim X$.

(b) Let $X = \operatorname{Spec} \mathcal{O}_K[T]$ where $\mathcal{O}_K$ is a discrete valuation ring with uniformizing parameter $t \neq 0$. Let $f = tT - 1$. Show that the ideal generated by f is maximal. Let $x \in X$ be the corresponding point. Show that X is irreducible, $\dim \mathcal{O}_{X,x} = 1$, and $\operatorname{codim}(\{x\}, X) + \dim\{x\} < \dim X$.

Solution. (a) For an irreducible closed set Y, $\operatorname{codim}(Y, X) = 0$ means there is no chain $Y \subsetneq Z_0 \subsetneq Z_1$ of irreducible closed sets, i.e. Y is maximal among irreducible closed subsets of X, i.e. Y is an irreducible component of X. (If $Y \subsetneq W$ with W irreducible closed, take $Z_0 = Y$, $Z_1 = W$; conversely a chain gives such a W.) Since $\operatorname{codim}(Z, X)$ is the infimum of the $\operatorname{codim}(Z_j, X)$ over the components Z_j of Z, and these are integers ≥ 0 or ∞, it is 0 iff some component Z_j of Z is an irreducible component of X.

If Z contains an irreducible component C of X, then C is an irreducible subset of Z, hence lies in a component Z_j of Z. Since Z_j is an irreducible closed subset of X and C is maximal, $C = Z_j$, so $\operatorname{codim}(Z, X) = 0$. Conversely, if $\operatorname{codim}(Z, X) = 0$, some component Z_j of Z is a component of X, so Z contains one.

Example. Let $X = \operatorname{Spec} k[x, y, z]/(xz, yz) = V(z) \cup V(x, y) \subseteq \mathbb{A}^3_k$, the union of the plane $P = V(z) \simeq \operatorname{Spec} k[x, y]$ and the line $L = V(x, y) \simeq \operatorname{Spec} k[z]$. These are irreducible closed (their ideals (z), (x, y) are prime), neither contains the other, so they are the irreducible components of X (Prop. 4.5(c)). By Cor. 5.17, $\dim P = 2$ and $\dim L = 1$, so $\dim X = 2$ (Prop. 5.5(c)). Take $Z = L$. It is an irreducible component of X, so $\operatorname{codim}(Z, X) = 0$, while $\dim Z = 1 < 2 = \dim X$.

(b) Put $A = \mathcal{O}_K[T]$ and $K = \operatorname{Frac} \mathcal{O}_K$. Every non-zero element of $\mathcal{O}_K$ is a unit times a power of t, so $\mathcal{O}_K[1/t] = K$. The ring homomorphism $\psi : A \to K$ which is the inclusion on $\mathcal{O}_K$ and sends $T \mapsto 1/t$ is surjective and has f in its kernel. In A/fA the element t is invertible with inverse $\bar{T}$, so the structure map $\mathcal{O}_K \to A/fA$ extends to $K = \mathcal{O}_K[1/t] \to A/fA$; this map is surjective because A/fA is generated over the image of $\mathcal{O}_K$ by $\bar{T} = t^{-1}$. As A/fA surjects onto K it is non-zero, and a ring homomorphism from the field K to a non-zero ring is injective. So $K \simeq A/fA$ and fA is maximal. Let x be the corresponding point; it is a closed point.

A is an integral domain, so X is irreducible. The local ring $\mathcal{O}_{X,x} = A_{fA}$ has dimension $\operatorname{ht}(fA)$ (Prop. 5.8(b)). By Thm. 5.12 (Krull), $\operatorname{ht}(fA) \leq 1$; and $0 \subsetneq fA$ since $f \neq 0$ in the domain A, so $\operatorname{ht}(fA) = 1$.

Finally $\dim \mathcal{O}_K = 1$ (its primes are $0 \subsetneq t\mathcal{O}_K$) and $\mathcal{O}_K$ is Noetherian, so $\dim X = \dim \mathcal{O}_K[T] = 2$ (Cor. 5.17). Using Ex. 5.2, $\operatorname{codim}(\{x\}, X) = \dim \mathcal{O}_{X,x} = 1$, and $\dim\{x\} = 0$ because it is a point. So $1 + 0 < 2$. □

Exercise 5.4

Problem. Let A be a ring, $\mathfrak{p}_1, \ldots, \mathfrak{p}_r$ prime ideals and I an ideal contained in none of the $\mathfrak{p}_i$. Show $I \not\subseteq \mathfrak{p}_1 \cup \cdots \cup \mathfrak{p}_r$.

(a) Show the property for $r \leq 2$.

(b) Assume the property for $r - 1$ and that $\mathfrak{p}_r$ does not contain any $\mathfrak{p}_i$, $i \leq r - 1$. Let $x \in I \setminus (\mathfrak{p}_1 \cup \cdots \cup \mathfrak{p}_{r-1})$. Show that there exists $y \in (I\mathfrak{p}_1 \cdots \mathfrak{p}_{r-1}) \setminus \mathfrak{p}_r$.

(c) Show that either x or $x+y$ is not in $\mathfrak{p}_1 \cup \cdots \cup \mathfrak{p}_r$.

Solution. (a) For $r = 0$ the union is empty and $I \ni 0$. For $r = 1$ this is the hypothesis. For $r = 2$: pick $a \in I \setminus \mathfrak{p}_1$ and $b \in I \setminus \mathfrak{p}_2$. If $a \notin \mathfrak{p}_2$ or $b \notin \mathfrak{p}_1$ we are done. Otherwise $a \in \mathfrak{p}_2 \setminus \mathfrak{p}_1$ and $b \in \mathfrak{p}_1 \setminus \mathfrak{p}_2$. Then $a + b \in I$, and $a + b \notin \mathfrak{p}_1$ (else $a \in \mathfrak{p}_1$) and $a + b \notin \mathfrak{p}_2$ (else $b \in \mathfrak{p}_2$). (Only the ideal property is used here.)

(b) The ideal $I\mathfrak{p}_1 \cdots \mathfrak{p}_{r-1}$ is not contained in $\mathfrak{p}_r$: if it were, since $\mathfrak{p}_r$ is prime one of the factors would be contained in $\mathfrak{p}_r$, i.e. $I \subseteq \mathfrak{p}_r$ (excluded) or $\mathfrak{p}_i \subseteq \mathfrak{p}_r$ for some $i < r$ (excluded). Pick y in the product but not in $\mathfrak{p}_r$. (The element x exists by the property for $r-1$ applied to $\mathfrak{p}_1, \ldots, \mathfrak{p}_{r-1}$.)

(c) Suppose $x \in \mathfrak{p}_r$ (otherwise $x \notin \mathfrak{p}_1 \cup \cdots \cup \mathfrak{p}_r$ and we are done). Then $x + y \in I$. For $i < r$ we have $y \in \mathfrak{p}_i$ and $x \notin \mathfrak{p}_i$, so $x + y \notin \mathfrak{p}_i$. For $i = r$ we have $x \in \mathfrak{p}_r$ and $y \notin \mathfrak{p}_r$, so $x + y \notin \mathfrak{p}_r$.

Conclusion (induction on r). The case $r \leq 2$ is (a). Let $r \geq 3$ and assume the result for fewer than r primes (and every ideal). If $\mathfrak{p}_i \subseteq \mathfrak{p}_j$ for some $i \neq j$, we may delete $\mathfrak{p}_i$ without changing the union, and we conclude by induction. Otherwise no $\mathfrak{p}_i$ is contained in another, in particular $\mathfrak{p}_r$ contains no $\mathfrak{p}_i$ with $i < r$. By induction there is $x \in I \setminus (\mathfrak{p}_1 \cup \cdots \cup \mathfrak{p}_{r-1})$, and (b), (c) produce an element of I outside $\mathfrak{p}_1 \cup \cdots \cup \mathfrak{p}_r$. □

Exercise 5.5

Problem. Let A be a graded ring, $\mathfrak{p}_1, \ldots, \mathfrak{p}_r$ homogeneous prime ideals and I a homogeneous ideal contained in none of the $\mathfrak{p}_i$. Show that there is a homogeneous element of I not in $\bigcup_i \mathfrak{p}_i$. (One may suppose $\mathfrak{p}_r$ contains no $\mathfrak{p}_i$, $i < r$.) (a) Show that there is a homogeneous element $a \in I\mathfrak{p}_1 \cdots \mathfrak{p}_{r-1} \setminus \mathfrak{p}_r$. (b) Let $b \in I$ be homogeneous, $b \notin \bigcup_{i \leq r-1} \mathfrak{p}_i$. Show that b or $a^{\deg b} + b^{\deg a}$ is not in $\mathfrak{p}_1 \cup \cdots \cup \mathfrak{p}_r$. Conclude.

Solution. *The statement as printed is false if I contains non-zero elements of degree* 0. Let $A = k[s,t,x]$ with $\deg s = \deg t = 0$, $\deg x = 1$, and $\mathfrak{p}_1 = (s)$, $\mathfrak{p}_2 = (t,x)$ (homogeneous primes, neither contains the other), $I = (s,x)$. Then $I \not\subseteq \mathfrak{p}_1$ (as $x \notin (s)$) and $I \not\subseteq \mathfrak{p}_2$ (as $s \notin (t,x)$). But $I_0 = s\,k[s,t] \subseteq \mathfrak{p}_1$ and $I_d = k[s,t]x^d \subseteq \mathfrak{p}_2$ for $d \geq 1$, so every homogeneous element of I lies in $\mathfrak{p}_1 \cup \mathfrak{p}_2$.

The statement is true (and this is what the hint's proof gives) under the hypothesis $I \subseteq A_+ = \bigoplus_{d>0} A_d$. This is automatic when A_0 is a field and $I \neq A$ (a proper homogeneous ideal has $I_0 = 0$), in particular for the homogeneous algebras B over k of Ex. 5.15; and if $I = A$ the element 1 works. We prove it under $I \subseteq A_+$.

Induction on r; the cases $r = 0$ (take 0) and $r = 1$ are clear (a homogeneous ideal not contained in $\mathfrak{p}_1$ has a homogeneous generator outside $\mathfrak{p}_1$). Let $r \geq 2$. As in Ex. 5.4 we may delete any $\mathfrak{p}_i$ contained in another $\mathfrak{p}_j$, so we may assume no inclusions among the $\mathfrak{p}_i$ and, after renumbering, that $\mathfrak{p}_r$ contains no $\mathfrak{p}_i$ with $i < r$.

(a) The ideal $J = I\mathfrak{p}_1 \cdots \mathfrak{p}_{r-1}$ is homogeneous, generated by products of homogeneous elements. As in Ex. 5.4(b), $J \not\subseteq \mathfrak{p}_r$. A homogeneous ideal is generated by its homogeneous elements, so some homogeneous $a \in J$ is not in $\mathfrak{p}_r$. Then $a \in I$, $a \in \mathfrak{p}_i$ for $i < r$, and $a \neq 0$, so $\deg a \geq 1$ since $I \subseteq A_+$.

(b) By the induction hypothesis (for $\mathfrak{p}_1, \ldots, \mathfrak{p}_{r-1}$) there is a homogeneous $b \in I$ with $b \notin \mathfrak{p}_1 \cup \cdots \cup \mathfrak{p}_{r-1}$; $b \neq 0$, so $\deg b \geq 1$. If $b \notin \mathfrak{p}_r$ we are done. Suppose $b \in \mathfrak{p}_r$ and put $c = a^{\deg b} + b^{\deg a}$. Both summands are homogeneous of degree $\deg a \deg b$, and both lie in I (exponents ≥ 1), so c is a homogeneous element of I. For $i < r$: $a \in \mathfrak{p}_i$ gives $a^{\deg b} \in \mathfrak{p}_i$, while $b \notin \mathfrak{p}_i$ gives $b^{\deg a} \notin \mathfrak{p}_i$ (prime), so $c \notin \mathfrak{p}_i$. For $i = r$: $b \in \mathfrak{p}_r$ gives $b^{\deg a} \in \mathfrak{p}_r$, while $a \notin \mathfrak{p}_r$ gives $a^{\deg b} \notin \mathfrak{p}_r$, so $c \notin \mathfrak{p}_r$. Hence b or c is a homogeneous element of I outside $\mathfrak{p}_1 \cup \cdots \cup \mathfrak{p}_r$. □

Exercise 5.6

Problem. Let A be a Noetherian ring and I an ideal.

(a) Let $x \in I$. Show $\operatorname{ht}(I/xA) + 1 \geq \operatorname{ht}(I)$, with equality if x lies in no minimal prime ideal of A. Show that a prime minimal among those containing x has height 1 if x is a regular element.

(b) Show $\operatorname{ht}(I) \geq 1$ iff I is not contained in the union of the minimal prime ideals of A.

(c) Let $r = \operatorname{ht}(I)$. Show that I contains an ideal J generated by r elements with $\operatorname{ht}(J) = r$ and $\operatorname{ht}(I/J) = 0$.

Solution. We assume $I \neq A$ (for $I = A$ all heights are the infimum over the empty set). Recall $\operatorname{ht}(I) = \min\{\operatorname{ht}(P) : P \supseteq I \text{ prime}\}$, and heights are finite in a Noetherian ring: $\operatorname{ht}(P) = \dim A_P < \infty$ by Cor. 5.14(b). Primes of A/xA containing I/xA are the P/xA with $P \supseteq I$ prime (note $x \in I$), and $(A/xA)_{P/xA} = A_P/xA_P$.

(a) Choose a prime $P \supseteq I$ with $\operatorname{ht}(P/xA) = \operatorname{ht}(I/xA)$. Apply Thm. 5.15 to the Noetherian local ring A_P and $x \in PA_P$: $\dim A_P/xA_P \geq \dim A_P - 1$, that is $\operatorname{ht}(P/xA) \geq \operatorname{ht}(P) - 1$. Hence $\operatorname{ht}(I/xA) + 1 \geq \operatorname{ht}(P) \geq \operatorname{ht}(I)$.

Suppose x lies in no minimal prime of A. Choose a prime $P \supseteq I$ with $\operatorname{ht}(P) = \operatorname{ht}(I)$. The minimal primes of A_P are the QA_P with $Q \subseteq P$ a minimal prime of A, and $x \notin Q$, so $x/1$ lies in no minimal prime of A_P. By the equality case of Thm. 5.15, $\dim A_P/xA_P = \dim A_P - 1$, i.e. $\operatorname{ht}(P/xA) = \operatorname{ht}(I) - 1$. Thus $\operatorname{ht}(I/xA) \leq \operatorname{ht}(P/xA) = \operatorname{ht}(I) - 1$, and with the first part $\operatorname{ht}(I/xA) + 1 = \operatorname{ht}(I)$.

Now let x be regular (a non-zero-divisor). First, a minimal prime $\mathfrak{q}$ of A consists of zero divisors: $\mathfrak{q}A_\mathfrak{q}$ is the only prime of $A_\mathfrak{q}$, so it is the nilradical; for $u \in \mathfrak{q}$ there are $s \notin \mathfrak{q}$ and $n \geq 1$ with $su^n = 0$. Take n minimal; then $s \neq 0$ and $u \cdot (su^{n-1}) = 0$ with $su^{n-1} \neq 0$ (for $n = 1$ this is $s \neq 0$), so u is a zero divisor. Hence the regular element x is in no minimal prime. Let p be minimal among the primes containing x. By Thm. 5.12, $\operatorname{ht}(p) \leq 1$, and $\operatorname{ht}(p) \neq 0$ since p is not a minimal prime of A. So $\operatorname{ht}(p) = 1$.

(b) $\operatorname{ht}(I) = 0$ iff some prime $P \supseteq I$ has height 0, i.e. is minimal, i.e. iff I is contained in a minimal prime. By Prop. 4.9 there are finitely many minimal primes $\mathfrak{q}_1, \ldots, \mathfrak{q}_s$. If $\operatorname{ht}(I) \geq 1$, then $I \not\subseteq \mathfrak{q}_j$ for all j, so $I \not\subseteq \bigcup_j \mathfrak{q}_j$ by Ex. 5.4. Conversely if $I \not\subseteq \bigcup_j \mathfrak{q}_j$, then I is in no minimal prime, so $\operatorname{ht}(I) \geq 1$.

(c) We construct $x_1, \ldots, x_r \in I$ such that $J_i = (x_1, \ldots, x_i)$ has $\operatorname{ht}(J_i) = i$ for $0 \leq i \leq r$, starting with $J_0 = 0$ (height 0 since $A \neq 0$). Let $0 \leq i < r$ and suppose J_i is built. The ideal J_i has finitely many minimal primes (Prop. 4.9 applied to A/J_i), all of height $\geq \operatorname{ht}(J_i) = i$; let $P_1, \ldots, P_t$ be those of height exactly i (there is at least one). Since $\operatorname{ht}(I) = r > i$, $I \not\subseteq P_j$. By

Ex. 5.4 there is $x_{i+1} \in I \setminus \bigcup_j P_j$. Put $J_{i+1} = J_i + x_{i+1}A$. Let $Q \supseteq J_{i+1}$ be prime. Then Q contains a minimal prime P of J_i. If P is one of the P_j, then $Q \neq P$ because $x_{i+1} \in Q \setminus P$, so $\operatorname{ht}(Q) \geq i+1$. Otherwise $\operatorname{ht}(Q) \geq \operatorname{ht}(P) \geq i+1$. So $\operatorname{ht}(J_{i+1}) \geq i+1$, and $\leq i+1$ by Cor. 5.14(a). Hence $\operatorname{ht}(J_{i+1}) = i+1$.

Put $J = J_r \subseteq I$: generated by r elements, $\operatorname{ht}(J) = r$. Let $P \supseteq I$ be a prime with $\operatorname{ht}(P) = r$. Then $P \supseteq J$, so it contains a minimal prime Q of J with $\operatorname{ht}(Q) \geq \operatorname{ht}(J) = r$. If $Q \subsetneq P$ then $\operatorname{ht}(P) \geq \operatorname{ht}(Q) + 1 > r$, impossible. So $P = Q$ is a minimal prime of J containing I. Then P/J is a minimal prime of A/J containing I/J, i.e. $\operatorname{ht}(I/J) = 0$. □

Exercise 5.7

> **Problem.** Let $f : X \to Y$ be a morphism of algebraic varieties with $f(X)$ dense in Y. Show that $\dim X \geq \dim Y$ (use Prop. 5.19).

Solution. Algebraic varieties are Noetherian (Rem. 3.48), so X and Y have finitely many irreducible components $X_1, \ldots, X_m$ and $Y_1, \ldots, Y_s$ (Prop. 4.9), with generic points ξ_i, η_j (Prop. 4.12(a)). By Prop. 5.5(c) it suffices to show that for each j there is an i with $\dim Y_j \leq \dim X_i$.

Since $f(X) = \bigcup_i f(X_i)$, we have $Y = \overline{f(X)} = \bigcup_i \overline{f(X_i)}$, and each $\overline{f(X_i)} = \overline{\{f(\xi_i)\}}$ is irreducible and closed (as $f(X_i) = f(\overline{\{\xi_i\}}) \subseteq \overline{\{f(\xi_i)\}}$ by continuity). The irreducible set Y_j is covered by these finitely many closed sets, so $Y_j \subseteq \overline{\{f(\xi_i)\}}$ for some i; the latter is irreducible closed, hence contained in a component of Y, and by maximality of Y_j we get $Y_j = \overline{\{f(\xi_i)\}}$. By uniqueness of the generic point (Prop. 4.12(a)), $f(\xi_i) = \eta_j$.

Give X_i and Y_j their reduced structures; they are integral algebraic varieties over k (closed subschemes of varieties), with function fields the residue fields $k(\xi_i)$ and $k(\eta_j)$ at the generic points. The morphism f is a morphism of k-schemes, so it induces a k-homomorphism of residue fields $k(\eta_j) = k(f(\xi_i)) \to k(\xi_i)$, which is injective. Thus $k(\eta_j)$ is isomorphic over k to a subfield of $k(\xi_i)$, and a transcendence basis of $k(\eta_j)$ over k gives algebraically independent elements of $k(\xi_i)$. So $\operatorname{trdeg}_k k(\eta_j) \leq \operatorname{trdeg}_k k(\xi_i)$. By Prop. 5.19, $\dim Y_j = \operatorname{trdeg}_k k(\eta_j) \leq \operatorname{trdeg}_k k(\xi_i) = \dim X_i \leq \dim X$. □

Exercise 5.8

> **Problem.** Let $\mathcal{O}_K$ be as in Ex. 5.3, $K = \operatorname{Frac}(\mathcal{O}_K)$, $\kappa = \mathcal{O}_K/t\mathcal{O}_K$ the residue field. Let $A = K \times \kappa$ and $\varphi : \mathcal{O}_K \to A$ induced by $\mathcal{O}_K \to K$ and $\mathcal{O}_K \to \kappa$. Show that $f_\varphi : \operatorname{Spec} A \to \operatorname{Spec} \mathcal{O}_K$ is surjective and $\dim \mathcal{O}_K > \dim A$. Also show that A is a finitely generated $\mathcal{O}_K$-algebra (a quotient of a polynomial ring over $\mathcal{O}_K$).

Solution. The ring $A = K \times \kappa$ has exactly two prime ideals, $\mathfrak{a}_1 = 0 \times \kappa$ (kernel of the projection to K) and $\mathfrak{a}_2 = K \times 0$ (kernel of the projection to κ). Indeed, if $\mathfrak{a}$ is prime then, with $e_1 = (1,0)$, $e_2 = (0,1)$, we have $e_1 e_2 = 0 \in \mathfrak{a}$ and $e_1 + e_2 = 1 \notin \mathfrak{a}$, so exactly one e_i lies in $\mathfrak{a}$, and then $\mathfrak{a}$ is the preimage of a prime of the field K or κ, i.e. $\mathfrak{a}_1$ or $\mathfrak{a}_2$. Neither contains the other, so $\dim A = 0$. On the other hand $\dim \mathcal{O}_K = 1$.

Now $\varphi^{-1}(\mathfrak{a}_1) = \ker(\mathcal{O}_K \to K) = 0$ and $\varphi^{-1}(\mathfrak{a}_2) = \ker(\mathcal{O}_K \to \kappa) = t\mathcal{O}_K$. The spectrum of $\mathcal{O}_K$ is $\{0, t\mathcal{O}_K\}$, so f_φ is surjective. (This shows that the integrality hypothesis in Prop. 5.10

cannot be dropped, and that Ex. 5.7 fails for schemes: $\operatorname{Spec} A \to \operatorname{Spec} \mathcal{O}_K$ has dense image, and $\dim \operatorname{Spec} A = 0 < 1$.)

Finally, let $u = (t^{-1}, 0) \in A$ and consider the $\mathcal{O}_K$-algebra map $\theta : \mathcal{O}_K[T] \to A$, $T \mapsto u$. For $p = \sum_i c_i T^i$ we get $\theta(p) = \big(\sum_i c_i t^{-i},\ \bar{c}_0\big)$. Given $(\alpha, \beta) \in K \times \kappa$, lift β to $c_0 \in \mathcal{O}_K$. Since $\mathcal{O}_K[1/t] = K$, write $\alpha - c_0 = \sum_{i=0}^{m} d_i t^{-i}$ with $d_i \in \mathcal{O}_K$; then $d_0 = (td_0)t^{-1}$, so $\alpha - c_0 = \sum_{i=1}^{m} e_i t^{-i}$ with $e_1 = d_1 + td_0$ and $e_i = d_i$ for $i \geq 2$ (all in $\mathcal{O}_K$). Hence $p = c_0 + \sum_{i \geq 1} e_i T^i$ satisfies $\theta(p) = (\alpha, \beta)$. So θ is surjective and A is a finitely generated (indeed monogenic) $\mathcal{O}_K$-algebra. □

Exercise 5.9

> **Problem.** Let X be a scheme over a field k. Show that the points of $X(k)$ are closed in X. If X is an algebraic variety over k, show that $x \in X$ is closed iff $k(x)$ is algebraic over k.

Solution. Here $X(k)$ is the set of points x with $k(x) = k$ (the images of the k-morphisms $\operatorname{Spec} k \to X$).

Claim: if $x \in X$ and $k(x)$ is algebraic over k, then x is closed in X (for any k-scheme X). Let $V = \operatorname{Spec} B$ be an affine open subset of X containing x, and let x correspond to the prime $\mathfrak{p}$ of B. Then $k \subseteq B/\mathfrak{p} \subseteq k(x)$, so $B/\mathfrak{p}$ is an integral domain which is algebraic over k, hence a field: for $0 \neq b \in B/\mathfrak{p}$, the ring $k[b]$ is a domain which is a finite dimensional k-vector space, so it is a field and b is invertible in $k[b]$. Thus $\mathfrak{p}$ is maximal, and $\{x\}$ is closed in V. For an affine open V not containing x, $\{x\} \cap V = \emptyset$ is closed in V. Since the affine opens cover X, the complement of $\{x\}$ is open, i.e. $\{x\}$ is closed.

Points of $X(k)$ have $k(x) = k$, which is algebraic over k, so they are closed. This also proves ($\Leftarrow$) in the variety case.

($\Rightarrow$) Let X be an algebraic variety, x closed. Choose an affine open $V = \operatorname{Spec} B \ni x$ with B a finitely generated k-algebra (Def. 3.47), $x \leftrightarrow \mathfrak{p}$. Since $\{x\}$ is closed in X it is closed in V, so $\mathfrak{p} = \mathfrak{m}$ is maximal (Lemma 4.11(a): $\overline{\{x\}} = V(\mathfrak{p})$). Then $k(x) = B/\mathfrak{m}$ is a field which is a finitely generated k-algebra, hence a finite algebraic extension of k (Cor. 1.12). □

Exercise 5.10

> **Problem.** Let k be an uncountable field. Let X be an algebraic variety over k with $\dim X \geq 1$ and $(Y_n)_n$ a sequence of closed subsets with $\dim Y_n < \dim X$. Show $\bigcup_n Y_n \neq X$.
>
> (a) Show the result for $X = \mathbb{A}^1_k$, then for $\mathbb{A}^m_k$. (b) Using Noether normalization, show the result for an arbitrary affine variety. Deduce the general case.

Solution. (a) *Claim: for $m \geq 1$ and non-zero polynomials $g_n \in k[T_1, \ldots, T_m]$ ($n \in \mathbb{N}$) there is $a \in k^m$ with $g_n(a) \neq 0$ for all n.* For $m = 1$ each g_n has finitely many roots, so the set of all roots of all g_n is countable, while k is not; pick $a \in k$ outside it. For $m \geq 2$, write $g_n = \sum_j h_{n,j}(T_1, \ldots, T_{m-1}) T_m^j$ and choose j_n with $h_n := h_{n,j_n} \neq 0$. By induction there is $a' \in k^{m-1}$ with $h_n(a') \neq 0$ for all n. Then the one-variable polynomials $g_n(a', T_m)$ are non-zero, and the case $m = 1$ gives $a_m \in k$ with $g_n(a', a_m) \neq 0$ for all n.

Now let $Y_n \subseteq \mathbb{A}_k^m$ be closed with $\dim Y_n < m = \dim \mathbb{A}_k^m$ (Cor. 5.17), so $Y_n \neq \mathbb{A}_k^m$. Write $Y_n = V(J_n)$; $J_n \neq 0$ (as $V(0)$ is everything), so choose $0 \neq g_n \in J_n$; then $Y_n \subseteq V(g_n)$. With a as in the claim, the maximal ideal $\mathfrak{m}_a = (T_i - a_i)_i$ is not in any $V(g_n)$, since $g_n \in \mathfrak{m}_a \iff g_n(a) = 0$. So $\mathfrak{m}_a \notin \bigcup_n Y_n$.

(b) *Affine case.* Let $X = \operatorname{Spec} A$, A a finitely generated k-algebra, $\dim X = d \geq 1$ (so $A \neq 0$). By Prop. 1.9 there is a finite injective homomorphism $\phi : A_0 = k[T_1, \ldots, T_e] \to A$. By Prop. 5.10(b) and Cor. 5.17, $\dim A = \dim A_0 = e$, so $e = d$. Let $\pi : X \to \mathbb{A}_k^d$ be the induced morphism, $Y_n = V(J_n)$ and $Z_n = V(\phi^{-1}(J_n)) \subseteq \mathbb{A}_k^d$. We claim $\pi(Y_n) = Z_n$. The inclusion $\subseteq$ is clear. For $\supseteq$: the induced map $A_0/\phi^{-1}(J_n) \to A/J_n$ is injective and finite, so by Prop. 5.10(b) $\operatorname{Spec}(A/J_n) \to \operatorname{Spec}(A_0/\phi^{-1}J_n)$ is surjective, i.e. every prime of A_0 containing $\phi^{-1}(J_n)$ is the inverse image of a prime of A containing J_n. Moreover Prop. 5.10(b) and Prop. 5.8(a) give $\dim Z_n = \dim A_0/\phi^{-1}(J_n) = \dim A/J_n = \dim Y_n < d$. By (a) there is a point $z \in \mathbb{A}_k^d$ outside $\bigcup_n Z_n$. Since ϕ is injective and integral, π is surjective (Prop. 5.10(b)); take $x \in X$ with $\pi(x) = z$. If $x \in Y_n$ then $z \in \pi(Y_n) = Z_n$, a contradiction. So $x \notin \bigcup_n Y_n$.

General case. X is covered by finitely many affine open subvarieties $U_1, \ldots, U_s$, and $\dim X = \max_i \dim U_i$ (Rem. 5.6), so there is an affine open $U = U_i$ with $\dim U = \dim X =: d \geq 1$. The sets $Y_n \cap U$ are closed in U with $\dim(Y_n \cap U) \leq \dim Y_n < d$ (Prop. 5.5(a)). By the affine case some point of U is outside all $Y_n \cap U$, hence outside $\bigcup_n Y_n$. □

Exercise 5.11

Problem. (Schemes of dimension 0.)

(a) Let X be a scheme which is a (finite or not) disjoint union of open subschemes X_i. Show $\mathcal{O}_X(X) \simeq \prod_i \mathcal{O}_X(X_i)$.

(b) Show that any scheme of finite cardinality and dimension 0 is affine.

(c) Let $X = \operatorname{Spec} A$ be a scheme of finite cardinality and dimension 0. Show that every point $x \in X$ is open. Deduce that $A \simeq \bigoplus_{\mathfrak{p} \in \operatorname{Spec} A} A_{\mathfrak{p}}$.

(d) Show that (c) is false if we do not assume $\operatorname{Spec} A$ of dimension 0.

Solution. (a) The X_i form an open cover of X with $X_i \cap X_j = \emptyset$ for $i \neq j$, and $\mathcal{O}_X(\emptyset) = 0$. By the sheaf axioms, $s \mapsto (s|_{X_i})_i$ is injective (a section with all restrictions zero is zero), and any family $(s_i)_i$ with $s_i \in \mathcal{O}_X(X_i)$ is compatible (restrictions to $X_i \cap X_j = \emptyset$ are both 0), so it glues to a section of $\mathcal{O}_X(X)$. Hence the ring homomorphism $\mathcal{O}_X(X) \to \prod_i \mathcal{O}_X(X_i)$ is an isomorphism.

(b),(c) Let X be a scheme with finitely many points and $\dim X = 0$. *Every point is closed:* let $y \in \overline{\{x\}}$. Then $\overline{\{y\}} \subseteq \overline{\{x\}}$ are irreducible closed sets, hence equal because $\dim X = 0$. So x and y have the same closure, hence the same open neighborhoods (an open set contains x iff it meets $\overline{\{x\}}$). By the T_0 property of schemes (Ex. 4.7), $x = y$. So $\overline{\{x\}} = \{x\}$. As X is finite, every subset is a finite union of closed points, hence closed; so every subset is also open, and X is discrete. This proves the first assertion of (c) (in particular for $X = \operatorname{Spec} A$).

Each one-point open subscheme $\{x\}$ is affine: choose an affine open $U = \operatorname{Spec} A' \ni x$; the non-empty open subset $\{x\}$ of U contains some non-empty principal open $D(g)$, which must equal $\{x\}$; hence $\{x\} = D(g) \simeq \operatorname{Spec} A'_g$. So $X = \{x_1\} \sqcup \cdots \sqcup \{x_n\}$ with $\{x_i\} = \operatorname{Spec} A_i$.

Put $A = \prod_i A_i$ with idempotents e_i. A prime ideal of A omits exactly one of the e_i: at most one, because $e_i e_j = 0$ for $i \neq j$, and at least one, because $\sum_i e_i = 1$. Hence $\operatorname{Spec} A$ is the disjoint union of the open sets $D(e_i) = \operatorname{Spec} A_{e_i} = \operatorname{Spec} A_i$. The morphism from $\bigsqcup_i \operatorname{Spec} A_i$ to $\operatorname{Spec} A$ which is, on each piece, the open immersion onto $D(e_i)$ is therefore an isomorphism. So $X \simeq \operatorname{Spec} A$ is affine, proving (b).

For the second assertion of (c), apply (a) to $X = \bigsqcup_{\mathfrak{p}} \{\mathfrak{p}\}$: $A = \mathcal{O}_X(X) \simeq \prod_{\mathfrak{p}} \mathcal{O}_X(\{\mathfrak{p}\})$. Since $\{\mathfrak{p}\}$ is the smallest open neighborhood of $\mathfrak{p}$, $\mathcal{O}_X(\{\mathfrak{p}\}) = \mathcal{O}_{X,\mathfrak{p}} = A_{\mathfrak{p}}$, the restriction map being $a \mapsto a/1$. As the product is finite it equals the direct sum: $A \simeq \bigoplus_{\mathfrak{p}} A_{\mathfrak{p}}$, via $a \mapsto (a/1)_{\mathfrak{p}}$.

(d) Let $A = \mathcal{O}_K$ be a discrete valuation ring (e.g. $\mathbb{Z}_{(p)}$). Then $\operatorname{Spec} A = \{0, \mathfrak{m}\}$ is finite of dimension 1. The map $A \to A_{(0)} \times A_{\mathfrak{m}} = K \times A$, $a \mapsto (a, a)$, is not surjective (its image is the diagonal, which does not contain $(1, 0)$); indeed A is a domain, so it is not a non-trivial product at all. Also the closed point $\mathfrak{m}$ is not open. So (c) fails. □

Exercise 5.12

Problem. Let k be a field. We determine the affine open subsets of $\mathbb{A}^n_k$ and $\mathbb{P}^n_k$.

(a) Show that the principal open subsets of $\mathbb{A}^n_k$ and of $\mathbb{P}^n_k$ (not equal to $\mathbb{P}^n_k$) are affine.

(b) Let $X = \bigcup_i D(f_i)$ be a finite union of principal open subsets of $\mathbb{A}^n_k$. Show $\mathcal{O}_{\mathbb{A}^n_k}(X) = k[T_1, \ldots, T_n]_f$, where $f = \gcd\{f_i\}$. Show that every affine open subset of $\mathbb{A}^n_k$ is principal.

(c) Show that every irreducible closed subset of $\mathbb{P}^n_k$ of dimension $n-1$ is principal.

(d) Let X be an affine open subset of $\mathbb{P}^n_k$. Show that the irreducible components of $\mathbb{P}^n_k \setminus X$ have dimension $n-1$, and that X is a principal open subset of $\mathbb{P}^n_k$.

Solution. Let $R = k[T_1, \ldots, T_n]$, a UFD with fraction field K. In $\mathbb{P}^n_k = \operatorname{Proj} B$, $B = k[T_0, \ldots, T_n]$, the principal open sets are the $D_+(F)$ with F homogeneous (those different from $\mathbb{P}^n_k$ have $\deg F \geq 1$ or $F = 0$), and a principal closed subset is read as a set $V_+(F)$ with F homogeneous (global sections of $\mathcal{O}_{\mathbb{P}^n_k}$ are only the constants, so this is the meaning in $\mathbb{P}^n_k$).

(a) $D(f) \subseteq \mathbb{A}^n_k$ is isomorphic to $\operatorname{Spec} R_f$. For $F \in B$ homogeneous of positive degree, $D_+(F) \simeq \operatorname{Spec} B_{(F)}$ (Lemma 3.36 and the construction of the scheme structure on $\operatorname{Proj} B$, Prop. 3.38). Both are affine. (If $F = 0$ the open set is empty, which is affine.)

(b) We may drop the f_i equal to 0 (as $D(0) = \emptyset$); if none is left, $X = \emptyset$, and $f = 0$, $R_0 = 0$. So assume some $f_i \neq 0$; then $f = \gcd\{f_i\}$ is defined up to a unit. Since $\mathbb{A}^n_k$ is integral, Prop. 4.18(c) gives, inside K,

$$\mathcal{O}(X) = \bigcap_{x \in X} \mathcal{O}_x = \bigcap_i \bigcap_{x \in D(f_i)} \mathcal{O}_x = \bigcap_i \mathcal{O}(D(f_i)) = \bigcap_i R_{f_i}.$$

We show $\bigcap_i R_{f_i} = R_f$. Since $f \mid f_i$, inverting f_i inverts f, so $R_{f_i} \supseteq R_f$. Conversely let $u \in \bigcap_i R_{f_i}$ and write $u = a/b$ with $a, b \in R$ coprime. For each i, $u = c/f_i^m$ for some $c \in R$, $m \geq 0$, so $a f_i^m = bc$, hence $b \mid a f_i^m$ and so $b \mid f_i^m$ (coprimality). Thus every prime factor π of b divides every f_i, hence divides f. Write $b = \epsilon \prod_j \pi_j^{e_j}$ with ϵ a unit; with $N = \max e_j$ we get $b \mid f^N$, so $u = a(f^N/b)/f^N \in R_f$. This proves $\mathcal{O}(X) = R_f$.

Now let $X \subseteq \mathbb{A}^n_k$ be an affine open subset. If $X = \emptyset$ then $X = D(0)$. Otherwise $X = \mathbb{A}^n_k \setminus V(I)$ with $I = (f_1, \ldots, f_s)$ (R is Noetherian), so $X = \bigcup D(f_i)$ and $\mathcal{O}(X) = R_f$ with $f = \gcd\{f_i\}$.

Since $f \mid f_i$, $D(f_i) \subseteq D(f)$, so $X \subseteq D(f)$, and the restriction map $\mathcal{O}(D(f)) = R_f \to \mathcal{O}(X) = R_f$ is the identity (both are subrings of K). The open immersion $j : X \to D(f) = \operatorname{Spec} R_f$ is a morphism of affine schemes, so by Prop. 3.25 it is the morphism induced by the ring homomorphism $\mathcal{O}(D(f)) \to \mathcal{O}(X)$, which is an isomorphism; hence j is an isomorphism, and in particular surjective. So $X = D(f)$ is principal.

(c) Let $Z \subseteq \mathbb{P}^n_k$ be irreducible closed of dimension $n-1$. Then $Z = V_+(\mathfrak{p})$ for a homogeneous prime $\mathfrak{p} \not\supseteq B_+$ (the usual correspondence between irreducible closed subsets of $\operatorname{Proj} B$ and such primes), and Z is homeomorphic to $\operatorname{Proj}(B/\mathfrak{p})$. As $\mathfrak{p} \subseteq B_+$ the quotient $B/\mathfrak{p}$ is a homogeneous k-algebra, and Cor. 5.21 gives $\dim B/\mathfrak{p} = \dim Z + 1 = n$. Since $\dim B = n+1$ (Cor. 5.17) and B is a finitely generated domain, Prop. 5.23(a) gives $\operatorname{ht}(\mathfrak{p}) = (n+1) - n = 1$. Take a non-zero $g \in \mathfrak{p}$ and an irreducible factor π of g lying in $\mathfrak{p}$ ($\mathfrak{p}$ is prime). Then (π) is a non-zero prime of the UFD B with $0 \subsetneq (\pi) \subseteq \mathfrak{p}$, so $\mathfrak{p} = (\pi)$ because $\operatorname{ht}(\mathfrak{p}) = 1$. The generator π is homogeneous: let $\pi_a \neq 0$ be its lowest homogeneous component; since $\mathfrak{p}$ is homogeneous, $\pi_a \in \mathfrak{p} = (\pi)$, so $\pi_a = \pi h$ with $h \neq 0$. Comparing total degrees in the domain B, $a = \deg \pi_a = \deg \pi + \deg h \geq \deg \pi \geq a$, so $\deg \pi = a$ and $\pi = \pi_a$. Hence $Z = V_+(\pi)$ with π homogeneous, i.e. Z is principal.

(d) Let $n \geq 1$ and let $X \neq \emptyset$ be an affine open subset. (For $X = \emptyset$ the first assertion is false, since the complement is $\mathbb{P}^n_k$ of dimension n, but $X = D_+(0)$ is principal anyway; for $n = 0$ everything is trivial.) Put $Z = \mathbb{P}^n_k \setminus X$. Then $Z \neq \emptyset$: otherwise $X = \mathbb{P}^n_k$ would be affine with $\mathcal{O}(\mathbb{P}^n_k) = k$, i.e. a point, which is absurd for $n \geq 1$. Let $W_i = D_+(T_i) \simeq \mathbb{A}^n_k$. The scheme $\mathbb{P}^n_k$ is separated (Example 3.3.7, which works over k as over $\mathbb{Z}$), so the intersection of the affine open sets X and W_i is affine (Prop. 3.3.6). By (b), $X \cap W_i = D(g_i)$ for some $g_i \in \mathcal{O}(W_i)$. As $\mathbb{P}^n_k$ is irreducible and $X \neq \emptyset$, $X \cap W_i \neq \emptyset$ (non-empty opens meet), so $g_i \neq 0$, and g_i is not nilpotent in the domain $\mathcal{O}(W_i)$. Hence $Z \cap W_i = V(g_i)$ is either empty or, by Cor. 5.26, every irreducible component of it has dimension $n-1$.

Let Z_1 be an irreducible component of Z. It meets some W_i, and $Z_1 \cap W_i$ is an irreducible component of $Z \cap W_i$ (Prop. 4.5(b)). So $\dim(Z_1 \cap W_i) = n-1$, and since a non-empty open subset of the irreducible variety Z_1 (reduced structure) has the same dimension (Prop. 5.19), $\dim Z_1 = n-1$.

Let $Z_1, \ldots, Z_s$ be the (finitely many) components of Z. By (c), $Z_j = V_+(F_j)$ with F_j homogeneous of positive degree. With $F = F_1 \cdots F_s$ we get $Z = \bigcup_j V_+(F_j) = V_+(F)$ and $X = \mathbb{P}^n_k \setminus V_+(F) = D_+(F)$, a principal open subset. $\square$

Exercise 5.13

> **Problem.** Let k be a field. A function field in n variables over k is a field K finitely generated over k with $\operatorname{trdeg}_k K = n$. Show that an extension K of k is a function field in n variables over k if and only if $K = K(X)$ for an integral algebraic variety X over k of dimension n. If so, X can be taken projective.

Solution. ($\Leftarrow$) Let X be an integral algebraic variety of dimension n, with a non-empty affine open $U = \operatorname{Spec} A$, A a finitely generated k-algebra which is a domain. Then $K(X) = \mathcal{O}_{X,\xi} = \operatorname{Frac}(A)$ (Prop. 4.18(a)), which is generated as a field by the finitely many generators of A over k, and $\operatorname{trdeg}_k K(X) = \dim X = n$ by Prop. 5.19.

($\Rightarrow$) Let $K = k(f_1, \ldots, f_r)$ with $\operatorname{trdeg}_k K = n$. Let $\mathfrak{q}$ be the kernel of $k[T_1, \ldots, T_r] \to K$, $T_i \mapsto f_i$, and $A = k[T_1, \ldots, T_r]/\mathfrak{q} \simeq k[f_1, \ldots, f_r] \subseteq K$. It is a domain with $\operatorname{Frac}(A) = K$. Let

$C = V(\mathfrak{q}) \simeq \operatorname{Spec} A$, a closed irreducible subset of $\mathbb{A}^r_k = D_+(T_0) \subseteq \mathbb{P}^r_k$. Let X be the closure of C in $\mathbb{P}^r_k$ with its reduced induced closed subscheme structure (Section 2.4): it is irreducible and reduced, hence integral, and it is a closed subscheme of $\mathbb{P}^r_k$, so a projective (hence algebraic) variety. The intersection $X \cap D_+(T_0)$ is the closure of C in $D_+(T_0)$, which is C (closed). The reduced closed subscheme of $\mathbb{A}^r_k$ with underlying set C is $\operatorname{Spec} k[T]/\mathfrak{q} = \operatorname{Spec} A$ since $\mathfrak{q}$ is prime, so $X \cap D_+(T_0) \simeq \operatorname{Spec} A$ is a non-empty open subscheme of X. By Prop. 4.18(a), $K(X) = \operatorname{Frac}(A) = K$, and by Prop. 5.19, $\dim X = \operatorname{trdeg}_k K = n$. □

Exercise 5.14

Problem. Let L/K be a finite field extension. For $x \in L$, multiplication by x is a K-linear endomorphism of L; $\operatorname{Norm}_{L/K}(x)$ is its determinant.

(a) Show that $\operatorname{Norm}_{L/K}$ is a multiplicative map $L \to K$.

(b) Let $A \subseteq B$ be rings with $K = \operatorname{Frac}(A)$, $L = \operatorname{Frac}(B)$, B integral over A (A, B domains, L/K finite). Show that for any $b \in B$, $\operatorname{Norm}_{L/K}(b)$ is integral over A.

(c) Suppose moreover that A is a polynomial ring over a field k. Show $\operatorname{Norm}_{L/K}(B) \subseteq A$.

Solution. (a) Let $m_x : L \to L$ denote multiplication by x. Then $m_{xy} = m_x \circ m_y$, so $\operatorname{Norm}(xy) = \det(m_x)\det(m_y) = \operatorname{Norm}(x)\operatorname{Norm}(y)$; also $m_1 = \mathrm{id}$ gives $\operatorname{Norm}(1) = 1$.

(b) Let $b \in B$ and let $\mu \in K[T]$ be its minimal polynomial over K, of degree $d = [K(b) : K]$, and $e = [L : K(b)]$. Choosing the K-basis $1, b, \ldots, b^{d-1}$ of $K(b)$ and a $K(b)$-basis of L with e elements, the matrix of m_b is block diagonal with e copies of the companion matrix of μ. So its characteristic polynomial is $\chi(T) = \mu(T)^e$, and $\operatorname{Norm}(b) = \det(m_b) = (-1)^{de}\chi(0) = (-1)^{de}\mu(0)^e$. Let Ω be an algebraic closure of K and $\beta_1, \ldots, \beta_d \in \Omega$ the roots of μ (with multiplicity), so $\mu(0) = (-1)^d \prod_j \beta_j$ and $\operatorname{Norm}(b) = \pm(\prod_j \beta_j)^e$. Since b is integral over A, it is a root of a monic $P \in A[T]$; μ divides P in $K[T]$, so each β_j is a root of P and thus integral over A. The elements of Ω integral over A form a ring, so $\operatorname{Norm}(b)$, a product of β_j's up to sign, is integral over A.

(c) A polynomial ring A over k is a UFD, hence integrally closed in K: if $u = p/q$ with p, q coprime satisfies $u^m + a_{m-1}u^{m-1} + \cdots + a_0 = 0$ with $a_i \in A$, then $p^m = -q(a_{m-1}p^{m-1} + \cdots + a_0 q^{m-1})$, so any prime factor of q divides p^m, hence p, a contradiction; so q is a unit and $u \in A$. By (b), and since $\operatorname{Norm}(b) \in K$, we conclude $\operatorname{Norm}_{L/K}(b) \in A$ for all $b \in B$. □

Exercise 5.15

Problem. Let $X = \operatorname{Proj} B$ be a projective variety over a field k.

(a) Let $f \in B_+$ be homogeneous and $\mathfrak{p}$ a prime ideal of B minimal among those containing f. Show that $\mathfrak{p}$ is homogeneous, contained in B_+, and $\operatorname{ht}(\mathfrak{p}) \leq 1$. Show that if $V_+(f) = \emptyset$ then $B_+ = \mathfrak{p}$ and $\dim X \leq 0$.

(b) Let Y be a closed subvariety of X of dimension r. Show that for any homogeneous $f_1, \ldots, f_r \in B_+$, $V_+(f_1, \ldots, f_r) \cap Y \neq \emptyset$.

Solution. Throughout, B is a homogeneous k-algebra: $B = \bigoplus_{d \geq 0} B_d$, $B_0 = k$, generated by finitely many elements of B_1; in particular B is Noetherian and B_+ is a maximal ideal. Recall

that the irreducible closed subsets of $\operatorname{Proj} B$ are the sets $V_+(\mathfrak{q})$ with $\mathfrak{q}$ a homogeneous prime ideal not containing B_+, that $V_+(\mathfrak{q})$ is homeomorphic to $\operatorname{Proj}(B/\mathfrak{q})$, and that $B/\mathfrak{q}$ is again a homogeneous k-algebra and a domain.

(a) *Homogeneity.* The ideal $\mathfrak{p}^h = \bigoplus_d(\mathfrak{p} \cap B_d) \subseteq \mathfrak{p}$ is prime (Lemma 3.35(a)) and contains f (a homogeneous element of $\mathfrak{p}$). By minimality $\mathfrak{p} = \mathfrak{p}^h$ is homogeneous. *Containment.* A proper homogeneous ideal $\mathfrak{a}$ has $\mathfrak{a}_0$ a proper ideal of the field $B_0 = k$, so $\mathfrak{a}_0 = 0$ and $\mathfrak{a} \subseteq B_+$. Hence $\mathfrak{p} \subseteq B_+$. *Height.* f is not invertible (it lies in B_+), so $\operatorname{ht}(\mathfrak{p}) \leq 1$ by Thm. 5.12.

Suppose $V_+(f) = \emptyset$. If $\mathfrak{p} \neq B_+$ then $\mathfrak{p} \subsetneq B_+$ does not contain B_+, so $\mathfrak{p} \in \operatorname{Proj} B$ and contains f, i.e. $\mathfrak{p} \in V_+(f)$, a contradiction. Hence $\mathfrak{p} = B_+$, and $\operatorname{ht}(B_+) \leq 1$.

To show $\dim X \leq 0$, let $Z = V_+(\mathfrak{q})$ be an irreducible component of X and $D = B/\mathfrak{q}$, a homogeneous k-domain with $\operatorname{Proj} D \simeq Z$ non-empty. The image $\bar{f} \in D_+$ satisfies $V_+(\bar{f}) = Z \cap V_+(f) = \emptyset$. By what we just proved applied to D, any prime of D minimal over $\bar{f}D$ equals D_+, and $\operatorname{ht}(D_+) \leq 1$. By Prop. 5.23(a) (for the finitely generated domain D), $\dim D = \operatorname{ht}(D_+) + \dim D/D_+ = \operatorname{ht}(D_+) \leq 1$, since $D/D_+ = k$. By Cor. 5.21, $\dim Z = \dim D - 1 \leq 0$. Since $\dim X$ is the supremum of the dimensions of its irreducible components (Prop. 5.5(c)), $\dim X \leq 0$.

(b) *Lemma.* Let $W \subseteq X$ be an irreducible closed subset with $\dim W \geq 1$ and let $g \in B_+$ be homogeneous. Then $W \cap V_+(g)$ contains a non-empty irreducible closed subset of dimension $\geq \dim W - 1$.

Proof of the Lemma. Write $W = V_+(\mathfrak{q}) \simeq \operatorname{Proj} D$ with $D = B/\mathfrak{q}$ a homogeneous domain over k, and let $\bar{g} \in D_+$ be the image of g. Then $W \cap V_+(g) \simeq V_+(\bar{g}) \subseteq \operatorname{Proj} D$. By Cor. 5.21, $\dim D = \dim W + 1 \geq 2$. If $\bar{g} = 0$ then $W \cap V_+(g) = W$ and we are done. Otherwise, let $\mathfrak{p}_1, \ldots, \mathfrak{p}_t$ be the minimal primes of $\bar{g}D$ (finitely many, Prop. 4.9); they are homogeneous and contained in D_+ by (a) applied to D. Each has height exactly 1 (height ≤ 1 by Thm. 5.12, and $\neq 0$ because D is a domain and $\bar{g} \neq 0$). If all $\mathfrak{p}_j$ were equal to D_+ then $V_+(\bar{g}) = \emptyset$ (a homogeneous prime containing $\bar{g}$ contains some $\mathfrak{p}_j = D_+$), and part (a) applied to D would give $\dim W \leq 0$, contrary to the hypothesis. So some $\mathfrak{p}_j \neq D_+$; then $\mathfrak{p}_j \subsetneq D_+$, so $\mathfrak{p}_j \in \operatorname{Proj} D$ and $V_+(\mathfrak{p}_j) \simeq \operatorname{Proj}(D/\mathfrak{p}_j)$ is a non-empty irreducible closed subset of $V_+(\bar{g})$. By Prop. 5.23(a), $\dim D/\mathfrak{p}_j = \dim D - \operatorname{ht}(\mathfrak{p}_j) = \dim D - 1$, and by Cor. 5.21 (as $D/\mathfrak{p}_j$ is a homogeneous algebra with non-empty Proj), $\dim V_+(\mathfrak{p}_j) = \dim D/\mathfrak{p}_j - 1 = \dim D - 2 = \dim W - 1$. This proves the Lemma.

Now let Y be a closed subset of X of dimension r and $f_1, \ldots, f_r \in B_+$ homogeneous. If $r = 0$ the empty sequence gives $V_+() = X$ and $Y \neq \emptyset$, so the claim holds. Let $r \geq 1$. Y has finitely many irreducible components and $\dim Y$ is the maximum of their dimensions, so Y contains an irreducible closed subset W_0 with $\dim W_0 = r$. Inductively for $i = 1, \ldots, r$, given an irreducible closed $W_{i-1} \subseteq Y \cap V_+(f_1, \ldots, f_{i-1})$ with $\dim W_{i-1} \geq r - i + 1 \geq 1$, the Lemma applied to W_{i-1} and f_i gives a non-empty irreducible closed $W_i \subseteq W_{i-1} \cap V_+(f_i)$ with $\dim W_i \geq r - i$. Finally W_r is non-empty and contained in $Y \cap V_+(f_1, \ldots, f_r)$. Hence $V_+(f_1, \ldots, f_r) \cap Y \neq \emptyset$. $\square$